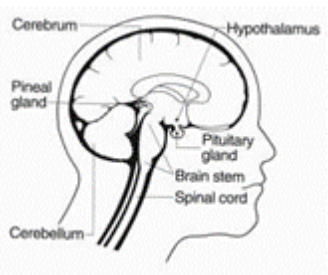
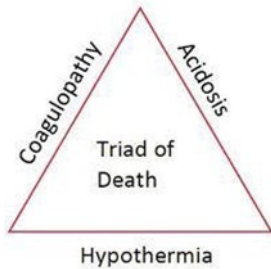


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ADULT – EMERGENCY SURGICAL CRICOTHYROTOMY

OVERVIEW

A cricothyrotomy is a surgical procedure to establish an emergency airway. It is an invasive procedure with multiple inherent complications and should be performed only on patients that are at high risk of death if an immediate airway is not established. One must first consider/attempt all alternative airway measures (e.g. OPA/NPA, ET intubation (direct line of site or video laryngoscope), Supraglottic device). While every attempt should be made to transport to the closest emergency department for a more controlled setting, but no patient under the care of Sumner County EMS should die secondary to airway obstruction.

REQUIREMENTS:

- Be a licensed paramedic credentialed through the Deputy Chief of Training.
- Must have completed bi-annual training sessions as required by Sumner County EMS.

INDICATIONS:

- **Attempt of visualization of the airway by direct line of site or video laryngoscope.**
- **Inability to intubate and/or ventilate.**
- Patient may include those with severe facial trauma or total airway obstruction not relieved by other methods.

COMPLICATIONS:

- Bleeding (can be severe).
- Misplacement (esophageal or soft-tissue placement).
- Damage to surrounding structures such as vocal cords, esophageal or tracheal damage.
- Infection.

CONTRAINDICATIONS:

- Given that you will only be performing this procedure on patients who have a very high probability of dying without it, most contraindications would therefore be relative. The one absolute contraindication would be pediatric patients ≤ 10 years old as the available equipment is too large for this population. Pediatric patients may be treated under the pediatric emergency cricothyrotomy protocol. The following are examples of patients with a high risk of complications:
 - Age ≤ 10 years old (Peds patients – Refer to Pediatric NEEDLE Cricothyrotomy Protocol).
 - Bleeding disorder.
 - Massive neck swelling (blood/tumor).

EQUIPMENT NEEDED	
Full BSI / PPE is recommended	Bougie stylet
Sharps container	6.0 ET Tube (other sizes may be used PRN)
Suction device	10 mL syringe
BVM with O2 supply	EtCO2 monitoring
Antiseptic / Betadine swabs	Securing device
#10 scalpel	Bandaging materials

ADULT – EMERGENCY SURGICAL CRICOTHYROTOMY (Procedure)

Position the patient supine and extend the neck as needed to improve anatomic view.

Prep neck with antiseptic.

Using your non-dominant hand, stabilize the larynx and locate the following landmarks:

- Thyroid cartilage (Adam's apple) and cricoid cartilage.
- The cricothyroid membrane lies between these cartilages.

Make an approximate 3 cm vertical incision 0.5 cm deep through the skin and fascia, over the cricothyroid membrane. With a finger, dissect the tissue and locate the cricothyroid membrane.

Make an approximate 1.5 cm horizontal incision through the cricothyroid membrane.

With your finger, bluntly dilate the opening through the cricothyroid membrane.

Insert the bougie curved tip first through the incision and angled towards the patient's feet.

Advance the bougie into the trachea feeling for "clicks" of tracheal rings and until "hang-up" when it cannot be advanced any further. This confirms tracheal position.

Advance a 6.0 mm ET Tube (ensure all air is aspirated out of the cuff) over the bougie and into the trachea.

Remove the bougie while stabilizing ET Tube, ensuring it does not become dislodged.

Inflate the cuff with 5-10 mL of air.

Confirm appropriate proper placement by:

- Symmetrical chest wall rise and fall.
- Auscultation of equal breath sounds and NO epigastric sounds upon ventilating.
- Condensation in the ET Tube.
- Capnography.

Secure the ET Tube.

Bandage if necessary.

Reassess tube placement frequently, especially after movement of the patient.

Ongoing monitoring of ET Tube placement and ventilation status using **waveform capnography is absolutely required.**

FACILITATED INTUBATION AND RSI

OVERVIEW:

Rapid sequence induction/intubation (RSI) is a series of maneuvers utilizing sedation and paralysis to establish an advanced airway in a critically ill patient. This is an advanced procedure with a potential for high-risk complications and should only be performed as an absolute life-saving procedure. It should only be performed after all other less invasive forms of airway control have been attempted or considered. At no time should a paramedic feel pressured to perform this procedure if he or she is not comfortable with its application on a given patient. Updated 2019 – ***All paramedics may initiate this procedure with the administration of choice sedatives (chemically facilitated intubation). However, you must meet the requirements below in order to perform full RSI, to include administration of paralytics.***

REQUIREMENTS TO PERFORM RSI (Use of paralytics):

- Be a licensed paramedic and employee of Sumner County EMS for at least 6 months, or experienced personnel *may* be approved by the Medical Director.
- Be in good standing with the service regarding clinical issues.
- Complete initial AND bi-annual airway, RSI and cricothyrotomy training courses.

INDICATIONS:

To establish an airway in a patient who is at risk of death secondary to loss of airway or inability to ventilate, and the airway cannot be controlled by conventional means. Examples of patients in which pre-hospital RSI might be indicated include, but are not limited to the following:

- Facial or head trauma patients with loss of airway control.
- Severe respiratory distress with hypoxia and/or respiratory exhaustion.
- Burn patients with airway involvement and respiratory distress.
- Overdose with loss of airway protection and hypoxia.

CONTRAINDICATIONS:

- Allergy to any one of the agents.

CONTRAINDICATIONS TO SUCCINYLCHOLINE:

- History of malignant hyperthermia.
- Renal failure.
- Spinal cord injury greater than 24 hours old or neuromuscular disease.
- Severe burns greater than 8 hours old.
- Massive crush injuries.
- Pesticide poisoning.
- Penetrating eye injuries.

- **Initiate standard treatment as indicated (ABC's, cardiac monitor, pulse ox, IV access, etc.).**
- **Attempt less invasive airway control and determine the need for RSI.**

Continued on Next 2 pages...

FACILITATED INTUBATION AND RSI (Procedure)**Preparation**

- Assemble and check all needed equipment and medications and anticipate a difficult airway.

**Preoxygenate**

- Allow patient to breathe high flow O₂, ventilate only as needed to increase SpO₂ (avoid gastric distention). Place on a nasal cannula @ 15 lpm and leave in place until the procedure is completed.

**Pretreatment**

- Attach capnography and begin monitoring early into the procedure.
- If hypotensive, attempt fluid resuscitation and consider use of push-dose pressors (page 9).

**Give SEDATIVE (Induction) – Use appropriate, available induction agent:**

- Ketamine 1-2 mg/kg IV/IO.
 - First choice induction agent (sedative).
 - Avoid if patient is hypertensive, tachyarrhythmias or acute MI.
- Etomidate 20 mg IV/IO, when Ketamine is contraindicated (**first choice alternate to Ketamine**).
- Versed 5 mg IV/IO (**only used as an alternate to Etomidate**).

*****CONSIDER ATTEMPTING FACILITATED INTUBATION AFTER AVAILABLE INDUCTION AGENT IS GIVEN*****

******REFER TO CONTRAINDICATIONS FOR SUCCINYLCHOLINE ON PAGE 6 PRIOR TO GIVING!!!!******

Give PARALYTIC (short-acting)

- Administer Succinylcholine 1 mg/kg IV/IO (**Standard Adult dose: 100 mg**) or Rocuronium 1 mg/kg IV/IO (**Standard Adult dose: 100 mg**)

**Placement and Proof**

- Intubate when patient becomes flaccid, often after fasciculations. If the patient cannot be intubated after 2 attempts, then use an alternative airway such as the iGel airway or basic airway adjuncts and continue to ventilate patient until the Succinylcholine wears off.
- Confirm placement with EtCO₂ detector, Capnography/Capnometry, auscultation, etc. (**minimum of 5 verification techniques**).



Continued on Next page...

FACILITATED INTUBATION AND RSI (Continued) – POST INTUBATION MANAGEMENT

For any patient who is already intubated (interfacility transfers), ALL PARAMEDICS can begin with this section to maintain sedation, go to long-acting paralytics as needed, see below...

Post-Intubation Management

- 100% O₂, and titrate to > 92% when possible.
- Maintain EtCO₂ of 35-45 mmHg when possible.
- Secure ET Tube.
- DO NOT overinflate with the BVM for risk of causing barotrauma.
- Use a PEEP valve (5-10 cmH₂O) in patients with pulses and a stable BP.

Give maintenance SEDATION and PAIN CONTROL... This MUST BE DONE!

Ketamine 0.5-1 mg IV/IO, one time dose, if Ketamine was used for induction prior to RSI

Versed 2-5 mg IV/IO

- Repeat 2-5 mg as necessary and titrate to desired effect.

Fentanyl 100 mcg IV/IO may also be given

- Repeat as necessary and titrate to desired effect.

Administer long-acting paralytic as indicated after correct placement is assured.

- Rocuronium (Zemuron) 1 mg/kg or 50 mg IV/IO (maximum).
 - To be used as alternate, if Norcuron (Vecuronium) is unavailable...
 - Actual dose range is 0.6-1 mg/kg. Per MD1, we can give 1 mg/kg for ease of administration.
- Norcuron (Vecuronium) 0.1 mg/kg or 10 mg IV/IO (maximum).
 - Second choice if Rocuronium is unavailable...

Transport without delay, as safely as possible...

DOCUMENT WELL... Include in your documentation the reason the procedure was required, the procedure used, intubation verification **(5)** methods, EtCO₂ must be used and documented, and the patient's response.

PUSH-DOSE PRESSORS

Indications:

- **Post-ROSC Hypotension**
- **RSI / Management of peri-intubation hypotension**
- **Bridge to continuous Epinephrine infusion**
- **Management of specific shock states including:**
 - Anaphylactic Shock
 - Septic Shock (After 30mL/Kg fluid bolus with ongoing S/S of shock)
 - Neurogenic Shock
 - Cardiogenic Shock
 - Hypovolemic Shock (Refractory to fluid resuscitation)

Contraindications:

- Tachydysrhythmias
- Hypotension responsive to fluid resuscitation

Mixing instructions:

- Take a 10 mL syringe filled with 9 mL of normal saline. If premade saline syringes are available, eject 1 mL of saline from the 10 mL saline syringe.
- Into this syringe, draw up 1 mL of Epinephrine from the cardiac Epinephrine ampule (cardiac amp contains 10 mL of epinephrine with a concentration 100 mcg/mL or 1:10,000).
- Shake well. Now you have 10 mL of Epinephrine 10 mcg/mL (1:100,000).
- **Label the syringe.**
- Confirm medication and concentration with partner.

Dosage instructions:

- Administer 5-10mcg (0.5mL – 1mL) doses via rapid IV push every 2-5 minutes, as needed, to achieve hemodynamic goal.
 - Systolic BP >90mmHg
 - MAP $\geq$ 65mmHg

Precautions:

- May cause or worsen tachycardia / tachydysrhythmias
- May cause acute hypertension, and thus possibly reflex bradycardia

OROGASTRIC (OG) TUBE INSERTION PROTOCOL

Paramedics may use this protocol to facilitate gastric decompression of the intubated patient after endotracheal intubation has been performed and verified, OR with use of the King LTS-D Airway.

Contraindications:

- Known esophageal disease (such as varices).
- Gastric bypass.
- Penetrating neck trauma.

Especially consider use in: ↓

- Submersion injuries (near drowning).
- Intubated Pediatric Patients.
- Any patient who has been aggressively artificially ventilated.

Equipment:

1. 18 French Sump Catheter.
2. Lubricant (water-soluble).
3. 60 mL catheter tip syringe.
4. Tape.

Procedure:

1. Estimate the length of the tube needed to reach the stomach by measuring the tube from the corner of the mouth to the earlobe and down to the xiphoid process. Mark the length with tape.
2. Lubricate the Salem sump tube with a water-soluble lubricant (e.g. KY Jelly).
3. Insert the tube through the oropharynx OR through the gastric access lumen on the KING LTS-D Airway until the marked depth is reached.
4. If the tube coils in the posterior pharynx, direct laryngoscopy can be utilized to place the tube in the esophagus. If the patient begins to cough, you may be advancing into the trachea. Reattempt insertion...
5. Verify placement (see OG Placement Verification).

Placement Verification:

1. Using a 60 mL catheter tip syringe, instill 30 mL of air into tube and auscultate over epigastrium for air sounds.
2. Aspirate for gastric contents and assess for cloudy, green, tan, brown, bloody or off-white color contents consistent with gastric contents.
3. Secure tube with tape.

Gastric Decompression:

1. Attach the tube to suction (and suction intermittently as needed). You can also provide continual suction by setting the suction device at 20-60 mmHg negative pressure (TORR).
2. If suction is not readily available, connect the 60 mL syringe to the tube with keeping the (blue) air vent patent. This will allow the sump function of the tube to work until suction can be applied and will also prevent gastric contents from leaking from the tube.

Special Note:

- If you cannot place the OG tube quickly (no more than 2 attempts), forego the procedure – **DO NOT** delay transport.
- The blue air vent must remain patent to ensure proper sump function and the prevent damage to the gastric lining during suction.

PAIN MANAGEMENT (ACUTE TRAUMATIC INJURIES)

- Assure ABC's are intact, stabilize as necessary, Oxygen as needed.
- Pulse oximetry / EtCO₂ – MUST BE USED – Cardiac Monitor is at the Paramedic's Discretion.

Obtain IV access (Critical patients may have IO access).

If there is an obvious fracture, refer to "Fractures General Care" protocol.

Consider Toradol (Ketorolac) 30 mg IV or 60 mg IM (DO NOT give if > 65 years old) for the following:

- Isolated Musculoskeletal injuries.
- Kidney Stones.
- Back pain.

No standing orders for repeat doses of Toradol.

Contraindications for Toradol

- Anticoagulants / blood thinners of any type.
- < 16 years of age or > 65 years of age
- Major trauma or head injuries
- Chronic kidney disease
- Allergies to NSAIDS

For acute traumatic injuries in the absence of hypotension and altered mental status (AMS) due to head injury, administer:
Fentanyl 1.0 mcg/kg slow IV/IO, Max initial dose of 50 mcg in geriatric patients.

- If hypotension or AMS due to head injury, consult with on-line medical control prior to giving any analgesics.

If patient is allergic to Fentanyl, or if Fentanyl is unavailable, consider:

Morphine 0.05 – 0.1 mg/kg IV/IO/IM, max initial dose of 5 mg in geriatrics.

If unable to obtain an IV, give either Fentanyl or Morphine IM once.

If pain is not relieved with Morphine or Fentanyl alone, then consider giving **Ketamine 20 mg IV/IO**

- The typical supply of Ketamine comes 500 mg/10 ml, so 20 mg would be drawn up at 0.4ml.
- To be given after initial Fentanyl or Morphine dose (not given by itself for pain, per standing orders).
- Standing orders are for **adults only**.

Fentanyl may be **repeated** at 0.5 mcg/kg increments, titrated to effect, given at least 5 minutes apart.

- Max 2nd dose of 25 mcg in patients > 65 years' old

Morphine may be repeated one time if needed, 5 minutes after the initial dose.

- Max repeat dose of 2 mg in geriatric patients.

****Use extreme caution when administering narcotics to geriatric population**** (Typically, 65 years of age and older)

Transport as indicated – **MUST** be transported ALS with Paramedic providing care.

THERE ARE NO STANDING ORDERS FOR ANALGESICS IN ABDOMINAL PAIN, OR OTHER MEDICAL COMPLAINTS...*CONSULT ON-LINE MEDICAL CONTROL AS NEEDED*

INTRAOSSEOUS (IO) INFUSION PROCEDURE

Protocol / Standing Order:

AEMTs or Paramedics working under SCEMS Protocols may perform emergency vascular access using approved IO infusion procedures. Personnel will only use devices approved by SCEMS MD1.

Approved Sites for AEMTs or Paramedics, **only use sites you are trained in identifying:**

Adults – Proximal tibia or humeral head. Humeral head may deliver a faster rate of infusion in critical shock / trauma patients who are in need of emergent fluid administration.

Pediatrics – Proximal tibia will be the preferred site, and humeral head access may be utilized for patients > 5 years of age.

Indications:

- Any patients who is in need of emergency vascular access where peripheral IV access is difficult.
- IO may be the initial vascular access attempted in critical (non-responsive to pain) pediatric patients.
- IO may be the initial vascular access attempted in adult cardiac arrest patients. However, peripheral IV access still remains an industry standard initial vascular access. (EJ access is considered peripheral).

Contraindications:

- Fracture in target bone.
- Previous, significant orthopedic procedures at insertion site, prosthetic limb or joint.
- IO access (or attempted IO access) in the targeted bone within the past 48 hours.
- Infection at the area of insertion.
- Excessive tissue (severe obesity) and/or absence of adequate anatomical landmarks.

Procedure:

1. Gain consent, inform and explain the need for IO to the conscious patient who is in need of the IO access.
2. Prepare the appropriate site with the proper placement of the extremity, select the proper needle size.
3. Apply and maintain aseptic technique throughout the procedure.
4. Properly insert the IO in the target bone/site. Note that the inability to aspirate (withdraw) bone marrow does not necessarily mean insertion was unsuccessful. Verify placement with the aspiration of marrow if possible.
---(Go to step 6 if unresponsive)---
5. **For patients responsive to pain, administer Lidocaine 2% (anesthetic) ASAP following IO insertion.**
 - **ADULT** – Administer 40 mg slow push, over 2 minutes.
 - **PEDIATRICS** – Administer 0.5 mg/kg (up to a max of 40 mg) slow push, over 2 minutes.
 - Allow initial doses of Lidocaine to dwell in the site for additional 1 minute before flushing with NS.
 - Provide a rapid (steady) flush with NS, 10 ml flush in adults, 5 ml flush in pediatrics.
 - After the initial flush, **repeat** Lidocaine 2% at HALF the initial dose and allow to dwell again for 1 minute. (Repeat doses of Lidocaine 2% would be 20 mg in adults or 0.25 mg/kg in peds)
 - Proceed with fluids or medications administration.
6. If the IO will not flush or flow, it may be against the inner wall inside the medullary space. Consider retracting the needle very slightly to clear the distal end from against the inside of the bone and reattempt to flush.
7. **WARNING** - When using a pressure bag or the other pressure infusion device, be careful to not allow air from the IV bag to enter the drip chamber of the tubing and inadvertently be infused into the patient.
8. Secure the IO needle at the site using tape, specifically designed devices, or other bulky dressings.
9. Observe for any signs of swelling with infusion or other indication of non-patency.
10. If required, IO needles can be removed by attaching a 10 mL syringe to the hub of the IO needle and pulling out with a gentle twisting motion. Cover the site with a dry sterile dressing and secure as indicated.

Note that any medication that is given via IV may also be given via the IO route as well.

VENTILATOR PROTOCOL (ADULTS ONLY)**INDICATIONS:**

1. Patient must be intubated.
2. Transfers.
3. Prolonged transports.
4. CPAP (See CPAP protocol).
5. BiPAP (See BiPAP protocol).

CONTRAINDICATIONS:

1. Endotracheal tube not in position and confirmed.
2. Pulseless patient/code situation.
3. Brief transport/delay in transport.

QUALIFICATIONS/TRAINING REQUIREMENTS:

1. Paramedics with at least 2 years of experience, and have completed the state mandated Ventilator Class.
2. Renewal requires a 4-hour ventilator update class annually.

INITIAL SETTINGS:

IF PATIENT IS TRANSFERRED FROM A HOSPITAL ON A VENT, USE THE SAME SETTINGS THE HOSPITAL IS USING.

1. Assist Control.
2. Rate – 12.
3. Tidal Volume – 400 mL or 5-8 mL/kg of lean body mass.
4. FiO₂ – 100%.
5. PEEP – 5.
6. I/E ratio – 1:2.

REQUIRED MONITORING:

1. Cardiac Monitoring.
2. O₂ Sat Monitoring.
3. End-Tidal CO₂ Monitoring.
4. Blood Pressure every 5 minutes.
 - *On prolonged transfers of hemodynamically stable patients, blood pressure can be done as indicated.*

VENTILATOR ADJUSTMENTS:

1. PEEP
 - If the patient is hypotensive, then PEEP = 0.
 - If O₂ Sat < 90% and systolic blood pressure is > 100, may increase PEEP by 5.
2. FiO₂ – May wean to keep O₂ Sats > 90%.
3. I:E ratio – Change to 1:4 for COPD or Asthma patients.

DISCONTINUATION OF VENTILATOR:

IF AT ANY TIME THE PATIENT DEVELOPS PULMONARY OR HEMODYNAMICALLY COMPROMISE, TAKE THE PATIENT OFF THE VENTILATOR AND VENTILATE WITH A BVM AND 100% OXYGEN!!!!

If current sedation is ineffective, and there is risk of failure to maintain endotracheal tube placement in the patient, **consider physician consult** to seek orders to discontinue current sedation, and default to our standing orders on page 7 of this section. Continue from that point, to possibly include long-acting paralytics. Remember, you MUST also be sure to give sedation as well (Versed, Fentanyl).

BiPAP PROTOCOL (ADULTS ONLY)**INDICATIONS:**

1. In-hospital patient receiving BiPAP.
2. Acute respiratory failure.
3. Non-intubated patient requiring pressure support.
4. Respiratory muscle exhaustion.

CONTRAINDICATIONS

1. Intubated patient or unable to maintain airway.
2. Hypotension or hemodynamically compromise.
3. Altered Mental Status or decreased LOC.
4. Inability to maintain adequate mask seal.
5. Facial trauma or abnormalities / deformities.
6. High-risk of aspiration or excessive secretions.

QUALIFICATIONS/TRAINING REQUIREMENTS:

1. Paramedics with at least 2 years of experience, and have completed the state mandated Ventilator Class.
2. Renewal requires a 4-hour ventilator update class annually.

DISCONTINUATION OF BiPAP:

IF AT ANY TIME THE PATIENT DEVELOPS PULMONARY OR HEMODYNAMICALLY COMPROMISE, TAKE THE PATIENT OFF THE VENTILATOR AND VENTILATE WITH A BVM AND 100% OXYGEN!

INITIAL SETTINGS:

- Use the settings used by the hospital, and adjust as needed, or if necessary, revert to the following:

Normal Initial Settings:

- Select Non-invasive CPAP mode on ventilator.
- Inspiratory Positive Airway Pressure (IPAP): 10 cmH2O
- Expiratory Positive Airway Pressure (EPAP): 5 cmH2O
- FiO2: 100%, then titrated down to minimum required.
- Pressure support is the value determined by the difference between the IPAP and EPAP. E.g., 10/5 = Pressure Support of 5 cmH2O.

REQUIRED MONITORING:

- Cardiac monitoring.
- SpO2 monitoring.
- End-Tidal CO2 monitoring.
- Mental Status Monitoring.
- Blood Pressure every 5 minutes.
 - *On prolonged transfers of hemodynamically stable patients, blood pressure can be done as indicated.*

VENTILATOR ADJUSTMENTS:

- Inspiratory Positive Airway Pressure (IPAP): Increase in increments of 2 cmH2O, not to exceed 20 cmH2O.
- Attempt to correct by increasing IPAP only before adjusting EPAP, except in CHF on pulmonary edema.
- Expiratory Positive Airway Pressure (EPAP): Maintain starting pressure of 5 cmH2O unless persistent hypoxia is present. Increase in increments of 2 cmH2O not to exceed 10 cmH2O, and adjust IPAP by equal amounts to maintain the same pressure support (minimum of 5 cmH2O) and an adequate tidal volume.
- FiO2: Titrate down to maintain target SpO2 of 91-99%.

If patient is unable to tolerate BiPAP after changing to the transport ventilator, but is maintaining equal and adequate ventilation and oxygenation, and patient comfort cannot be established by adjusting settings, consider consulting the sending physician for on scene administration of anxiolytic medication or for mild sedation orders prior to departure.

CHEST FINGER THORACOSTOMY

OVERVIEW:

- Chest Finger Thoracostomy is an action that is taken to make surgical incisions into a patient's chest to access the thoracic cavity in the event of traumatic death that may have resulted from a life-threatening tension pneumothorax. Then inserting the finger of the paramedic into the patient's chest cavity area to relieve pressure from air or fluid (blood) that has accumulated in the thoracic cavity.

REQUIREMENTS TO PERFORM PROCEDURE:

- Must be a paramedic
- Must be checked off by your Medical Director or Training Officer, and have paperwork on file to confirm competency.

INDICATIONS:

- Traumatic cardiac arrest from blunt force trauma or penetrating injuries to the thoracic cavity.

CONTRAINDICATIONS:

- Not checked off on the procedure
- If patient is conscious and not in traumatic arrest.

PROCEDURE:

PARAMEDIC

Paramedic – Standing Order with Special Credentialing

1. Position the arm in an abducted manner to easily locate the fifth rib along the midaxillary line, falling within the "triangle of safety." Mark this specific area for incision.
2. Sterilize the designated incision area using an antiseptic swab stick to minimize the risk of infection.
3. Carefully execute a transverse incision, approximately 1-2 inches (3-5 cm) in length, over the marked location on the fifth rib. The incision should be just anterior to the midaxillary line.
4. Utilize large forceps to quickly dissect the tissue above the rib, extending through the intercostal muscles.
5. Penetrate the pleura using the forceps. Once in position, expand the forceps to create a sufficient opening
6. Retract the expanded forceps from the chest, ensuring the created tract remains open.
7. Insert a finger along the dissected tract to reach the pleural cavity. Perform a sweeping motion to widen the opening.
8. Evaluate for any release of air or blood from the newly created opening, as this will guide further medical intervention.
9. For documentation and potential future investigations, encircle each incision with a surgical marker. Label these incisions as either "EMS-R" for the right side or "EMS-L" for the left side to identify procedures performed by EMS.

ADULT (ACLS) CARDIAC GUIDELINES

- Always treat the patient not the monitor.
- Cardiac arrest caused by trauma is treated by the correcting the underlying problem.
- Protocols for cardiac arrest situations, presumes that the condition under discussion continually persists, the patient on a cardiac monitor remains in cardiac arrest, and CPR is being performed.
- Chest compressions and **defibrillation** are more important than the administration of medications or establishment of an advanced airway.
- All attempts should be made to minimize interruptions in compressions which include allowing no more than 10 seconds for pulse check or no more than 5 seconds to deliver breaths. If an advanced airway is in place, CPR is not stopped to deliver breaths.
- Attempts should be made to use the mechanical CPR device.
- The cardiac monitor and other necessary equipment shall be taken to the immediate side of any unconscious, known cardiac arrest, or possible cardiac arrest patients. Perform a “quick-look” by applying pads and assessing the underlying rhythm.
- Patients presenting in cardiac arrest should receive ACLS care with an emphasis in high quality CPR and early defibrillation prior to being moved to an ambulance. Any scene safety concerns are justification to modify this to keep EMS personnel as safe as possible.
- IV or IO access is the preferred method of delivery of drugs. If an IV cannot be started in an arrest patient IMMEDIATELY, start IO line.
- **IO may be attempted initially in cardiac arrest patients.** Peripheral access should still be considered. Paramedics may perform EJ access, but no IV catheter larger than an 18 gauge may be used.
- Lidocaine, Epinephrine, Atropine, and Narcan (LEAN) can be administered via the endotracheal tube at 2 to 2 ½ times their regular doses if an IV or IO line cannot be established. This should be followed by 5-10 mL of NS, along with hyperventilation of the patient after each drug.
- After each IV medication, give a 20 to 30 mL bolus of IV fluid and elevate the extremity.
- The fluid of choice for administering medications to the patient in cardiac arrest is Normal Saline.
- 12-lead ECGs should be obtained on the scene for the following cardiac problems and then every 10 minutes until arrival at receiving facility:
 - Non-traumatic chest pain greater than 25 years of age.
 - Symptomatic tachyarrhythmias such as wide-complex tachycardia's, atrial fibrillations, SVT or frequent ectopy.
 - Symptomatic bradycardias or heart blocks (Type I and II and AV dissociation).
 - Congestive Heart Failure / Pulmonary edema.
- Induced Hypothermia is no longer a protocol.
- Routine use of Sodium Bicarb in cardiac arrest may not be warranted until ROSC is obtained.
- Consider reversible causes for PEA / Asystole:

• Hypoxia • Hypovolemia • Hydrogen ion (acidosis) • Hypo- / Hyperkalemia • Hypothermia	• Toxins • Tamponade (cardiac) • Tension Pneumothorax • Thrombosis (coronary, pulmonary) • Trauma
--	---

For example:

- If possible pre-existing acidosis, consider giving Sodium Bicarbonate.
- If renal failure w/ possible increased potassium or Calcium Channel Blocker OD, consider Calcium Chloride.
- If tension pneumothorax, needle decompression.
- If hypovolemia, Lactated Ringers bolus.
- If tricyclic overdose, give Sodium Bicarbonate.

If narcotic overdose, give Narcan.

12-LEAD ECG

Transmit all ECG'S that are concerning for possible MI

Upload ALL 12-Lead ECGs to PCR

Out of hospital 12-Lead ECG's and advance notification to the receiving facility speeds the diagnosis, shortens the time to fibrinolysis or catheterization, and may be associated with decreased mortality rates. Providers shall complete training for 12-Lead ECG's acquisition prior to utilizing protocol and ECG machines.

INDICATIONS:

- 12-Lead ECGs should be done on the scene for the following and then repeat every 10 minutes until arrival to the receiving facility:
 - Non-traumatic chest pain greater than 25 years of age.
 - Symptomatic tachyarrhythmia's such as wide-complex tachycardia's, atrial fibrillations, SVT or frequent ectopy.
 - Symptomatic bradycardias or heart blocks (Type I, II, and AV dissociation).
 - Congestive Heart Failure / Pulmonary edema.
 - CVA / Stroke.
 - Dyspnea, to rule out cardiac causes of breathing problems in adults.
 - Syncope (fainting) or near syncope – maintain high index of suspicion for underlying ACS/MI.
 - Patient who are likely to have atypical (silent) MI, females >45 yrs of age, diabetics, vague complaints of nausea, exertional dyspnea or other signs of underlying possible ACS.

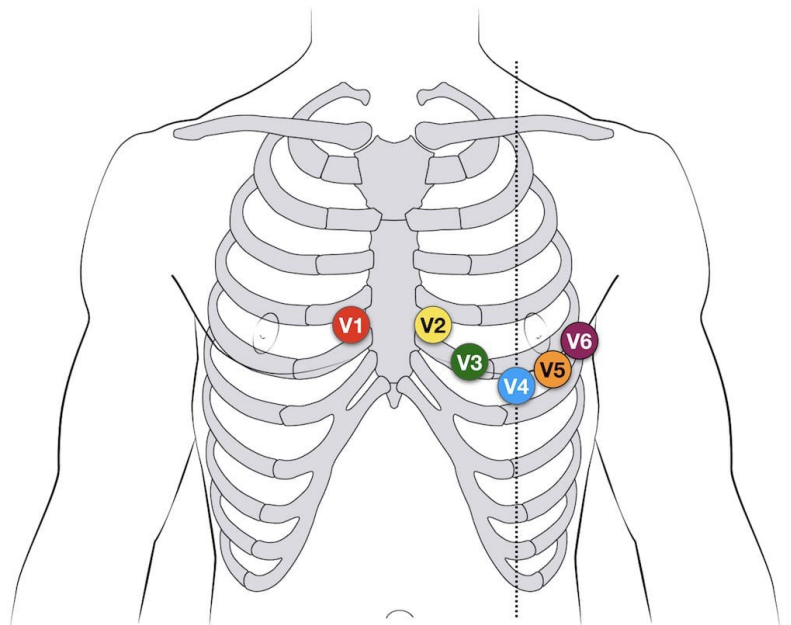
PRECAUTIONS:

- Ideally, 12-Lead ECG acquisition and treatment should occur concurrently but ultimately should not delay treatment of any life-threatening conditions.
 - Lethal dysrhythmias.
 - Respiratory emergencies.
 - Treatments such as O2, aspirin, and NTG.
 - Request for advanced life support.
- Scene time should not be prolonged by acquisition of ECG.
- Factors than can reduce quality of tracing include dirt, oil, sweat, and other material on the skin and patient/vehicle movement.

"ACS" – Acute Coronary Syndrome – includes MI, STEMI, or PE cases.

EMTs and AEMTs may acquire and transmit 12 lead ECGs. However, only paramedics may interpret ECGs and provide care specific to these findings.

- TN EMS State Rules and Regulations approved.



12-LEAD ECG (CONTINUED)**PROCEDURE:**

1. Prepare all the equipment and ensure the cable is in good repair. Check to make sure there are adequate leads and materials for prepping the skin.
2. Prep the skin by first drying sweat or water. Lightly buff the electrode placement areas with an alcohol prep if skin is dirty.
3. Place the four limb leads in accordance with the manufacturer's recommendations. Limb Lead electrodes are typically placed on the deltoid area and the lower leg or thigh. Avoid placing limb lead over body prominences.
4. Place the precordial leads (chest or V leads) in accordance with the manufacturer's recommendations. Proper placement is important for accurate diagnosis. Leads locations are identified as V1 through V6.
5. Locating the V1 position is critically important because it is the reference point for locating the placement of the remaining V leads. To locate the V1 position:
 - Place your finger at the notch in the top of the sternum.
 - Move your fingers slowly downward about 1.5 inches until you feel a slight horizontal ridge of elevation. This is the Angle of Louis where the manubrium joins the body of the sternum.
 - Locate the second intercostal space on the patient's right side, lateral to and just below the Angle of Louis.
 - Move your finger down two more intercostal spaces to the fourth intercostal space which is the V1 position.
6. Place V2 by attaching the positive electrode to the left of the sternum at the further intercostal space.
7. Place V4 by attaching the positive electrode at the midclavicular line at the fifth intercostal space. Note: V4 must be placed prior to V3.
8. Place V3 by attaching the positive electrode in the line midway between lead V2 and V4.
9. Place V5 by attaching the positive electrode at the anterior axillary line as the same level as V4.
10. Place V6 by attaching the positive electrode to the midaxillary line at the same level as V4.

CAUTION!

- NEVER USE THE NIPPLES AS REFERENCE POINTS FOR LOCATING THE ELECTRODS FOR MALE OR FEMALE PATIENTS BECAUSE NIPPLE LOCATION VARY WIDELY.
- WHEN PLACE ELECTRODES ON FEMALE PATIENTS, ALWAYS PLACE LEADS V3 THROUGH V6 UNDER THE BREAST RATHER THAN ON THE BREAST.

11. Ensure that all leads are attached.
12. Turn on the monitor.
13. Record the tracing by following the machines specific acquisition procedure and function.
14. Document on the tracing, the patient's name, date and time the tracing was obtained.
15. Refer to the ST-ELEVATION MYOCARDIAL INFARCTION (STEMI) TRIAGE as indicated.

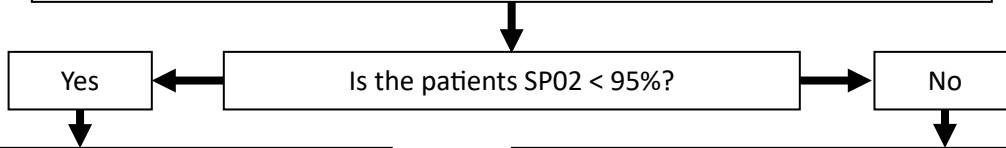
CONSIDERATIONS:

1. Acquire an additional 12-Lead ECG every 10 minutes or if the patient's clinical condition changes.

Personnel who are staffed as AEMTs on a BLS unit are encouraged to acquire 12-Lead ECGs on any patient who presents with a new-onset complaint of chest pain, shortness of breath, or altered mental status. **Transmit the ECG to the facility the patient will be transported too as soon as possible.** This is permitted in the TN Rules and Regulations and is allowed for the AEMT scope of practice. Only the paramedics may interpret an ECG. – Updated 2020.

CHEST PAIN (non-traumatic)

- Assure ABC's are intact.
- Obtain vital sign (baseline) with a manual cuff.
- ECG monitor and Pulse oximetry.



Administer O2 at 2 – 4 LPM nasal cannula or as appropriate for patient needs.

Apply cardiac monitor and **acquire 12-Lead ECG.**

- Obtain initial 12-Lead on scene and repeat every 10 minutes until arrival at the receiving facility.

- If the patient is less than 25 years old with a normal 12-Lead ECG, then transport without delay and treat as needed.
- If no ST segment elevation is found on the 12-Lead ECG, then consider a 15-Lead ECG and then notify the receiving ED and transmit the ECG.

Obtain IV access.

Administer 4 chewable 81 mg **Aspirin.**

- Withhold Aspirin if the patient is on anticoagulants – Consult on-line medical control if unsure.

If patient is having active chest pain and has a systolic BP > 100mmHg, proceed with giving Nitroglycerin.

Administer: Nitroglycerin sublingual, 1 tablet or spray, 0.4mg

- Once IV is obtained, may repeat twice as needed, for a maximum of 3 doses total, to be given 5 minutes apart PRN.
- Ensure that BP remains stable before giving any subsequent doses of NTG.

CAUTION!

DO NOT GIVE NTG IF THE PATIENT HAS TAKEN MEDICATIONS USED TO TREAT PULMONARY HYPERTENSION OR ERECTILE DYSFUNCTION, SUCH AS PHOSPHODIESTERASE INHIBITORS (PDEs), EXAMPLE INCLUDE: (DRUGS ENDING IN "FIL", VIAGRA (SILDENAFIL), CIALIS (TADALAFIL), ETC. IF UNSURE, CONTACT ON-LINE MEDICAL CONTROL.

If chest pain is not relieved after 3 doses of Nitroglycerin, consider **Morphine 2 mg IVP.**

- Blood pressure must still be above 100 systolic before giving Morphine (or Fentanyl).
- Repeat at 2 mg increments, every 5 minutes, as needed if pain persists, up to a **MAX OF 10 MG.**
- Fentanyl 50 mcg may be given and repeated once, if Morphine is contraindicated.

If chest pain is due to recent Cocaine usage, consider Versed 2 mg IV or 5 mg IM.

Transport as indicated.

CHEST PAIN / STEMI

- Assure ABC's are intact.
- Obtain vital signs (baseline) with manual cuff.
- Pulse oximetry.
- ECG Monitor

Oxygen 2-4 LPM via nasal cannula.
DO NOT WITHHOLD, if patient is in distress or hypoxic – give 100% via NRB.

Apply cardiac monitor, acquire 12-Lead ECG ASAP.

- Any signs of ST-elevation in 2 or more contiguous leads of ≥ 1 mm elevation, send STEMI alert to receiving facility.
- Transmit ECG without delay, transport emergency to nearest appropriate PCI capable facility.

Obtain IV access.

Administer 4 chewable 81 mg Aspirin, unless patient is allergic or currently on anticoagulants.

Give 1 NTG tablet or spray sublingual if systolic blood pressure is greater than 100, every 5 minutes up to a maximum of 3 doses, until pain is relieved or systolic blood pressure falls below 100.

- NTG IS CONTRAINDICATED WITH RIGHT SIDED INVOLVEMENT (INFERIOR MI).

CAUTION!

DO NOT GIVE NTG IF THE PATIENT HAS TAKEN MEDICATIONS USED TO TREAT PULMONARY HYPERTENSION OR ERECTILE DYSFUNCTION, SUCH AS PHOSPHODIESTERASE INHIBITORS (PDEs), EXAMPLE INCLUDE: (DRUGS ENDING IN "FIL", VIAGRA (SILDENAFIL), CIALIS (TADALAFIL), ETC. IF UNSURE, CONTACT ON-LINE MEDICAL CONTROL.

If chest pain is not relieved after 3 doses of Nitroglycerin, consider **Morphine 2 mg IVP.**

- Blood pressure must still be above 100 systolic before giving Morphine (or Fentanyl).
- Repeat at 2 mg increments, every 5 minutes, as needed if pain persists, up to a **MAX OF 10 MG.**
- Fentanyl 50 mcg may be given and repeated once, if Morphine is contraindicated.

Treat any dysrhythmias accordingly.

Transport without delay.

12-Lead and 15-Lead Reference Chart

Table 1. Infarction Distribution With ST-Segment Elevation Myocardial Infarction And Consequences^{4,15-17}

ST Elevations	Affected Coronary Artery	Area of Damage	Complications
V ₁ through V ₄	Left coronary artery: Left anterior descending	Anterolateral heart wall Septum Left ventricle His bundle Bundle branches	Left ventricular dysfunction: Decreased carbon dioxide, congestive heart failure Left bundle-branch block Right bundle-branch block Left posterior fascicular block Infranodal block (2° or 3°)
V ₅ through V ₆ , I, aVL	Left coronary artery: Left circumflex branch	Left lateral heart wall	Left ventricular dysfunction: Decreased carbon dioxide, congestive heart failure Infranodal block (2° or 3°)
II, III, aVF, V ₄ R	Right coronary artery: Posterior descending branch	Inferior heart wall Right ventricle	Hypotension (particularly with nitroglycerin and morphine, which can decrease preload) Supranodal 1° heart block Atrial fibrillation/flutter, premature atrial contractions Infranodal block (2° and 3°) Papillary muscle rupture (murmur)
V ₈ and V ₉ (or ST depressions in V ₁ and V ₂)	90% Right coronary artery: Posterior descending branch 10% Left coronary artery: Left circumflex branch (will see elevations in V ₅ through V ₆)	Posterior heart wall	Hypotension Supranodal 1° heart block Infranodal block (2° and 3°) Atrial fibrillation/flutter, premature atrial contractions Papillary muscle rupture (murmur)

15-Lead Myocardial Infarction Reference Chart

How to do:
"15-Lead ECG"



Posterior ECG lead placement

Figure 1 - Posterior ECG lead placement

V7
Left posterior axillary line:
in the same horizontal plane as V4-V6

V8
Tip of the left midscapula:
in the same horizontal plane as V7-V9

V9
Left paraspinal region:
in the same horizontal plane as V4-V6

V1-V3
Should remain unchanged from standard 12-lead ECG

V4 Lead

V5 Lead

V6 Lead

Posterior MI is suggestive by the following changes in V1-V3:

- *Horizontal ST depression
- *Tall, broad R waves (>30ms)
- *Upright T waves
- *Dominant R wave (R/S ratio >1) in V2

Please note that V6 is a good reference point for the horizontal placement of the posterior electrodes V7-9.

If you don't have access to a 15 or 18 lead ECG machine, then leave V1-3 in their normal position and use V4-6, these leads will then become V7-9.

VENTRICULAR FIBRILLATION / PULSELESS V-TACH

Per the AHA Guidelines, you no longer have to perform CPR prior to defibrillation. **Shock immediately!**

Focus on HIGH QUALITY CPR

- Compressions at 100-120 / min.
- At least 2 – 2.4 inches deep.
- Allow adequate chest recoil.
- Minimize interruptions, no more than 10 secs without compressions.

Defibrillate as soon as possible!

Consider placing Defib pads Anterior & Posterior.

RESUME CPR

Reassess every 2 minutes and repeat shocks as needed, no limit on defibrillation attempts.

Initiate shocks at 200 Joules, stay at 200 Joules for other shocks.

Perform rhythm checks every 5 cycles of CPR!

Can patient be effectively ventilated with a BVM and oral airway?

NO

YES

Insert OPA (prn) and Ventilate with BVM attached to high flow O₂.

Oropharyngeal intubate the patient ASAP and assess **capnography** for confirmation of tube placement.

- Supraglottic airways may be used if unable to intubate due to difficult airway or COVID.
- Provide 1 breath every 6-8 seconds once a secured airway is in place. (Titrate to effect for a target ETCO₂ upon ROSC).
- Once a secured airway is in place, chest compressions may be continuous.

Establish Vascular Access IV/IO.

Administer **Epinephrine 1 mg 1:10,000** IV/IO.

- Repeat every 3-5 minutes as shockable cardiac arrest persists.

Administer **Amiodarone 300 mg** IV/IO bolus.

- Repeat once in 3-5 minutes at 150 mg IV/IO if arrhythmia persists.

Or, if Amiodarone is unavailable, use **Lidocaine 1-1.5 mg/kg** IV/IO bolus max total of 3 mg/kg.

- Repeat once in 3-5 minutes at half the initial dose of 0.5-0.75 mg/kg if arrhythmia persists.
- Maintenance drip of 1-4 mg/min.

Or, if Torsades de' Pointes (TDP) is present, consider **Magnesium Sulfate 2 gm mixed in 10 ml NS bolus infusion IV/IO** for pulseless TDP.

- TDP may be defibrillated in cardiac arrest when it presents on the ECG.

Consider Underlying Causes and Treat ASAP:

Hypoxia	Tamponade
Hypovolemia	Thrombosis
Hydrogen Ion	Tension Pneumothorax
Hypo/Hyperkalemia	Trauma
Hypothermia	Toxins

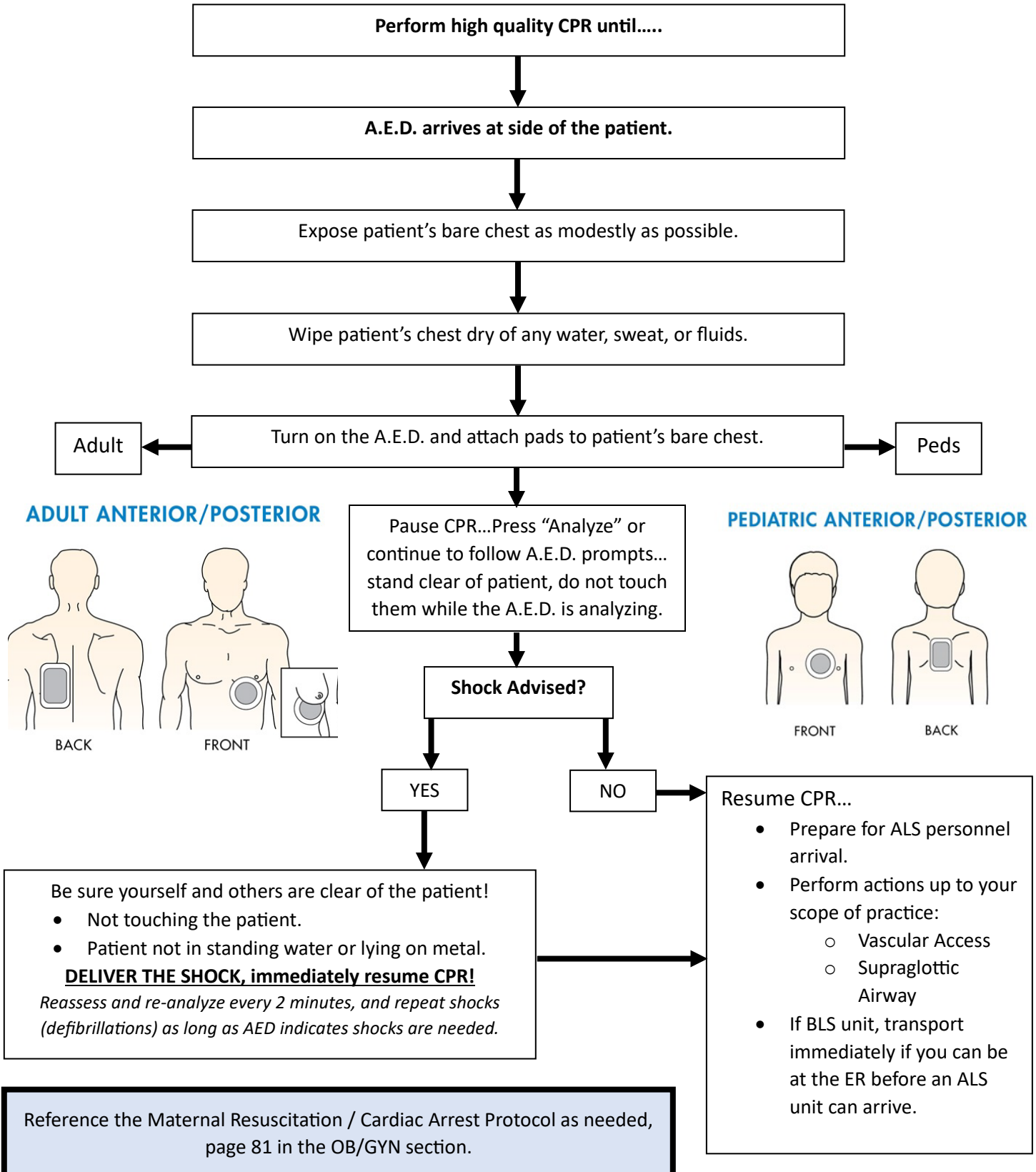
Transport Emergency.

Follow the Return of Spontaneous Circulation (ROSC) protocol as needed.

Reference the Maternal ACLS / Cardiac Arrest Protocol as needed, page 81 in the OB/GYN section.

BLS CARDIAC ARREST MANAGEMENT AND A.E.D. PROTOCOL

Per the current AHA Guidelines, you may now deliver a shock/defibrillation as soon as one is indicated, regardless of how long you have been doing CPR



LUCAS 3 Mechanical CPR Device

INDICATIONS:

- The LUCAS device is used to perform mechanical chest compressions on patients in cardiac arrest.
- The LUCAS is intended for use on patients' chest depth from 6.7 inches to 11.9 inches with a maximum chest width of 17.7 inches.

CONTRAINDICATIONS:

- If there is no indication that chest compression is likely to help the patient.
- Patients who do not fit within the LUCAS device.
- Patients chest is too wide or tall to fit in the LUCAS device.
- Patients chest is too small (pressure pad does not contact the sternum when the sternum pad is in its lowest position).

APPLICATION:

PRECAUTIONS!

Bruising, injury, vomiting, fractured or broken ribs, and soreness of the chest can be possible side effects of performing CPR. Side effects of CPR are common and considered acceptable ramifications given that the alternative is clinical death. If the patient is resuscitated, evaluate and assess for post CPR related injuries.

Power on the LUCAS device:

- Push ON/OFF to start self-test and power up the LUCAS device.
- The device will be ready and in the ADJUST mode.
- A second rescuer continues manual CPR.

Place the LUCAS BACK PLATE:

- Pause manual CPR briefly.
- Put the BACK PLATE under the patient, immediately below the armpits.
- A second rescuer needs to assist placing the BACK PLATE and then continuing CPR once the BACK PLATE is placed.

Attach the UPPER PART:

- Pull the RELEASE RINGS once to open CLAW LOCKS, then let go of the rings.
- Stop manual CPR briefly while attaching the UPPER PART to the BACK PLATE, and listen for the "CLICK" sound during attachment.
- Pull up once to assure attachment.
- A second rescuer continues CPR as long as possible, and help to attach the UPPER PART when needed.

Push down SUCTION CUP. Adjust position if needed.

- Push down the SUCTION CUP.
- The lower edge of the SUCTION CUP should immediately above the end of the sternum.
- Adjust if necessary (stay in ADJUST mode).
- A second rescuer can help assist.

Lock position. Start compressions.

- Push PAUSE to lock START POSITION.
- Push ACTIVE (continuous) or ACTIVE (30:2) to start compressions.
- A second rescuer can help assist.

- **Attach stabilization strap to prevent downward movement of the LUCAS device during use.**
- **Secure patient's wrist with patient straps on LUCAS device for easier movement of patient.**

Follow CPR protocols.

ASYSTOLE / PULSELESS ELECTRICAL ACTIVITY**Focus on HIGH QUALITY CPR!**

- Compressions at 100-120/min.
- At least 2 - 2.5 inches deep.
- Allow adequate chest recoil.
- Minimize interruptions, no more than 10 secs without compressions.

*Perform rhythm checks every 5 cycles of CPR.
(Every 2 minutes)*

Can patient be effectively ventilated with a BVM and oral airway?

NO

YES

Insert OPA (prn) and Ventilate with BVM attached to high flow O₂.

Oropharyngeal intubate the patient ASAP and assess capnography for confirmation of tube placement.

- Supraglottic airway may be used if unable to intubate due to difficult airway or COVID.
- Provide 1 breath every 6-8 seconds once a secured airway is in place.
- Once a secured airway is in place, chest compressions may be continuous.

Establish Vascular Access IV/IO.

Administer **Epinephrine 1 mg 1:10,000 IV/IO.**

- Repeat every 3-5 minutes as shockable cardiac arrest persists.

Transport Emergency

Follow the Return of Spontaneous Circulation (ROSC) protocol as needed.

Consider Underlying Causes and Treat ASAP:

Hypoxia	Tamponade
Hypovolemia	Thrombosis
Hydrogen Ion	Tension Pneumothorax
Hypo/Hyperkalemia	Trauma
Hypothermia	Toxins

- Can consider Sodium Bicarb 1 mEq/kg of 8.4% for adults in unwitnessed arrests or witnessed arrests > 15 minutes.
- Consider Calcium Chloride 1 gram IV / IO in renal patients.

Reference the Maternal ACLS / Cardiac Arrest Protocol as needed, page 81 in the OB/GYN section.

- If an IV or IO cannot be established, give Epinephrine 1:10,000 2 mg (double the normal dose) via ETT.
- Waveform Capnography with a reading of less than 10 mmHg may indicate poor CPR or suggest consultation of on-line medical control to terminate resuscitation efforts.
- Persistent Asystole despite > 20 minutes of resuscitation may also suggest a consult with on-line medical control to consider termination of efforts.

POST RESUSCITATION (R.O.S.C.) Return of Spontaneous Circulation

The primary focus is to optimize oxygenation and ventilation.

Be careful to ventilate the patient with appropriate rates and volumes.

Assess BP – If systolic <90 mmHg, administer 250 mL NS or LR bolus (**peds systolic BP 70 + (2x age in years), 20 cc/kg bolus**) repeat until BP >90 mmHg or appropriate for pediatric age. Consider use of push-dose pressors (give 1 mL).

Chest the patient's **blood glucose level**. If low, titrate dextrose 50% prn slowly until normal levels achieved. Try to avoid large swings in serum glucose levels (**peds – see glucose dosing chart**).

Glucose (Dextrose)	D50 1-2 mL/kg	> 8 years
	D25 2-4 mL/kg	6 months - 8 years
	D10 2-4 mL/kg	neonate - months
	Max Rate 2mL/kg/Min	
If D25 or D10 are not available, utilize a syringe of D50. To make D25, expel 25 mL of D50 and draw up 25 mL of NS. To make D10, expel 40 mL of D50 and draw up 10 mL of NS.		
*Reminder IO is appropriate after 2 failed IV attempts or 90 seconds		

Ensure head of bed is elevated 30 degrees, if possible.

12 Lead EKG, TRANSMIT.

If anti-arrhythmic administered:

If the ROSC patient who has received an initial dose of anti-arrhythmic medications goes into cardiac arrest again, proceed to second dose for the antiarrhythmic given...Example: Amiodarone given at 300 mg in V-fib, patients get ROSC, if condition changes and Amiodarone is needed again, it would be at the subsequent dose of 150 mg.

Initiate Maintenance Drips for Antiarrhythmic:

- If the patient converted with Amiodarone, start an Amiodarone drip at 150 mg and run over 10 minutes.
- If the patient converted with Lidocaine, start a Lidocaine drip at 1-4 mg/min, reference the formulary as needed for instructions.

Continue ventilator support to maintain ETCO₂>20.

- ADULTS – Respirations <12 ideally.
- SCHOOL AGE – min resp. rate should be 20-30.
- INFANT – PRESCHOOL – min resp. rate should be 30.

Titrate the artificial ventilator rate to yield an ETCO₂ range of 35-45 mmHg.

If patient does not tolerate ET tube or other advanced airway, contact Medical Control for Versed 2-5 mg IV/IO for patient sedation. Reference the "Post Intubation Management" protocol, as needed.

* For continued hypotension not responsive to fluid bolus, consider use of Epinephrine push-dose pressors (page 9) or continuous Epinephrine infusion (reference shock protocol on page 37).

TACHYCARDIA – WIDE COMPLEX WITH A PULSE**Initial Steps:**

- Assess ABC's stabilize as necessary.
- Pulse oximetry and cardiac monitor, waveform capnography can help assess perfusion status.
- Obtain IV access, refer to vascular access protocol if needed (for IO infusion).
- If patient is stable, acquire 12 Lead ECG to confirm V-tach prior to treating.

Unstable Patient?

Heart rate of 150/min, AND the patient exhibits the following:

- Hypotension.
- Altered Mental Status.
- Acute heart failure.
- Ischemic chest discomfort.

Yes

If the patient is conscious / semi-conscious: give available sedation agent prior to cardioversion / defibrillation.

(Only one sedative may be administered)

First choice: Etomidate (Amidate) 10 mg IV/IO.

- If Etomidate is unavailable, then give 2nd choice: Versed 5 mg IV/IO.
- If no IV/IO is established, consider Versed 5 mg IM.

Proceed to Synchronized Cardioversion:**Monomorphic (regular)**

Give synchronized shock (cardioversion) –

- 100 Joules, 150 Joules, 200 Joules

Polymorphic (irregular) / Torsades de Pointes

Give unsynchronized shock (defibrillations) –

- 120 Joules, 150 Joules, 200 Joules PRN

If in Torsades de Pointes:

- Administer Magnesium Sulfate 2 grams IV/IO.

Stable Patient?

- Symptomatic, however not yet critical... (treat with medication).

In a stable, yet symptomatic, wide-complex tachycardia that is regular (consistent R-R intervals), consider Amiodarone 150 mg mixed in 100 mL bag of NS infused IV slowly over 10 minutes. If Amiodarone is unavailable, Lidocaine 100 mg IV can be given. You can repeat at 50 mg in 5-10 minutes.

Refer to the Amiodarone drip instructions in the formulary section if needed.

Re-assess and treat appropriately, consult on-line medical control if further orders are needed.

TRANSPORT WITHOUT DELAY TO THE NEAREST APPROPRIATE FACILITY.

PREMATURE VENTRICULAR COMPLEXES (PVCs)

- Assess ABC's, stabilize as needed.
- Attach pulse oximetry and administer O2 as needed.
- Attach cardiac monitor and acquire 12-Lead ECG.
- Obtain IV access.

Transport without delay to the
nearest appropriate facility.

YES

Verify if PVC's are present after oxygenation.

NO

Apply treatment if patient exhibits symptoms that suggest hemodynamic compromise.

INDICATIONS:

- Lightheaded / dizziness.
- Syncope / near syncope.
- Ischemic heart disease (severe chest pain).

PVCs tend to be more symptomatic when multifocal and occurring with increased frequency.

Continue to monitor in transport and withhold further treatments unless the patient becomes more symptomatic.

Consult with on-line medical control if further orders are needed.

Administer Lidocaine 1 mg/kg IV/IO.

- May be repeated at 0.5 mg/kg as needed, up to a maximum of 3 mg/kg.

If Lidocaine is successful in suppressing ventricular ectopy, begin Lidocaine drip at 2-4 mg/min.

Infuse desired dose:

- 1 mg/min = 15 gtt/min.
- 2 mg/min = 30 gtt/min.
- 3 mg/min = 45 gtt/min.
- 4 mg/min = 60 gtt/min.

Reduced dose by HALF if:

- Patient is over 65 years of age.

Refer to the Lidocaine drip instructions in the formulary section if needed.

Transport without delay to the nearest
appropriate facility.

SUPRAVENTRICULAR TACHYCARDIA (SVT)

Transport without delay to the nearest appropriate facility.

- Assess ABCs, stabilize as needed.
- Attach pulse oximetry and cardiac monitor.
- Obtain IV access as soon as possible.

NO

Is the patient stable?

YES

Unstable Patient?

Heart rate over 150/min, AND the patient exhibits the following:

- Hypotension.
- Altered Mental Status.
- Acute Heart Failure.
- Ischemic Chest Discomfort.

YES

If the patient is conscious / semi-conscious: give available sedation agent prior to cardioversion / defibrillation.

(only one sedative may be administered)

First choice: Etomidate (Amidate) 10 mg IV/IO

- if unavailable then given 2nd choice: Versed 5 mg IV/IO.
- If no IV/IO is available, consider Versed 5 mg IM.

Proceed to Synchronized Cardioversion:

Place defibrillator in **synchronized mode** and shock in the following sequence until patient converts:

- 50 Joules, repeat PRN at 100 Joules.

Consult with on-line medical control if needed for further orders.

Transport without delay to the nearest appropriate facility.

ASAP, obtain 12-Lead ECG, confirm the following before treating as SVT: (transmit as needed)

- Is the rate over 150/min in the adult patient?
- R-R intervals regular? (if not, consider A-fib).
- QRS width is ≤ 1 mm (1 small block).
- Is there a history suggesting a compensatory tachycardia that just needs fluids?

For stable SVT, attempt vagal maneuvers and if these are ineffective:

- Begin recording ECG strip...
- Administer Adenosine:
 - Initial dose - 6 mg rapid IV
 - Immediately follow with a 10 mL NS flush after each dose.

Still no change... (approx. 1 minute later).

- Repeat Adenosine at 12 mg rapid IV immediately followed by 10 mL NS flush.
- Attempt to record on ECG strip (print).

If there is still no change following a second dose:

- Give a third dose of Adenosine, 12 mg immediately followed by 10 mL NS flush. The total of all doses should NOT exceed 30 mg.

**Note – If repeat doses of Adenosine do not convert the presumed SVT, it may not be SVT...*

ATRIAL FIBRILLATION / FLUTTER

- Assess ABC's, stabilize as needed.
- Attach pulse oximetry and cardiac monitor.
- Acquire 12-Lead ECG, transmit as needed.
- Establish IV Access as soon as possible.

STABLE Patient?

For Atrial Fibrillation or Flutter with a rate greater than 130, the paramedic may consider:

- **Cardizem** 10 mg IVP over 2 minutes for rate control.
- Same dose may be repeated every 5 minutes up to 30 mg's PRN.

Do not administer if the patient has the following:

- Hypotension.
- HR less than 130.
- Congestive Heart Failure (SPO2 less than 92% or rales noted on exam).
- Wide Complex Tachycardia.
- History of WPW.

UNSTABLE Patient?

Cardioversion of Atrial Fibrillation or Atrial Flutter requires On-line Medical Control order.

IF on-line medical control physician orders to proceed with cardioversion, these are the suggested settings:

Atrial Fibrillation – 120 Joules, 150 Joules, 200 Joules.

Atrial Flutter – 50 Joules, 100 Joules, 150 Joules, 200 Joules.

If patient becomes hypotensive after Cardizem:

- Administer 1 mL (1 cc) of Calcium Chloride (100 mg) slow IV/IO.
- Administer 500 mL Normal Saline bolus, reassess.

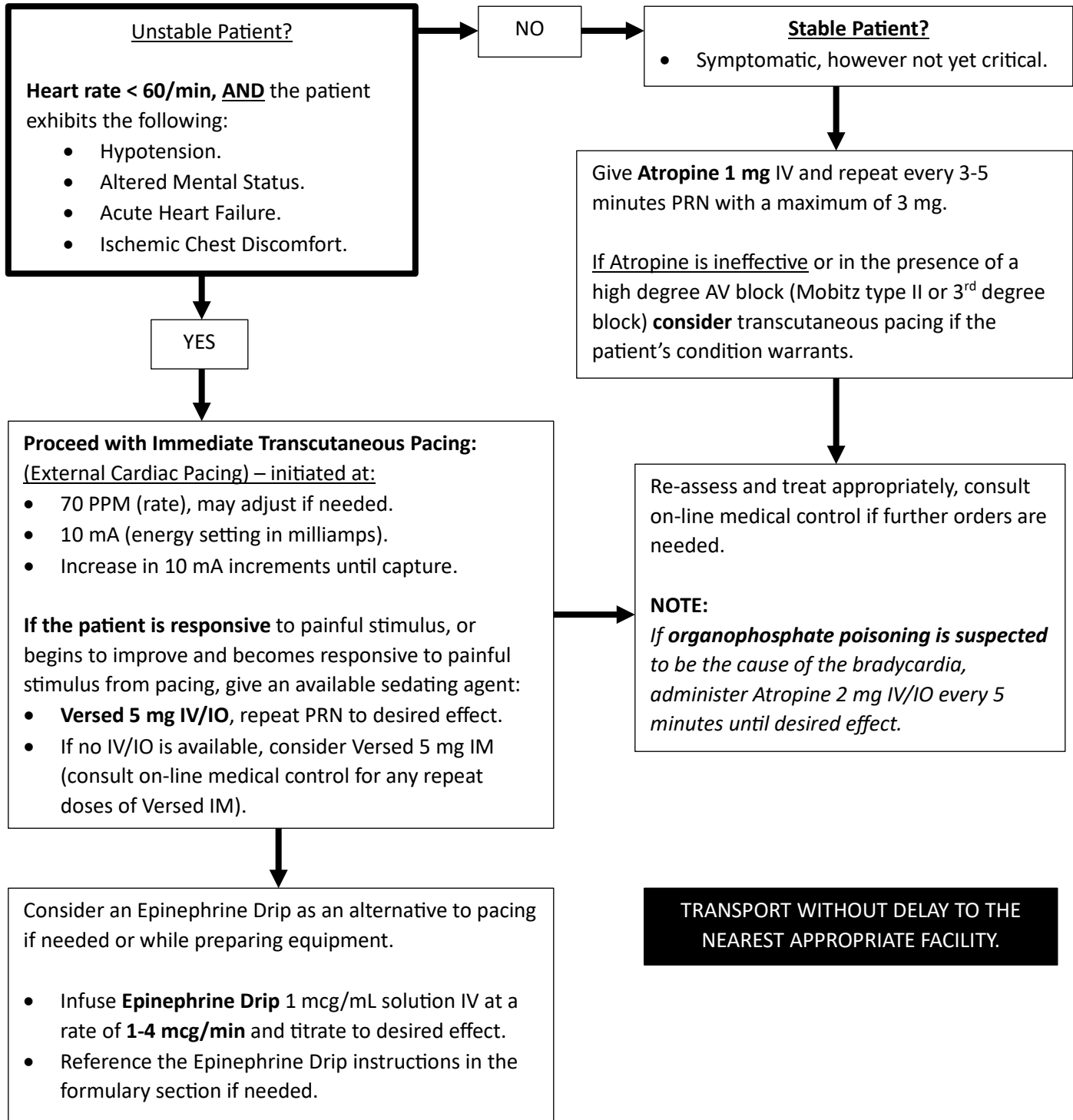
(This may be seen in dialysis / renal failure patients)

Consult On-Line Medical if further orders are needed.

Transport as indicated.

SYMPTOMATIC BRADYCARDIA

- Assess ABC's, stabilize as needed.
- Pulse oximetry and cardiac monitor, waveform can help assess perfusion status.
- Obtain IV access, refer to vascular access protocol if needed (IO infusion).
- If patient is stable, and time permits, acquire 12-Lead ECG every 10 minutes throughout transport.
- **Transport any unstable patient without delay to nearest appropriate facility.**



ABDOMINAL PAIN

- Assess / Stabilize ABC's as needed.
- Attach pulse oximetry.
- Cardiac monitor if indicated.

Administer O2 as needed.

Obtain IV access as indicated:

- Severe pain.
- Abnormal vital signs.
- Needed for administration of analgesia or antiemetic's.

**NO STANDING ORDERS FOR
ANALGESICS IN NON-TRAUMATIC
ABDOMINAL PAIN!**

Treat for shock with LR 500 mL bolus, repeat as necessary and titrate to effect.

- *Use with cation in CHF and renal failure patients.*

Place the patient in a position of comfort, and transport as indicated...

Clinical Note:

Abdominal pain of moderate to severe acuity should be considered a surgical emergency until proven otherwise and transport should be made without delay in the following:

- Females or childbearing age / pregnant patients.
- Patients presenting with S/S indicative of AAA.
- Recent post-operative complications.

NAUSEA / VOMITING

- Assess ABC's, stabilize as needed.
- Control airway and be prepared to suction.
- Pulse oximetry.

Give oxygen as indicated.

Cardiac Monitor, to rule out underlying acute coronary syndrome, especially in patients > 45 years of age. A patient who has been experiencing emesis may be hypovolemic and have electrolyte depletion as well.

- Acquire and transmit a 12 Lead ECG if you suspect any cardiac related illness.

Obtain IV access, administer fluids as needed for dehydration 500 mL LR (NS may be used as an alternative if LR is unavailable) bolus in adults, reassess and repeat as needed to restore normotensive BP, using caution in patients with renal failure or CHF.

Administer Zofran:

- Adult > 40 kg: 4 mg IV, IM or PO (repeat once if necessary).

Transport as indicated.

DEHYDRATION

- Assess ABC's, stabilize as needed.
- Pulse Oximetry.
- Cardiac Monitor.



Oxygen as indicated.

**Cardiac Monitor**

- Dehydration/hypovolemia can be associated with electrolyte imbalance which may lead to arrhythmias.
- Tachycardia in elderly patients is often associated with hypovolemia.

**Obtain IV access:**

- Adults – Administer LR 500 mL bolus, repeat PRN and titrate to effect.
- Use caution in patients with a history of CHF or renal failure.
- NS may be used as an alternative if LR is unavailable.
- Use 10 gtt/mL IV tubing to deliver fluids faster.

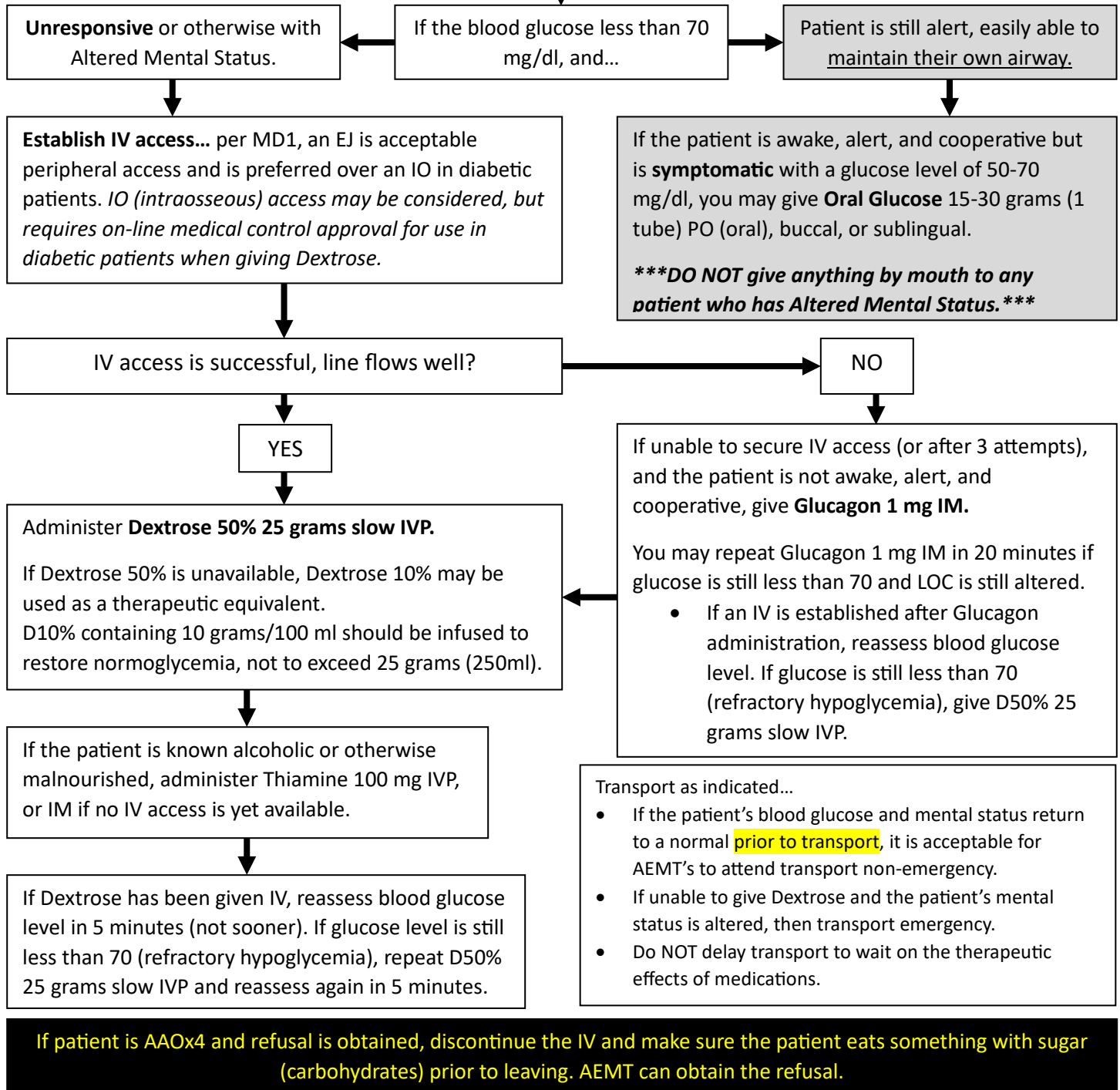


Transport as indicated.

DIABETIC EMERGENCY / HYPOGLYCEMIA

- Assess ABC's, stabilize as needed.
- Apply pulse oximetry, give oxygen as needed.
- Attach cardiac monitor.
- Obtain IV access if able to secure peripheral access.

The use of glucometers to assess blood glucose levels is within protocol for **EMT-Basics (service trained)** and levels **above**. (This is permitted by TN DOH, EMS, Feb. 2017 Clinical Issues Committee Update)



DIABETIC EMERGENCY / HYPERGLYCEMIA

- Assess ABC's, stabilize as needed.
- Apply pulse oximetry, give oxygen as needed.
- Attach cardiac monitor.
- Obtain IV access.
- EtCO2.



Determine glucose level:

- If > 70 and < 400, transport as indicated.

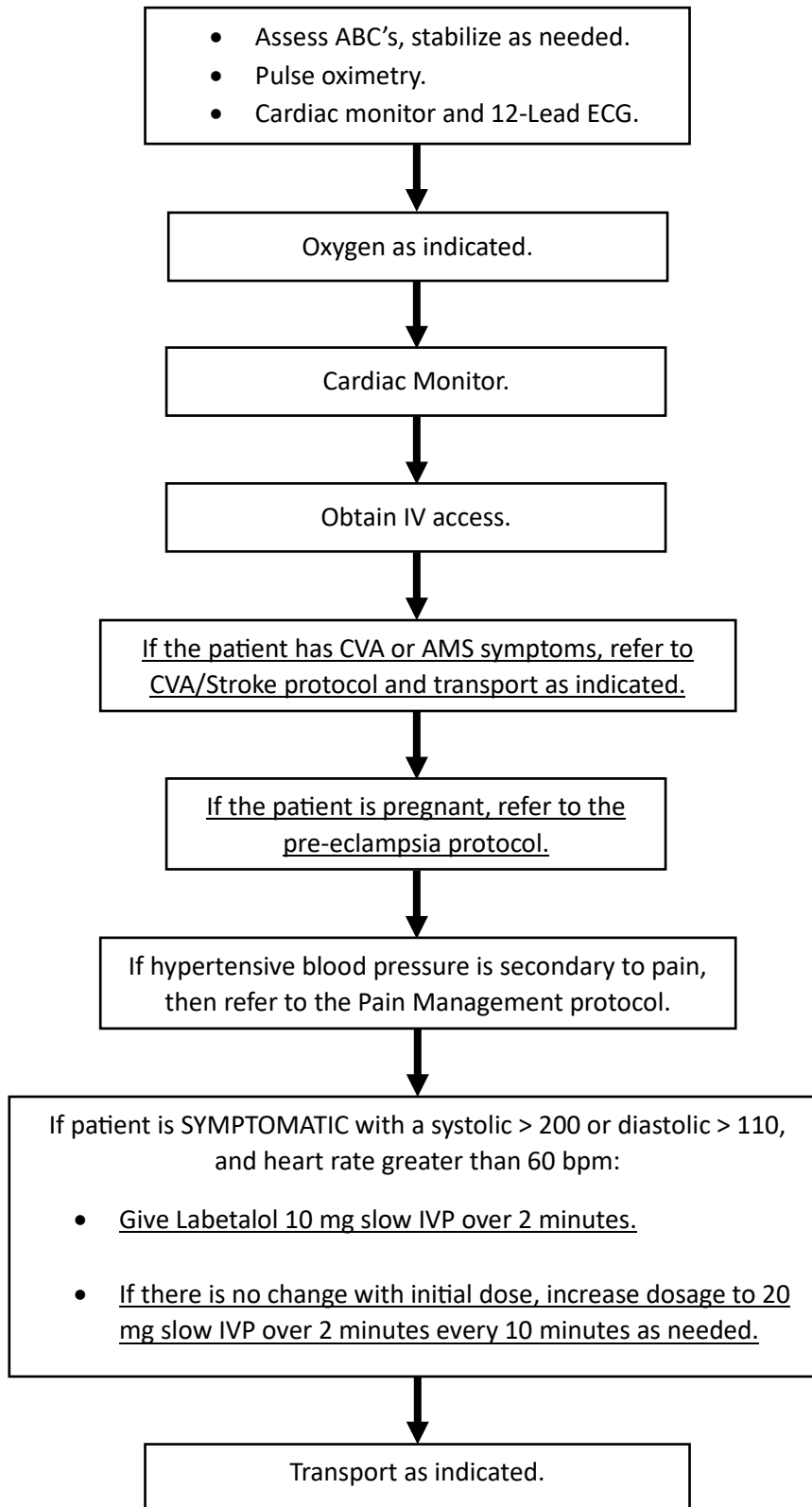


If blood glucose is > 400, administer IV fluids, 500 mL NS/LR bolus.

- Use caution with CHF and renal failure patients.



Transport as indicated.

HYPERTENSIVE EMERGENCY

SHOCK PROTOCOL (all types)

- Assure CAB's.
- Pulse oximetry.
- Oxygen via NRB.
- Cardiac monitor.

Place in supine position as tolerated.

Obtain IV access (2 large bore is preferred).

Give fluids 500 mL bolus may be repeated PRN:

- **Lactated Ringers is the preferred fluid.**
- Check lung sounds after each bolus.

Attempt to determine the etiology of shock by history and exam!

Hypovolemic – Hemorrhagic Shock:

- Continue with IV fluids bolus as necessary and titrate to effect to maintain a minimum systolic of 90 mmHg.
 - If hypotension is refractory to 30mL/Kg bolus, consider push-dose Epinephrine if required.

Anaphylactic Shock:

- Continue with IV fluid bolus and go to Anaphylaxis / Allergic Reaction Protocol.
 - Push-dose Epinephrine may be used temporarily until continuous Epinephrine infusion.

Cardiogenic Shock:

- Go to appropriate protocol.
- After the rate and rhythm normalize and the patient is still in shock, then administer either push-dose Epinephrine as needed or start a continuous Epinephrine infusion at 1-4 mcg/min carefully titrated to effect. Contact on-line medical control for any higher dosing.

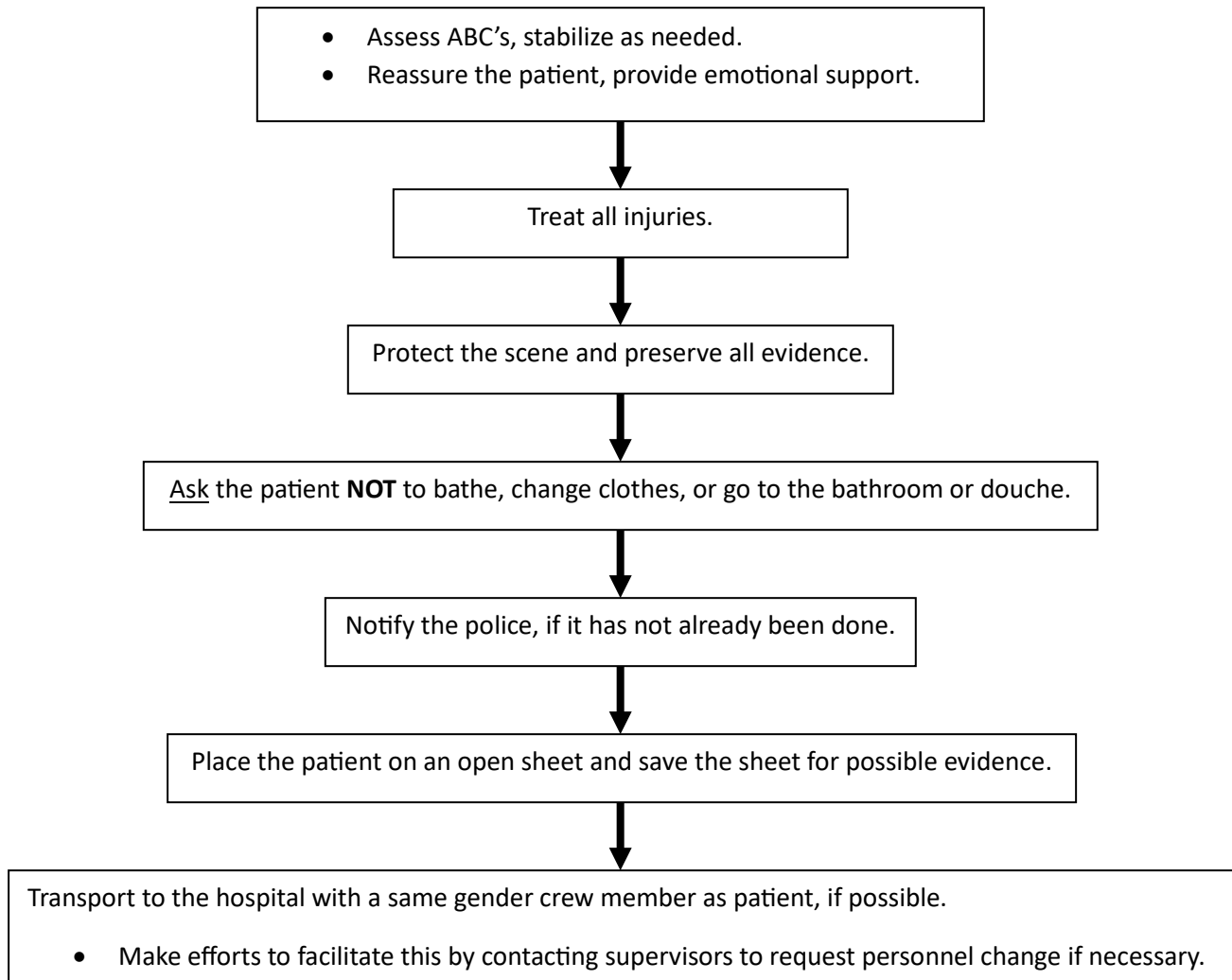
Spinal Shock (Neurogenic):

- Begin Epinephrine Drip at 1-4 mcg/min titrated to effect. Push-dose Epinephrine may be used as a bridge to continuous infusion. Contact on-line medical control for higher dosing.

Septic Shock:

- Initiate **fluids** at 30 mL/kg.
- Move to vasopressors after 30 mL/kg if there is no change.
 - Consider push-dose Epinephrine or continuous Epinephrine infusion at 1-4 mcg/min.
- Notify the receiving facility of a possible **sepsis alert** patient.

Transport Emergency.

SEXUAL ASSAULT**NOTE:**

- In situations where the patient displays acute emotional instability, it is not absolutely required to assess all vitals or other assessments that may require physical contact.
- Documentation should provide visual indicators of effective perfusion, and adequate work of breathing as best as possible given the situation at hand.

SICKLE CELL ANEMIA

Assessment / Indications supporting need for treatment:

- **History of Sickle Cell Anemia.**
- Signs of infection.
- Hypoxia.
- Dehydration.
- Painful Joint(s)
- Limited movement of joints.

- Assess ABC's, stabilize as needed.
- Obtain vital signs.
- Pulse oximetry.

Oxygen and airway maintenance appropriate to patient's condition.

Supportive Care.

IV access, consider IV bolus at 20 mL/kg (LR).

ECG, 12-Lead transmit, if indicated.

If pain persists / becomes acute:

Consult with on-line medical control for order to administer analgesics.

NO PAIN MEDS ARE TO BE GIVEN WITHOUT ORDERS.

- Attempt to verify history of actual disease process.

PSYCHIATRIC EMERGENCIES

- Assure personnel safety and involve law enforcement when needed.
- Approach the patient slowly.
- Talk in a calm and reassuring tone.
- Protect the patient's modesty and promote mutual respect to de-escalate a violent / combative individual.

Assess ABC's, stabilize as needed.
Apply O2 as indicated by patient's condition.

Physical Restraint may be needed to facilitate chemical restraint of a dangerous person:

- Restrain the patient as needed for patient care and safety.
- If restraining and securing is needed, then attempt to restrain the patient to a long spine board, if not then safely restrain them to the cot.
- Utilize law enforcement assistance whenever possible.

For the acutely agitated and dangerous patient:

Consider **Versed 2 mg IV or 5 mg IM:**

- Repeat once and only in 10 minutes at 2 mg IV or 5 mg IM.
- Further doses MUST be approved by on-line medical control.

Obtain **IV access** if possible, and if able to do so without risking safety of providers and/or patient.

Determine the patient's blood glucose level and treat as indicated.

Special Note:

Ketamine 2-4 mg/kg IM or 1-2 mg/kg IV/IO may be considered, if Versed doesn't work.

Transport as indicated.

If chemical restraint is used, once it is safe to do so for the provider, the patient should be assessed with a cardiac monitor, pulse oximetry, and oxygen applied as indicated.

DYSPNEA (ADULT)

- Assess ABC's, stabilize as needed.
- Pulse oximetry / Cardiac monitor / Side-stream ETCO₂ should be used.

Oxygen to keep O₂ sats > 90%.

Assess temperature for suspected pneumonia / sepsis.

Obtain IV access (may give one nebulizer treatment without IV access).

If patient has a history of **Asthma** or **COPD** with **wheezing** or poor air movement, then give:

Albuterol 2.5 mg in 3 mL via nebulizer and
Atrovent 0.5 mg in 2.5 mL or DuoNeb.

*If allergic to Albuterol or heart rate is > 130, then
administer nebulized Atrovent.*

Repeat Albuterol in 10 minutes if an IV is
successfully established.

In absence of IV access, contact on-line medical
control for orders to proceed with additional
Albuterol treatments.

Consider CPAP.

Magnesium Sulfate – 2 grams mixed in 100-150 mL
bag of Normal Saline infused IV/IO over 10
minutes, if severe difficulty breathing.

Solu-Medrol – 125 mg slow IV or IM.

Epinephrine 1:1,000 0.3 mg IM for Status
Asthmaticus.

If patient has **rales**, known history of **CHF**, on
diuretics and no recent fevers, then give:

If patient also has wheezing, give
Albuterol then continue from here.

Patient is not
wheezing.

Consider CPAP.

Nitroglycerin 0.4 mg SL (one sublingual tablet or
spray every 5 minutes to maximum of 3 doses), if
systolic blood pressure ≥ 130.

Consider Morphine 2 mg IV/ IO (no repeat
dosage).

Respiratory patients should
be positioned upright or
semi-fowlers when possible!

If NO RESPONSE, Intubate and/or ventilate as needed.

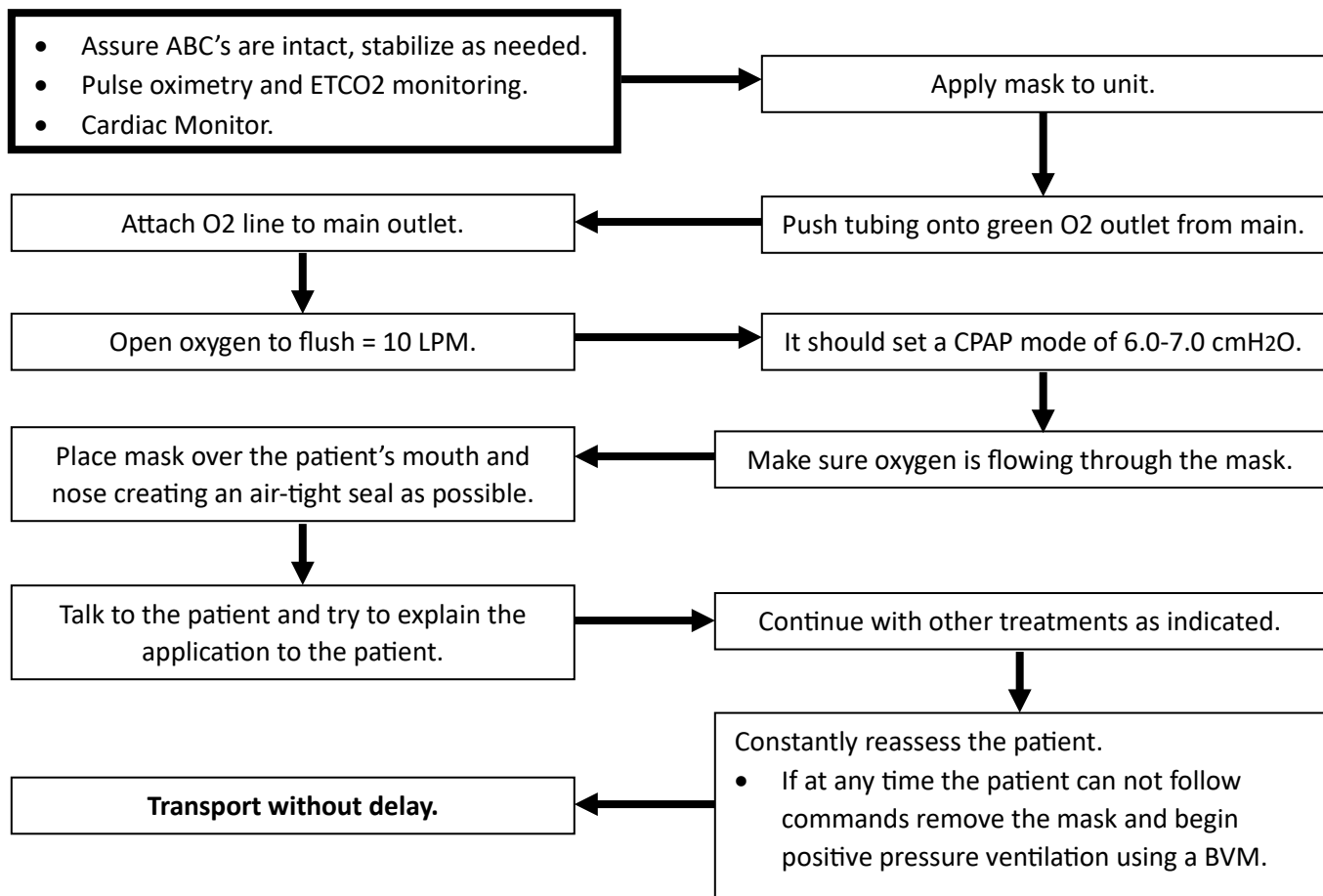
Transport as indicated.

CONTINUOUS POSITIVE AIRWAY PRESSURE (CPAP)**INDICATIONS:**

- Obvious signs in patient of moderate to severe respiratory distress (such as accessory muscle use or tripod position) from an underlying pathology, such as pulmonary edema or obstructive pulmonary disease.

CONTRAINDICATIONS:

- Respiratory Arrest.
- Signs and symptoms of a pneumothorax or chest trauma.
- Tracheotomy.
- Active gastrointestinal bleeding or vomiting.
- Patient unable to follow verbal commands.
- Inability to properly fit the CPAP system mask and strap.
- Overdoses.
- Altered Mental Status.

PROCEDURE**NOTE!**

- CPAP is only used for patients in respiratory distress, NOT FAILURE.
- Any patients with Altered Mental Status are likely in respiratory failure and needs more invasive treatments / therapies.

ALTERED MENTAL STATUS

Assess temperature if providers suspect sepsis, hypothermia, or head related illness.

- Assess ABC's, stabilize as needed.
- Attach pulse oximetry.
- Cardiac monitor if indicated.

Assess Carbon Monoxide levels, if CO exposure is suspected, give O2 NRB.

Acquire 12-Lead ECG.

Administer O2 as needed.

Obtain IV access.

If the patient is malnourished or has a history of alcoholism, administer:

- Thiamine 100 mg IV/IM.

Determine glucose level:

- If glucose is < 70 or > 400, go to appropriate Hypoglycemia / Hyperglycemia protocol.

If patient has a history of drug abuse, constricted pupils, or respiratory depression, administer Narcan.

Narcan (Naloxone) 0.4 – 4 mg IV/IM:

- EMR or EMTB personnel may administer Narcan via the intranasal route (IN).
- AEMT or Paramedic personnel may administer Narcan IN, if IV/IM supplied Narcan is unavailable.
- REPEATED as necessary or titrate to effect (maximum of 8 mg).
- Remember, Narcan treats only opiate overdose. **DO NOT** give Narcan to combative or active patients because these patients are likely not suffering from opiate overdose.

Consider Zofran 4 mg IV prophylactic administration for nausea / vomiting prior to giving Narcan.

For the agitated, violent patient, consider:

- Versed 5 mg IM.
- Repeat 2 mg IV / 5 mg IM every 5-10 minutes and titrate to effect.

Transport as indicated.

STROKE / CVA

- Assess ABC's, stabilize as needed.
- Obtain vital signs (baseline) with a manual cuff.
- Pulse oximetry, Acquire 12-Lead ECG in transport.

Transport without delay, ASAP (EMERGENCY).

Note: If known time since onset of signs/symptoms of CVA is greater than 6 hours, then non-emergency transport is acceptable (if the patient is stable).

Administer O2 at 2-4 LPM nasal cannula or NRB 15 LPM, if patient is hypoxic.

Apply cardiac monitor and acquire 12-Lead ECG.

- Upload to EPCR.

Obtain IV access.

Elevate the patient's head no higher than 30 degrees.

Check glucose:

- If glucose is less than 70, go to the hypoglycemia protocol.

Obtain history from the family, if the patient is unable to provide:

- Onset of symptoms?
- Seizure at onset of symptoms?
- Previous CVA w/ deficits?
- Previous neurologic surgery?
- Any antiplatelets or anticoagulants?
- Any recent trauma, bleeding, or surgery?

Perform Prehospital Stroke Scale:

- LAMS.
- MEND.
- Cincinnati Stroke Scale.

DO NOT TREAT HYPERTENSION without consultation of on-line medical direction.

Notify the receiving facility of a "STROKE ALERT" as soon as possible!

- When calling report, include onset of symptoms.
- Be prepared to go directly to CT scan.

If a Large Vessel Occlusion (LVO) stroke is suspected, and less than 6-hour onset:

Transport to a Comprehensive Stroke Center if transport time is NOT exceeding an additional 15 minutes.

If two or more of the following are present, the patient likely has a LVO Stroke:

- Patient states the incorrect month and/or age?
- Gaze palsy and/or deviation?
- Arm weakness (hemiplegia or hemiparesis)?

CINCINNATI PREHOSPITAL STROKE SCALE**Facial Droop** (have patient show teeth or smile):

- Normal: Both sides of face will move equally well.
- Abnormal: One side of face does not move as well as the other side.

Arm Drift (have patient close eyes and hold both arms out):

- Normal: Both arms move the same or both arms do not move at all. Findings, such as pronator grip may be helpful.
- Abnormal: One arm does not move or one arm drifts down compared with the other.

Speech (have the patient say “you can’t teach an old dog new tricks”):

- Normal: Patient uses correct words with no slurring.
- Abnormal: Patient slurs word(s), uses inappropriate words, or is unable to speak (For evaluation of acute, non-comatose, non-traumatic neurologic complaint).

Facial/Smile or Grimace:		
Have the patient show teeth or smile.	Normal: Both sides of the face move equally.	Abnormal: Left or right side of face does not move as well.
Arm Drift:		
Have the patient close both eyes and hold both arms straight out for 10 seconds.	Normal: Arms move equally or do not move.	Abnormal: Left or right arm does not move or drifts down.
Speech:		
Have the patient repeat a simple phrase such as “It is sunny outside today”.	Normal: Words stated correctly without slurring.	Abnormal: Patient slurs words or uses the wrong words, or is unable to speak.

MEND PREHOSPITAL STROKE SCALE

MIAMI EMERGENCY NEUROLOGIC DEFICIT (MEND) PREHOSPITAL CHECKLIST					
Date: _____		Name: _____		Age: _____ Sex: _____	
BASIC DATA			EXAMINATION		
WITNESS NAME: ★		WITNESS PHONE: ★		BP: L _____ / _____ R _____ / _____ Pulse: Rate & Rhythm: _____ Resp _____	
Dispatch time: _____		EMS arrival time: _____		MEND EXAM	
Departure to ED time: _____		ED arrival time: _____			
HISTORY			On scene: Perform LOC & basic exam (Cincinnati Prehospital Stroke Scale in shaded boxes) En route: If time allows, perform the complete MEND exam.		
LAST TIME PATIENT WITHOUT SYMPTOMS ★ DATE: _____ TIME _____					
T-PA EXCLUSIONS		ADDITIONAL HISTORY		MENTAL STATUS	
YES	NO			CHECK IF ABNORMAL	
☐	☐	Head trauma at onset ★		ON SCENE	EN ROUTE
☐	☐	Seizure (shaking or staring) at onset ★		☐	☐
☐	☐	Taking warfarin (Coumadin)			
☐	☐	History of bleeding problems			☐
☐	☐	Possible brain hemorrhage (severe headache, stiff neck, ↓LOC)		☐	
		Symptoms _____		CRANIAL NERVES	
		Allergies _____		B L B L	
		Medications _____		☐	☐
		Past History _____		LIMBS	
		Last Meal _____		B L B L	
		Events Prior _____		☐	☐
MANAGEMENT			STROKE-SPECIFIC ED REPORT (see starred items on checklist)		
☐ Do <u>NOT</u> treat hypertension ☐ Do <u>NOT</u> allow aspiration → Keep NPO, head up, O ₂ 2-4 L ☐ Do <u>NOT</u> give glucose (unless glucose <50) → IV NS; check fingerstick: _____ ☐ ECG rhythm _____ → If AMI, 12-lead time: _____					
SYMPTOM ONSET		NEUROLOGIC EXAM		WITNESS	
★ TIME (last time w/o sx's)		★ Level of consciousness		★ Name	
★ Trauma (history)		★ Speech/language		★ Contact info	
★ Seizure (staring, shaking)		★ Visual fields			
		★ Moto strength			

LAMS PREHOSPITAL STROKE SCALE

LAMS	Score the Affected Side	Description	Corresponding NIHSS score
Facial Droop	0 Absent	No facial asymmetry. Normal.	Facial Palsy 0-1
	1 Present	Partial or complete lower facial droop	Facial Palsy 2-3
Arm Drift	0 Absent	No drift. Normal.	Motor Arm 0 (Normal)
	1 Drifts down	Drifts down but does not hit the bed within 10 sec.	Motor Arm 1 (drift)
	2 Falls rapidly	Arm cannot be held up against gravity and falls to the bed within 10 sec.	Motor Arm 2-4
Grip Strength	0 Normal	Normal.	No NIHSS for this. Scored 0 if admission neuro exam rated grip strength as 5 (normal)
	1 Weak grip	Weak but some movement.	Admission neuro exam rated grip strength as 4 (weak), 3 (some movement against gravity), or 2 (some movement, but not against gravity)
	2 No Grip	No movement. Muscle contraction be seen but without movement.	Neuro exam rated grip strength as 1 (muscle contraction but no movement) or 0 (no muscle contraction)
TOTAL	= ____ (Range 0-5)		A score ≥ 4 is highly predicted of large artery occlusion

PREHOSPITAL SCREEN FOR THROMBOLYTIC THERAPY

Complete this assessment for all patients symptomatic for Acute Coronary Syndrome or CVA. Report to the Emergency Department Physician/Nurse any positive findings. Document all findings in the EPCR.

Witness/next of kin contact info: _____		
Time of onset of the symptoms: _____		
Systolic BP >240 mmHg	☐ Yes	☐ No
Diastolic BP >110 mmHg	☐ Yes	☐ No
Right arm vs. Left arm Systolic BP difference >15 mmHg	☐ Yes	☐ No
History of recent brain/spinal cord surgery, CVA, or injury	☐ Yes	☐ No
Recent trauma or surgery	☐ Yes	☐ No
Bleeding disorder that causes the patient to bleed excessive	☐ Yes	☐ No
Prolonged CPR (>10 minutes)	☐ Yes	☐ No
Pregnancy	☐ Yes	☐ No
Taking Coumadin, Aspirin, or other blood thinners	☐ Yes	☐ No

INTERFACILITY GROUND TRANSPORT PROTOCOL FOR PATIENTS DURING/AFTER IV tPA ADMINISTRATION FOR ACUTE ISCHEMIC STROKE

- Obtain and record vitals every 15 minutes.
- Obtain and record neurologic checks per the Cincinnati Prehospital Stroke Scale every 15 minutes.
- Follow blood pressure (BP) management per Medication Guide below.
- Patient must be kept strict NPO.
- Maintain head of cot at 30 degrees.

Acute Stroke Management:

• Maintain BP < 180/100. • If BP > 180/100, follow BP protocol below. • If Systolic BP <140 or Diastolic <80 and patient is on antihypertensive drip, titrate down and/or discontinue. • Total tPA infusion time should be 60 minutes. • Once tPA infusion completes, hang normal saline with existing tubing to infuse remaining tPA.	*** <u>No other medications are to run through tPA infusion line.</u>*** ***<u>STOP tPa if the patient develops the following symptoms:</u>*** • <u>Worsening LOC.</u> • <u>Severe headache.</u> • <u>Acute hypertension.</u> • <u>Nausea and Vomiting.</u>
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Medication Guide for controlling BP in patients during/after IV tPA administration for Acute Ischemic Stroke:

- If BP > 180/100 and Heart Rate > 60, give Labetalol 10 mg IVP slow x1 over 2 minutes; if no response after 10 minutes, may repeat x1.
- If BP > 180/100 and Heart Rate < 60, ask transferring MD for advice. **DO NOT GIVE LABETALOL!**

Potential Complications:

<u>Symptom(s):</u>	<u>Treatment(s):</u>
Hypotension (Systolic BP < 90):	• Head of bed flat. • Discontinue any antihypertensive drips. • Administer 500 mL normal saline bolus. • If major bleeding suspected, STOP tPA.
Hypertension (BP > 180/100):	• Per medication guide above.
Neurologic Deterioration:	• Assess circulation, airway, breathing (CAB). • Obtain full set of vitals and Neuro check. • Check glucose and treat if < 50.
Airway edema:	• STOP tPA if infusing. • Treat according to Allergic Reaction protocol.
Nausea and Vomiting:	• Treat according to protocol.
Bleeding:	• Apply direct pressure. • Treat according to protocol. • If major bleeding suspected, STOP tPA.

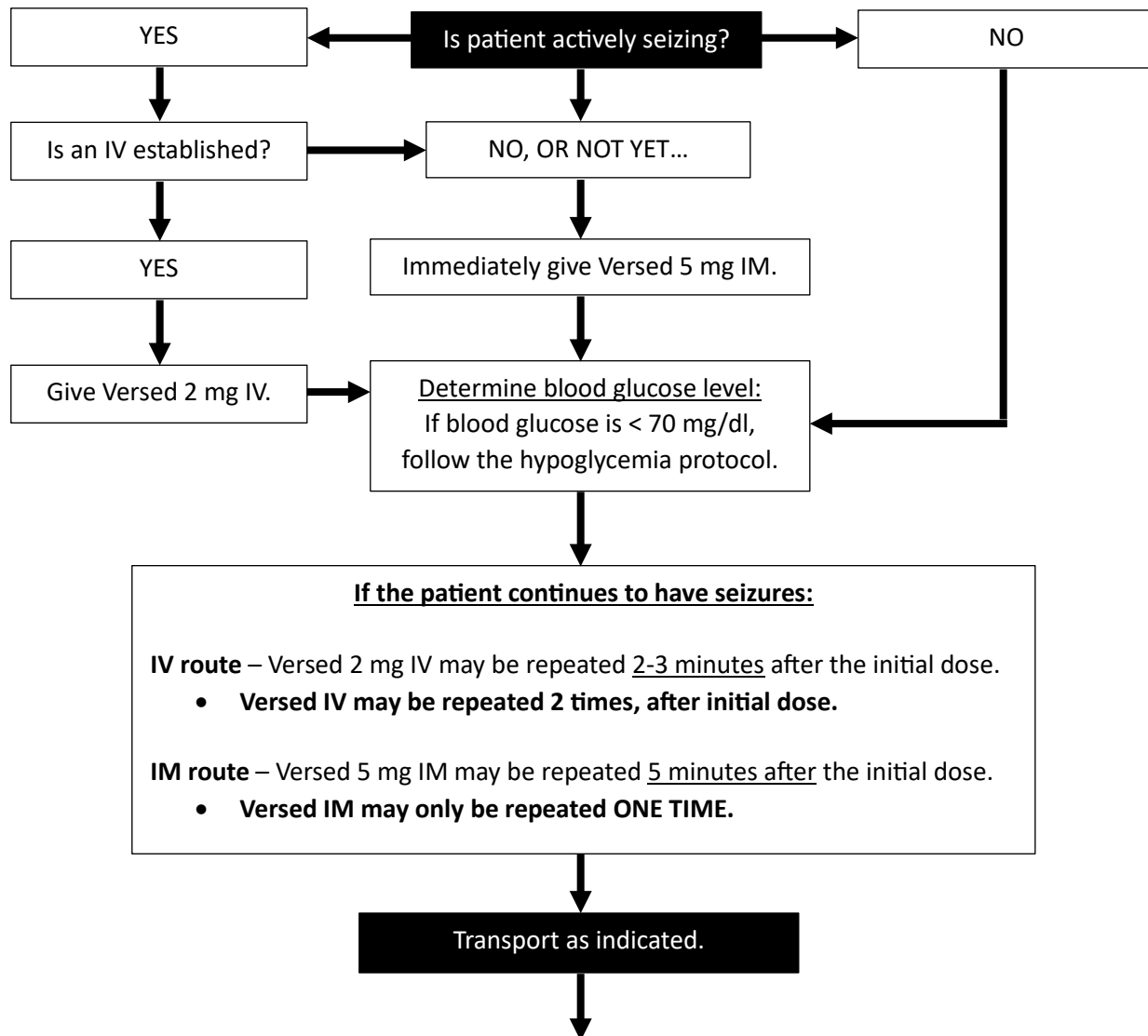
CONTACT SENDING OR RECEIVING FACILITY FOR QUESTIONS

SEIZURES**Primary Assessment:**

- Assess ABC's, stabilize as needed.
- Protect patient from injury.
- Suction as needed.
- Nasal Airway (NPA) as needed.
- Give O2 AND Assist Ventilations as needed.

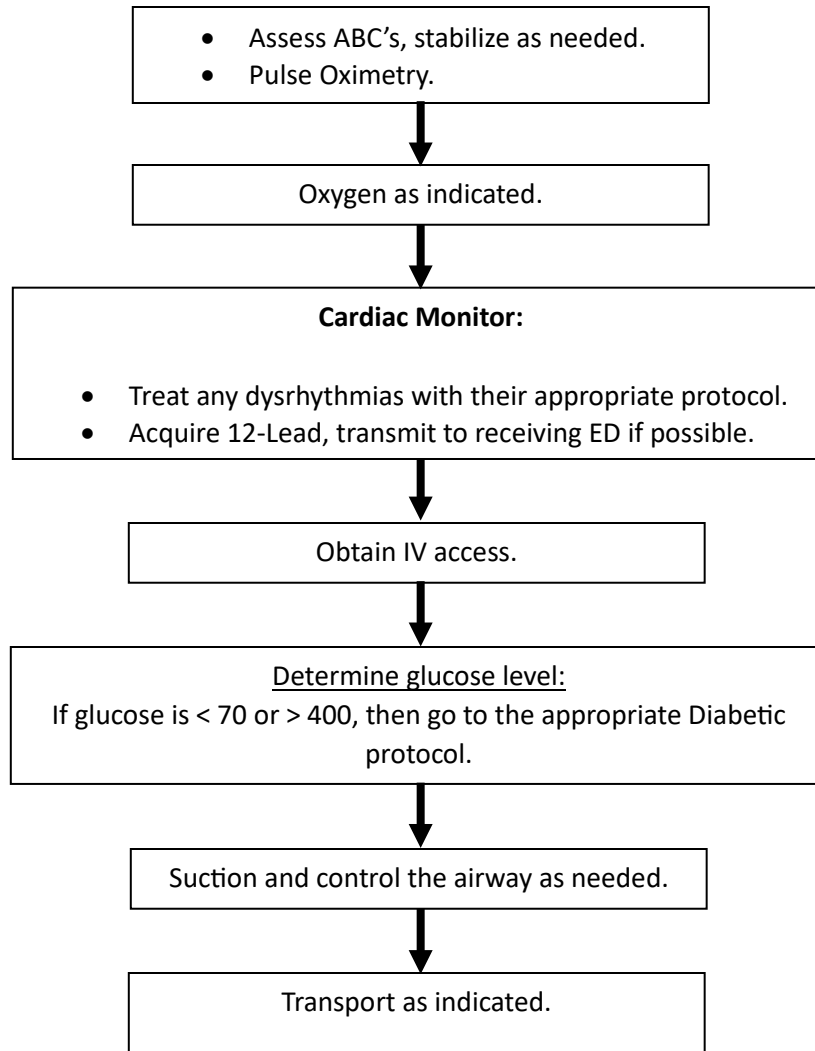
Secondary Assessment:

- Assess vital signs ASAP.
- Cardiac monitor as indicated.
- Pulse oximetry.
- Capnography (required if giving Versed).
- Assess temperature as indicated.

**SPECIAL CONSIDERATION IN PREGNANT / POST-PARTUM PATIENTS, (suspected eclamptic seizure):****If the patient is > 20 weeks pregnant OR < 2 weeks post-partum without a history of seizures:**

- Mix 4 grams of Magnesium Sulfate in a 100 mL bag of NS and infuse IV/IO over 20 minutes.

This can be given in conjunction with Versed

SYNCOPE (FAINTING)

ANAPHYLAXIS / ALLERGIC REACTION (ADULT)

- Assess ABC's, stabilize as needed.
- Pulse oximetry.
- Oxygen via Non-rebreather.
- Cardiac monitor.

Administer Epinephrine (1:1,000):
0.01 mg/kg IM, maximum each dose 0.5 mg.

- Do not delay Epinephrine, give ASAP.
- Can repeat once after 15 minutes.
- Use caution with known cardiac history or age greater than 60 years. If so, consider a maximum of 0.3 mg.

If wheezing is present:

- Administer **Albuterol 2.5 mg / 3ml** via Nebulizer.

Obtain IV access.

If hypotensive or inadequate tissue perfusion:

- Infuse 500 mL bolus of LR or NS, repeat as necessary and titrate to effect.

Administer Antihistamine:

H1 Blocker – Diphenhydramine (Benadryl)

- 25 mg IVP or 50 mg IM.

Administer Steroid:

Corticosteroid – Solu-Medrol (Methylprednisolone)

- 125 mg IV or IM.

If patient is still in extreme Anaphylaxis after treatment above, then initiate continuous Epinephrine infusion at 1-4 mcg/min in Adults. Push-dose Epinephrine may be used temporarily until continuous infusion.

Transport as indicated.

OVERDOSE – GENERAL / MEDICATIONS

- Assess ABC's, stabilize as needed.
- Pulse oximetry / Cardiac monitor/ use Capnography as indicated.

Suction as needed.

Oxygen via NRB.

Aggressive airway control
with ventilation if needed.Obtain IV access.

Any hypotension, then give a fluid bolus of NS 1 Liter.

Intubate as needed.Giving Narcan early in opioid overdose may
prevent the need to intubate the patient.Check blood glucose, if < 70 or > 400, go to the
appropriate hypo/hyperglycemia protocol.If the patient is seizing, then go to the
Seizure protocol.If a **Narcotic Opiate** overdose is suspected:

- Pinpoint pupils.
- Hypotension.
- Decreased Respirations.

Administer **Narcan 0.4-4 mg IV or IM**, repeat
PRN, titrate to effect up to 8 mg.For all personnel, **Intranasal (IN) Narcan
prefilled nasal spray may be administered at 4
mg in one nostril, repeated in 5 minutes as
needed. Total dose of 8 mg without further
orders.***Otherwise, Narcan should be given IM in the
absence of vascular access.*Obtain history:

- Type and amount of poison.
- If possible, bring the container with the patient.
- Route of intake.
- Time of intake.
- History of drug or alcohol usage.

If a **Beta Blocker** overdose is suspected and the
patient is bradycardic and/or hypotensive:

- Give **Glucagon 1 mg IVP.**

If the patient is agitated and a possible stimulant
overdose is suspected:Consider **Versed 5 mg IVP or IM.**

- Repeat 2 mg IVP or IM every 5 minutes
and titrate to effect.

*Ketamine can be considered but may only be
given with on-line medical control orders.*If a **Tricyclic** overdose is suspected:*(Contact medical control if not sure of the drug)*

- Give **Sodium Bicarb 8.4% 1 mEq/kg
IV/IO.**

If a **Calcium Channel Blocker** overdose is
suspected and the patient is bradycardic and/or
hypotensive:

- Give **Calcium Chloride 1 gram (10 mL) in
a 100 mL bag of NS and given IV/IO over
2-5 minutes.**

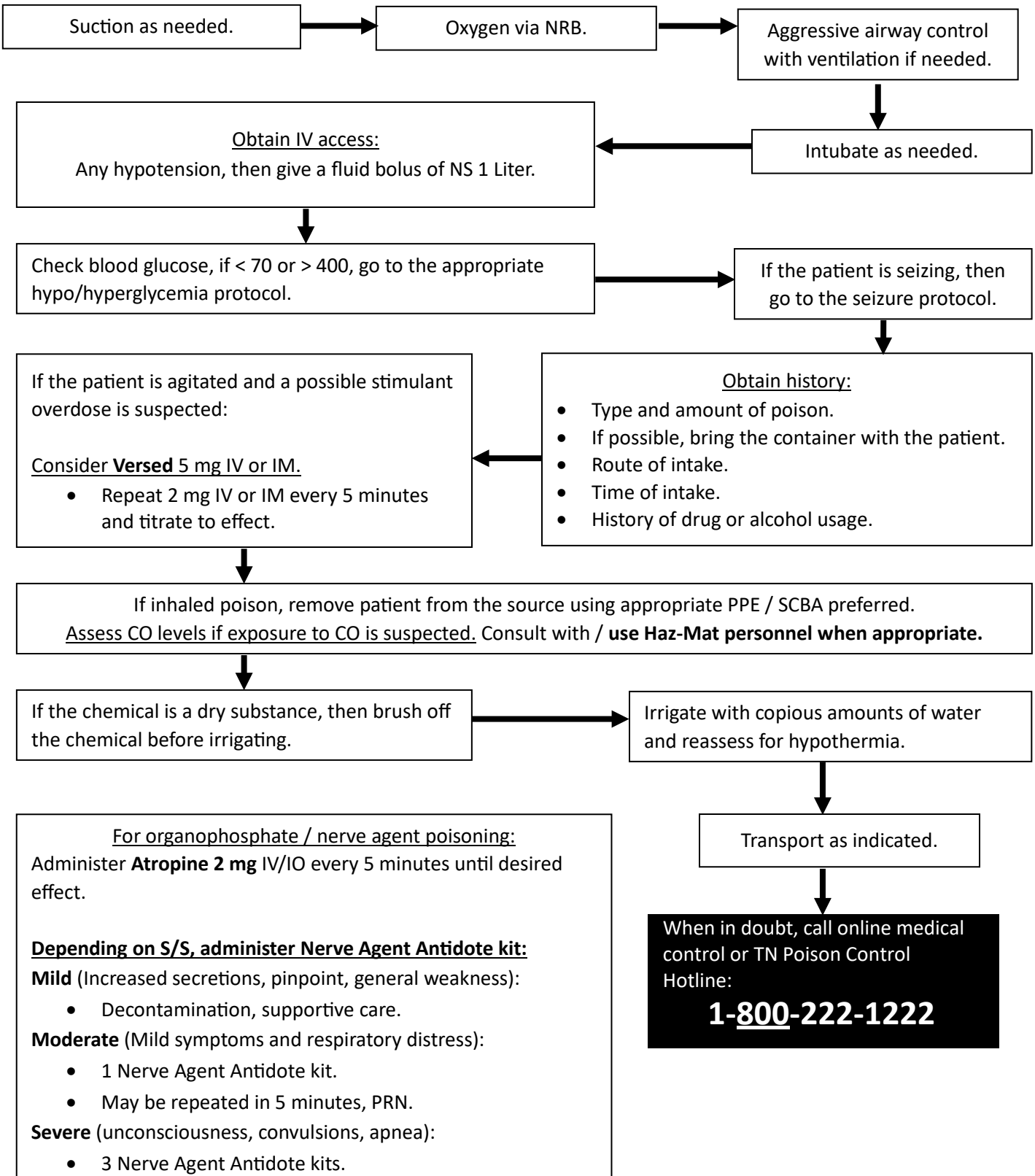
Transport as indicated.

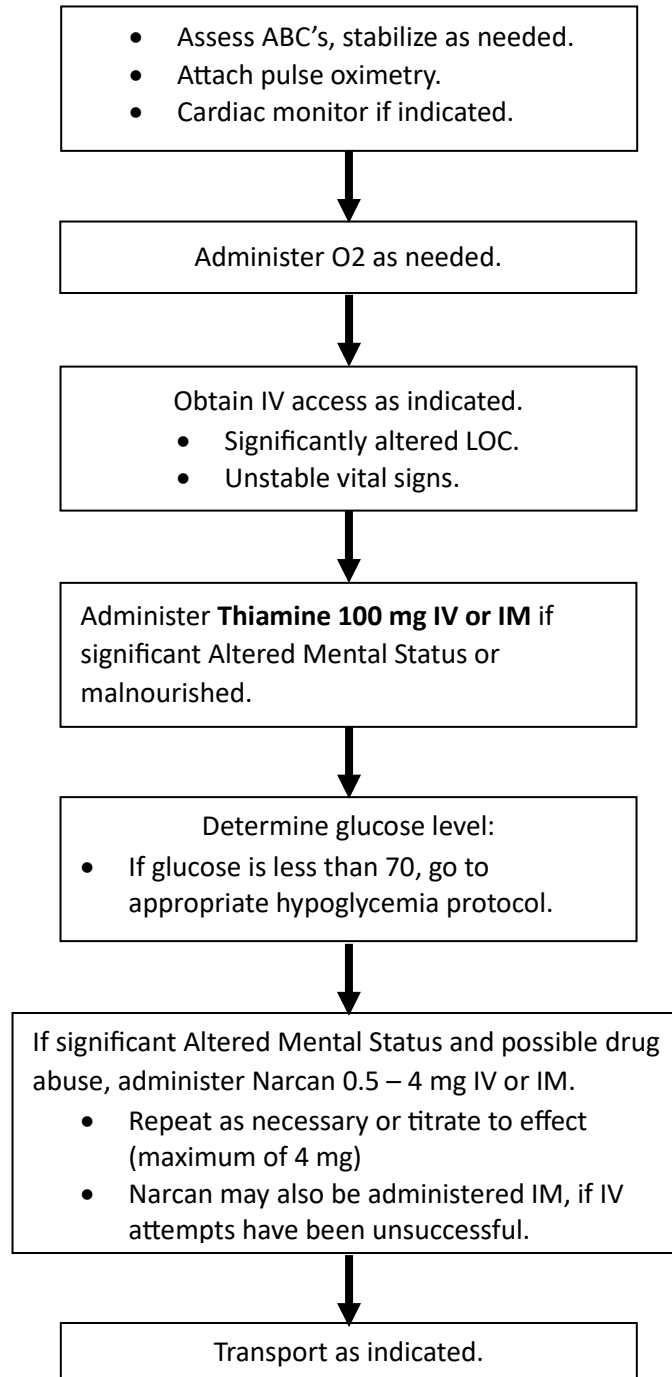
When in doubt, call online medical control or TN Poison Control Hotline:

1-(800)-222-1222 or 1-(888)-222-1222<https://www.vumc.org/poisoncenter/>

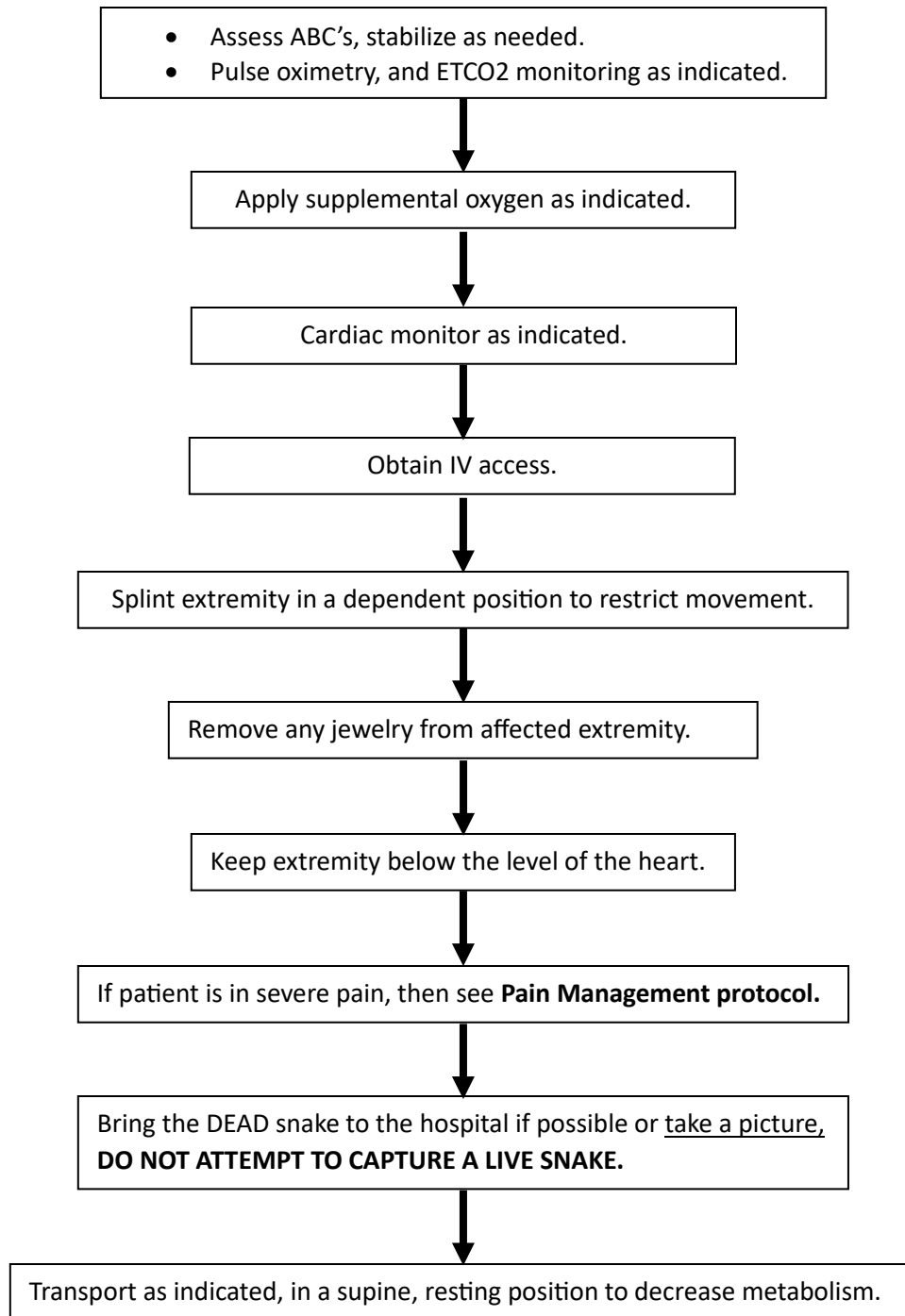
POISONING / CHEMICAL EXPOSURE / HAZ-MAT / NERVE AGENTS

Personnel safety is the highest priority. Do not handle the patient unless they have been decontaminated.
All EMS treatment should occur in the Support Zone after decontamination.



ALCOHOL EMERGENCIES**Clinical Note:**

If the patient does require medical attention yet is combative, refer to the restraint protocol and be prepared to use chemical restraint (preferred over physical) as necessary to best protect EMS personnel, the patient, and others from any harm. Sound clinical judgment shall be applied, consult on-line medical control as needed.

SNAKE BITE / ENVENOMATION

The most acute / **immediate life threat** from a snake bite or other envenomation is **anaphylaxis**.

- Immediately proceed to the allergic reaction / anaphylaxis protocol as needed.

NEAR DROWNING

- Assess ABC's with attention to C-spine, stabilize as needed.
- Use aggressive **airway control** and suction as needed.
- Pulse oximetry and EtCO₂ monitoring as needed.
- Consider placement of a gastric tube in patients suffering from submersion injuries / near drowning.



Administer oxygen by BVM or non-rebreather as necessary.



Cardiac monitor as indicated.



Obtain IV access.



Consider CPAP for fresh water and/or saltwater drowning.



If in cardiac arrest, got to the appropriate protocol.



Transport as indicated.

Transport should be made in all situations involving near drowning. Even in patients who present as stable upon scene, secondary drowning is still a threat to the patient's life. Refusals should involve physician consult via on-line medical control, and absolutely AGAINST MEDICAL ADVICE. Do your best to encourage any patient considering refusal for near drowning to allow transport to a hospital. If possible, assess and document O₂ Sat and EtCO₂ to ensure hypoxia is not present before allowing a patient to refuse transport.

If guardians of minors are not allowing transport of a child/minor who had a near drowning event, consult with supervisors, on-line medical control, and involve law enforcement as necessary.

HYPERTHERMIA / HEAT RELATED ILLNESS

Immediate cooling has been proven to be more beneficial if done prior to transport from sporting events. If athletic trainers have cooling capabilities, allow them to cool the patient prior to transport.

- Assess ABC's, stabilize as needed.
- Assess temperature.
- Cardiac Monitor.

If history is suggestive of heat exhaustion or heat stroke:

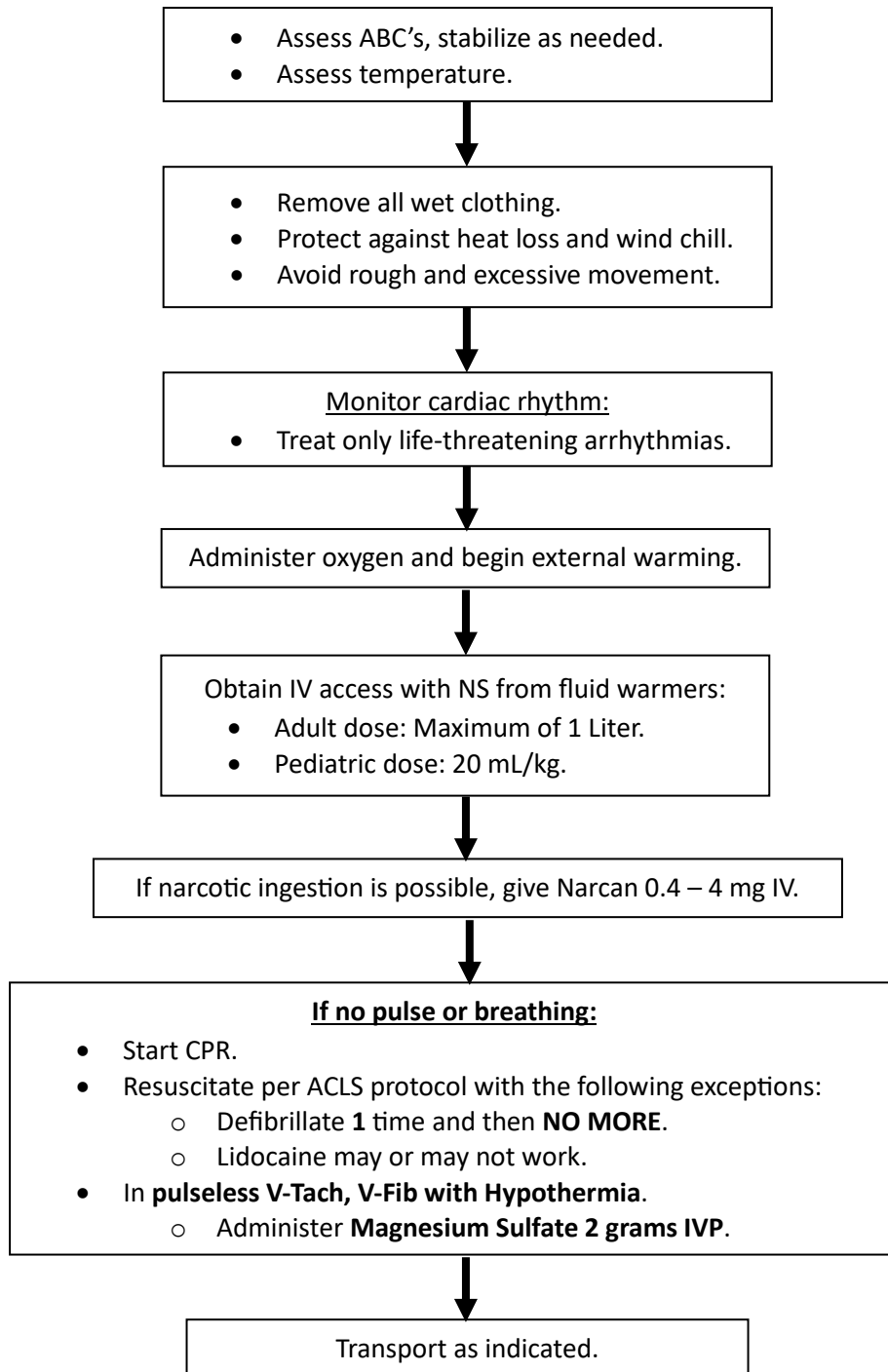
- Remove to cooler environment.
- Cool with moist sheets slowly so that the patient will not start to shiver.

Oxygen as indicated.

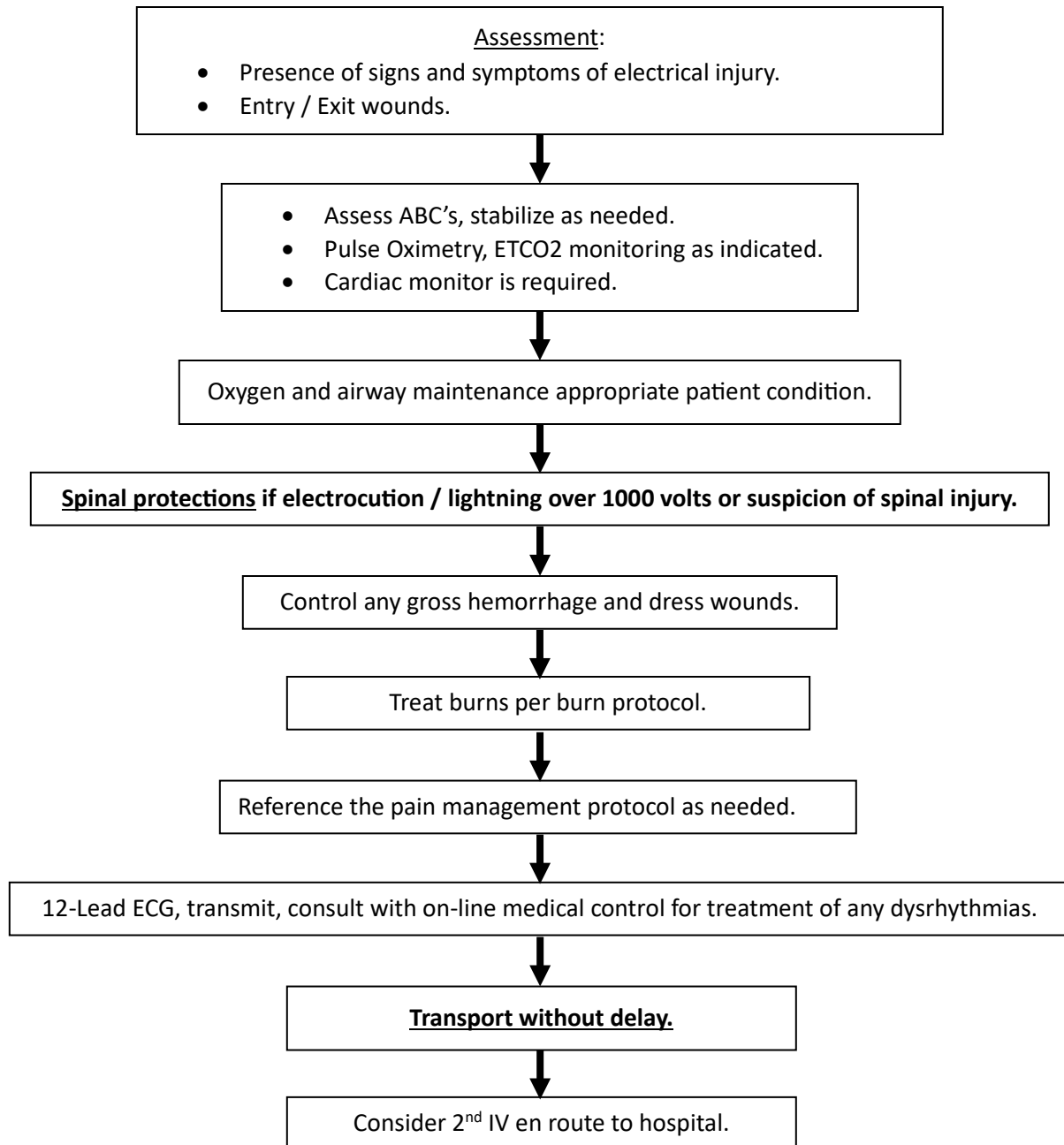
Obtain IV access and administer Lactated Ringers (LR) 20 mL/kg.

If seizures are present, then go to the Seizure protocol.

Transport as indicated.

HYPOTHERMIA

- Keep the doors of the ambulance CLOSED with heat on high, especially in the winter, 3 layers of blankets are recommended for re-warming, with at least one blanket under the patient between them and spine boards, etc.
- Note that with a hypothermia as subtle as 96 degrees. The body can lose the ability to form clots / slow bleeding in trauma patients.

ELECTROCUTION / LIGHTNING INJURIES

ABDOMINAL / PELVIC TRAUMA

- Assess ABC's, stabilize as needed.
- Pulse oximetry, ETCO2 monitoring, and ECG as indicated.



Oxygen and airway maintenance appropriate for the patient's condition.



C-Spine protection, as indicated.



Stop any life-threatening hemorrhaging.



Supportive care.



Place patient in position of comfort as best possible, with attention applied to C-spine needs.



Patient Pregnant:

If patient is past 1st trimester: place patient in left lateral recumbent position if placed supine or immobilized onto spine board, this is to minimize the risk of compression the inferior vena cava with the uterus in pregnant patients.



Penetrating Object(s):

- If no penetrating object: place patient supine with legs elevated and flexed at knees and hips. If no C-Spine concerns, contact medical control.
- If penetrating object present: stabilize object(s).



Evisceration:

- If present: place patient supine with legs elevated and flexed at knees and hips. If no C-Spine concerns, contact medical control. Cover evisceration(s) with saline soaked trauma dressing.



Establish Vascular Access – appropriate to patient's needs:

- If systolic BP < 90 mmHg, infused IV/IO Lactated Ringers 20 mL/kg bolus to titrate systolic BP at permissible hypotension to maintain mental status and distal pulses.
- **Consider TXA protocol for the unstable trauma patients.**



Transport without delay.

AMPUTATIONS

- Assess ABC's, cardiac monitor, and stabilize as needed.

**Control bleeding!**

- Direct Pressure
- **Tourniquet**, if needed (Document time of application)
 - **Bright red / spurting blood = Tourniquet!**
 - **Apply 1 on arm, 2 on leg – High and Tight!**
- Consider using hemostatic dressings / gauze as needed.



- Assess pulse oximetry.
- Apply O2 as indicated.

Obtain IV access.
(LARGE BORE PREFERRED)



Manage hypovolemic shock if present with:
Lactated Ringers 500 mL bolus

- Use warm fluids when possible.
- **Repeat as necessary and titrate to effect.**
- Use caution in patients with a history of CHF or renal failure.

**SEE PAIN MANAGEMENT PROTOCOL**

- Fentanyl is recommended, along with Ketamine for adults.



Transport as indicated.

Care of the amputated part:

1. Rinse the amputated part, DO NOT SCRUB!
2. Wrap part in moistened gauze and place in a plastic bag.
3. Place sealed bag in a container filled with ice water if available.
4. Label container with name, date and time.

Refer to TXA protocol for unstable trauma patients who have signs of shock secondary to blood loss.

AVULSED TEETH

- Assess ABC's, stabilize as needed.
- Pulse oximetry, ETCO2 monitoring, and ECG as indicated.

Oxygen and airway maintenance appropriate for the patient's condition.

C-Spine protection, as indicated.

Treat other associated injuries.

Pay attention to the airway, bleeding and avulsed teeth may cause obstruction.

Supportive care – Pain management.

Avulsed teeth may be handled in much the same manner as small body parts; i.e. rinse in normal saline.

- (Do not rub or scrub) and place in moistened gauze, but there is **no need to cool with ice.**

Consult On-Line Medical Control regarding re-implantation of avulsed teeth, proceed as follows if physician approves:

Re-implantation at the scene is recommended as this creates maximum possibility of reattachment. The following guidelines pertain to re-implantation at the scene:

- Applicable only for permanent teeth (i.e with patients over 6.5 years of age).
- Applicable when only one or two teeth are cleanly avulsed and the entire root is present.
- Applicable only to anterior teeth (front 6, upper and lower).
- **The patient must be conscious.**
- Should be attempted within the first 30 minutes. (The sooner performed the greater the success rate.)

Do not force re-implantation. Gentle insertion is all that is necessary. Slight incorrect positioning can be corrected later.

If re-implantation is not feasible and the patient is a fully conscious adult then the best procedure is to place the tooth in the mouth, either under the tongue or in the buccal vestibule. **This is not recommended for children.**

Transport without delay.

BURNS**Stop the burning!**

- Remove burned or smoldering clothes.
- Cool with cool (not cold), moist, sterile towels if available.
- Burns involving more than 10 percent body surface area should be covered with a dry sterile dressing, preserve heat loss when possible.

Remove dry chemicals by brushing off the substance, and remove liquid chemicals by flushing with large amounts of water unless contraindicated according to the ERG handbook.

Assess ABC's, stabilize as needed.

Oxygen via NRB and control airway as indicated.

Cardiac monitor as indicated.

Obtain IV/IO access if applicable (Large Bore).

Is the patient hypotensive?

NO

YES

****Only if patient is hypotensive,
Initiate a LR bolus of 500 mL in
adults.**

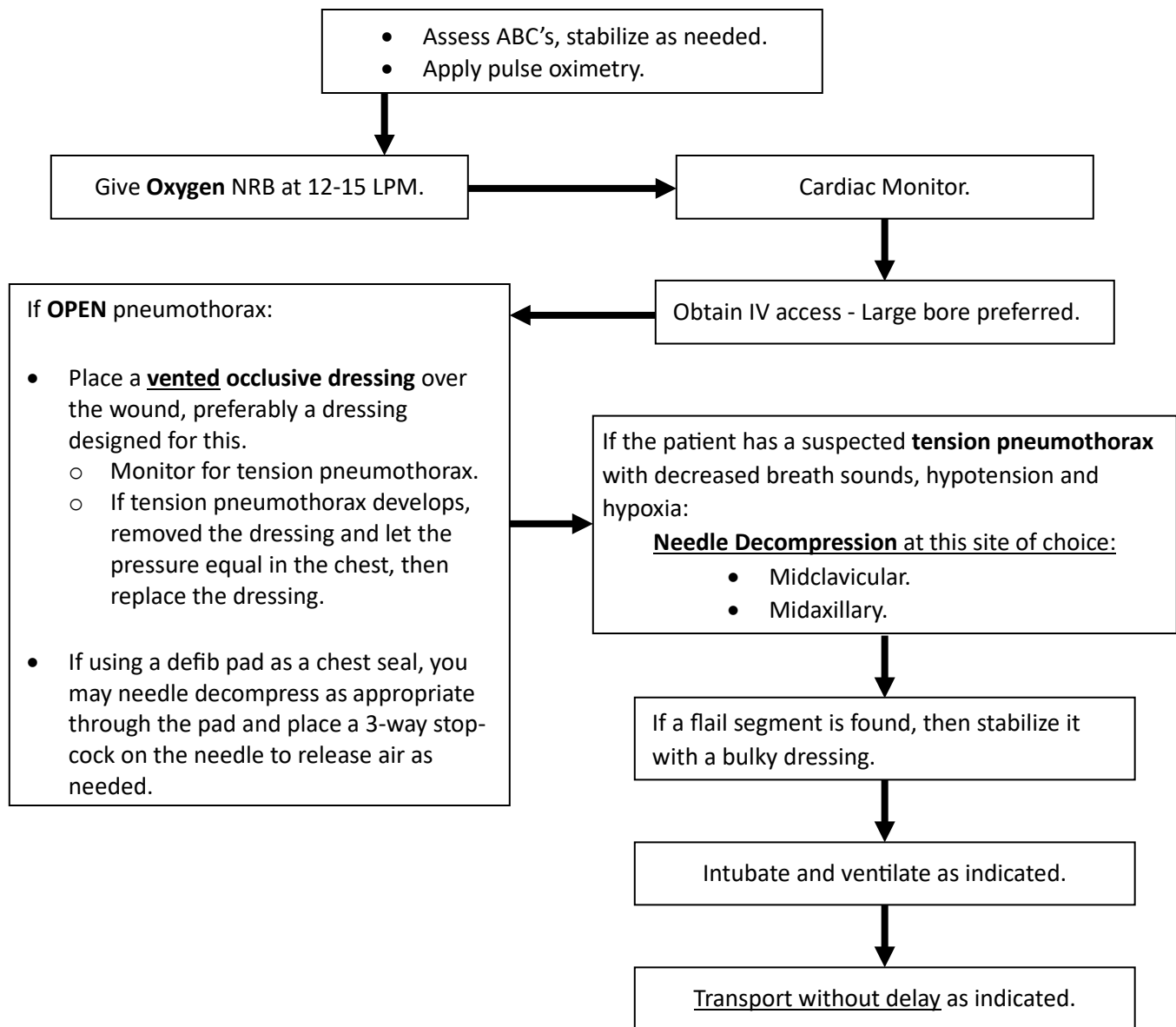
Reference the Pain Management Protocol as needed, for acute pain, Fentanyl may need to be administered in conjunction with Ketamine, this will also support BP.

Transport as indicated...Critical burns, which likely require a burn center (Vanderbilt) or (Skyline) would be:

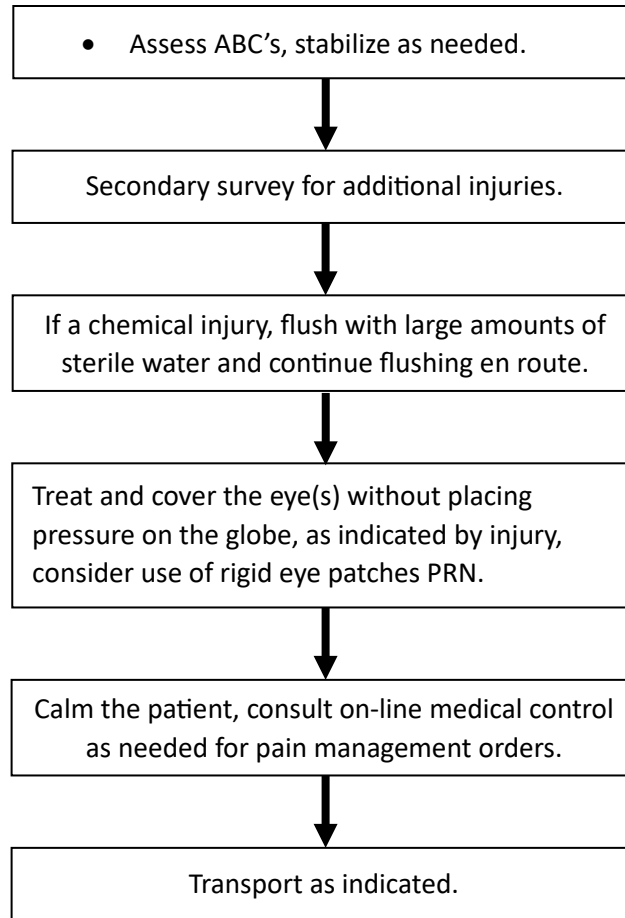
- Burns with > 20% BSA of partial thickness involvement or worse in adults.
- Any burns that involve airway.

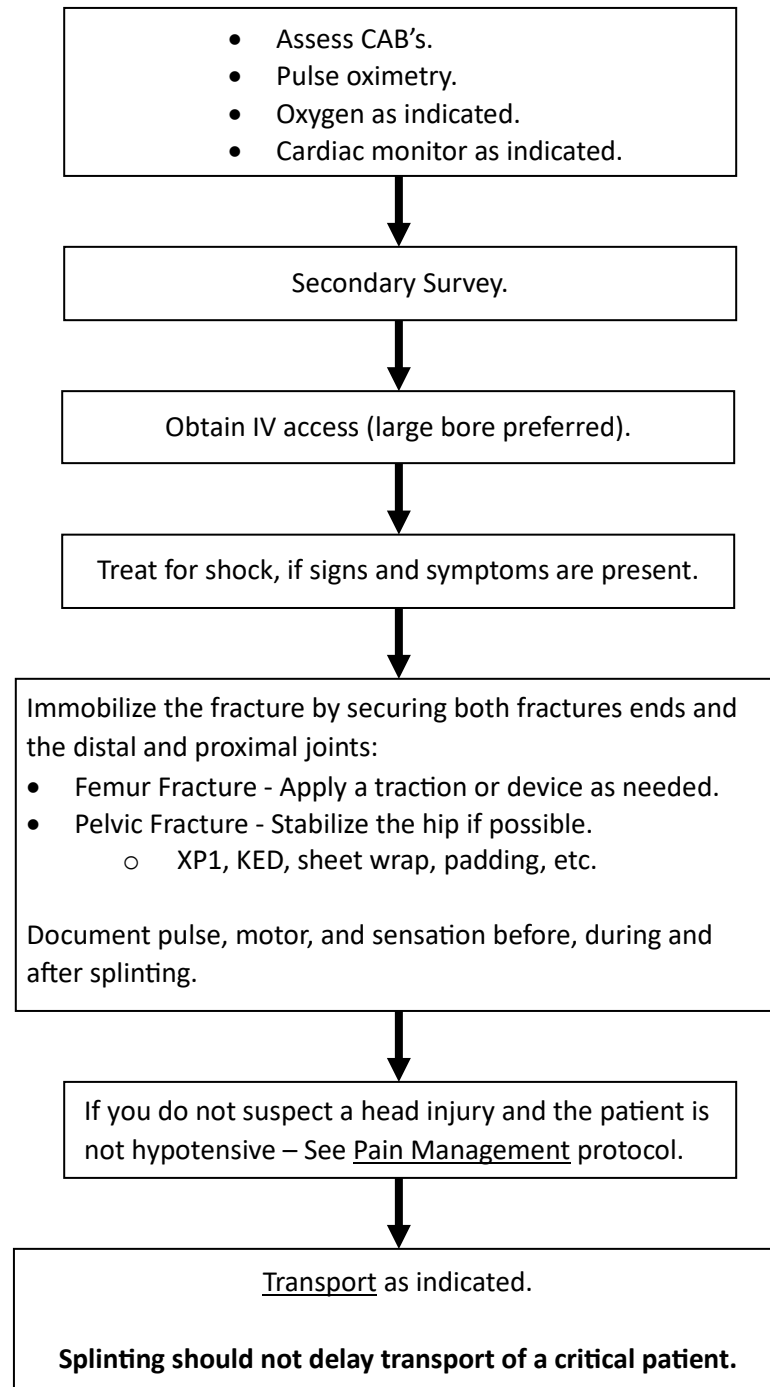
Treatment notes:

- Keep patient warm, hypothermia is a complication of critical burned patients.
- Focus to prevent infection, use dry sterile dressings (burn sheets) if critical.

CHEST TRAUMA

Reference the Pain Management Protocol as needed.

EYE INJURIES

FRACTURES (General Care)

HEAD INJURIES (Traumatic Brain Injury)

Maintain C-Spine precautions.

- Assess ABC's, stabilize as needed.
- Pulse oximetry.
- Cardiac monitor.

Oxygen as indicated.

Ventilate with 100% oxygen and intubate as soon as possible, if needed.

- Consider RSI, if needed.

IV access (vascular access).

Incline head of spine board or cot 15 degrees.

Refer to chemical restraint protocol and consult on-line medical control as needed.

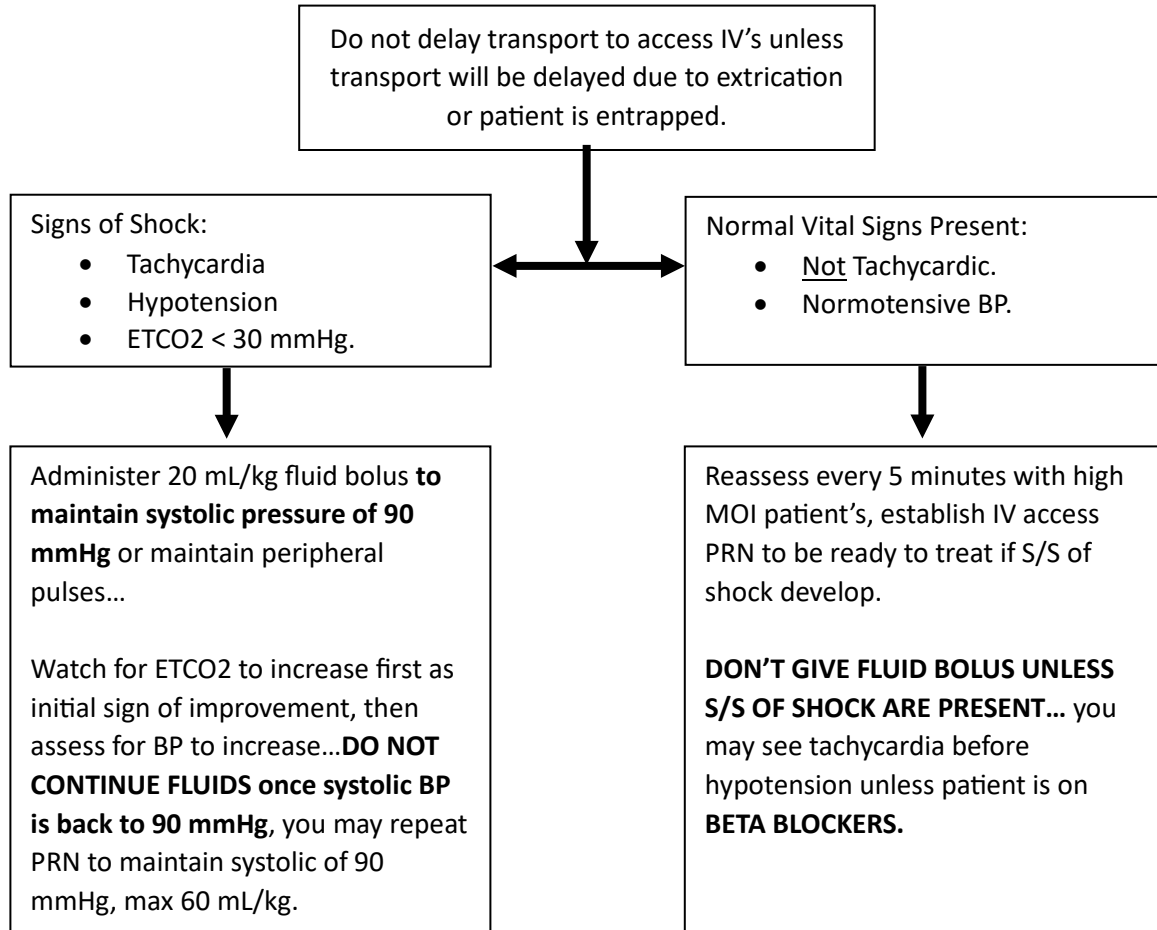
Restrain as needed to LSB for the combative patient.

(Avoid restraining the patient to the cot, as they will have to be transferred to the ER bed at the hospital.)

Refer to the **Pain Management Protocol** as needed.

- Patients may receive pain medication if necessary.
- If patient is altered due to head injury, then consult medical control.
- Document mental status before and after administration of pain medications.

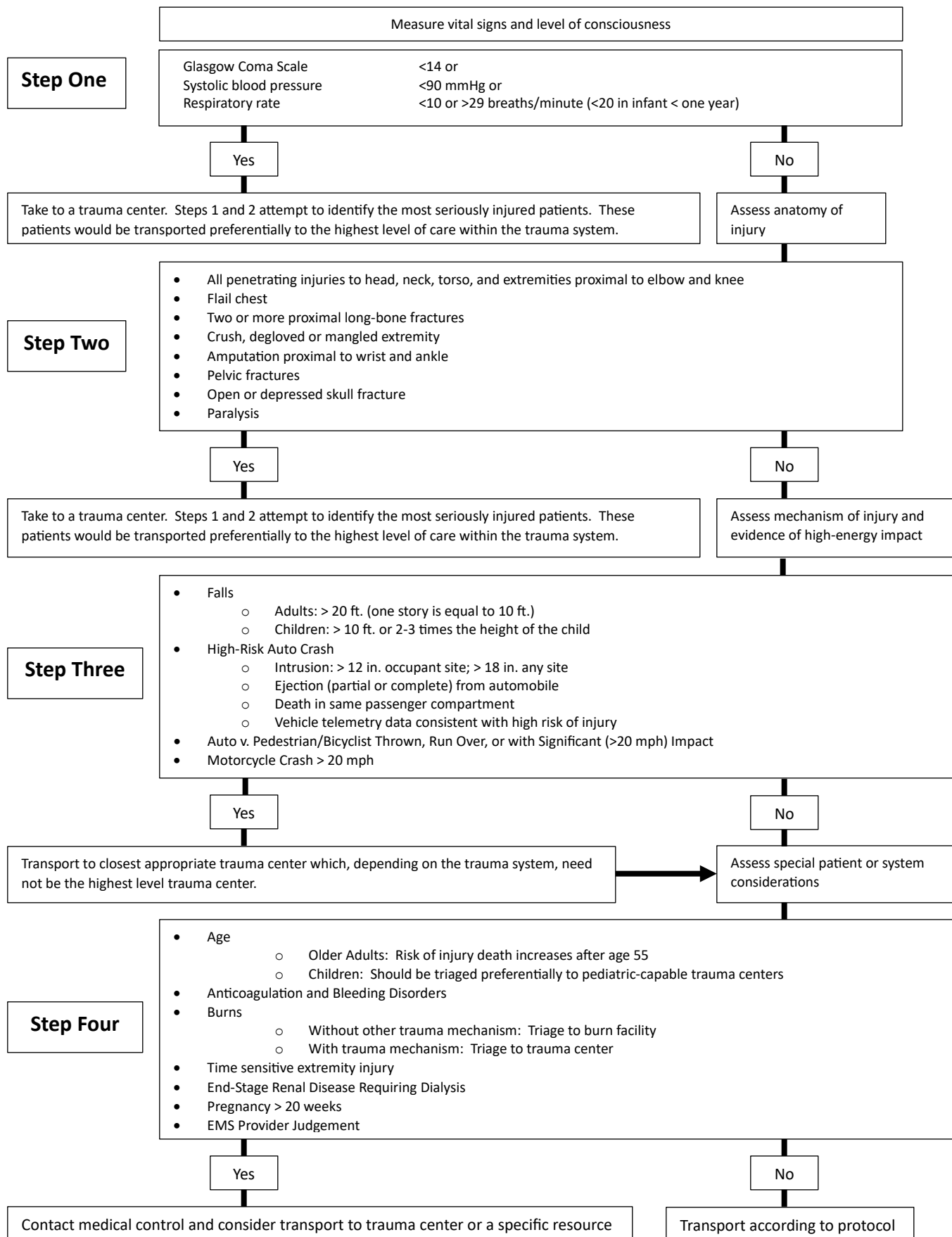
Transport as indicated.

PERMISSIVE HYPOTENSION / TRAUMA FLUID RESUSCITATION**Consideration:**

- Ensure IV fluids are warm when infusion into trauma patients.
- You may consider use of IO device unless area of injury prohibits IO placement when you are unable to obtain peripheral IV's.... fluid administration and rate are the same with IV/IO.
- **Lactated Ringers (LR) is the preferred IV fluid of choice for all patients receiving bolus of isotonic crystalloids. (Updated Jan 2020)**
- **TXA should be considered in shock trauma cases.**

HUMERAL HEAD IO PLACEMENT IS PREFERRED FOR A PATIENT THAT NEEDS FLUID RESUSCITATION.

Humeral head IO access is NOT permitted for pediatric patients less than 8 years old, unless authorized by on-line medical control. (Updated Jan 2021)

FIELD TRIAGE DECISION SCHEME: THE NATIONAL TRAUMA TRIAGE PROTOCOL

SOFT-TISSUE / CRUSH INJURIES

- Assess ABC's, stabilize as needed.
- Pulse oximetry, ETCO2 monitoring, and ECG, as indicated.



Oxygen and airway maintenance appropriate for the patient's condition.



C-Spine protection PRN.



Stop any life-threatening hemorrhaging.



Extremity trauma / exsanguinating hemorrhage – consider tourniquet use.



Apply splints as needed and stabilize penetrating objects:

- Splinting may help prevent secondary injury from dangerously sharp broken bone ends.
- Do not delay transport.
- Critical patients need to be placed in correct anatomical position on LSB/cot, and transport to a trauma center preferably.



Cover open fractures/lacerations, check PMS, avoid unnecessary movement.



Reference the **Pain Management Protocol** as needed.



IV Normal Saline/Lactated Ringer TKO

If systolic BP < 90 mmHg, infuse IV/IO Normal Saline 20 mL/kg bolus to titrate systolic BP at permissible hypotension, or to maintain mental status and distal pulses.

- **Consider TXA protocol for the UNSTABLE trauma patient.**

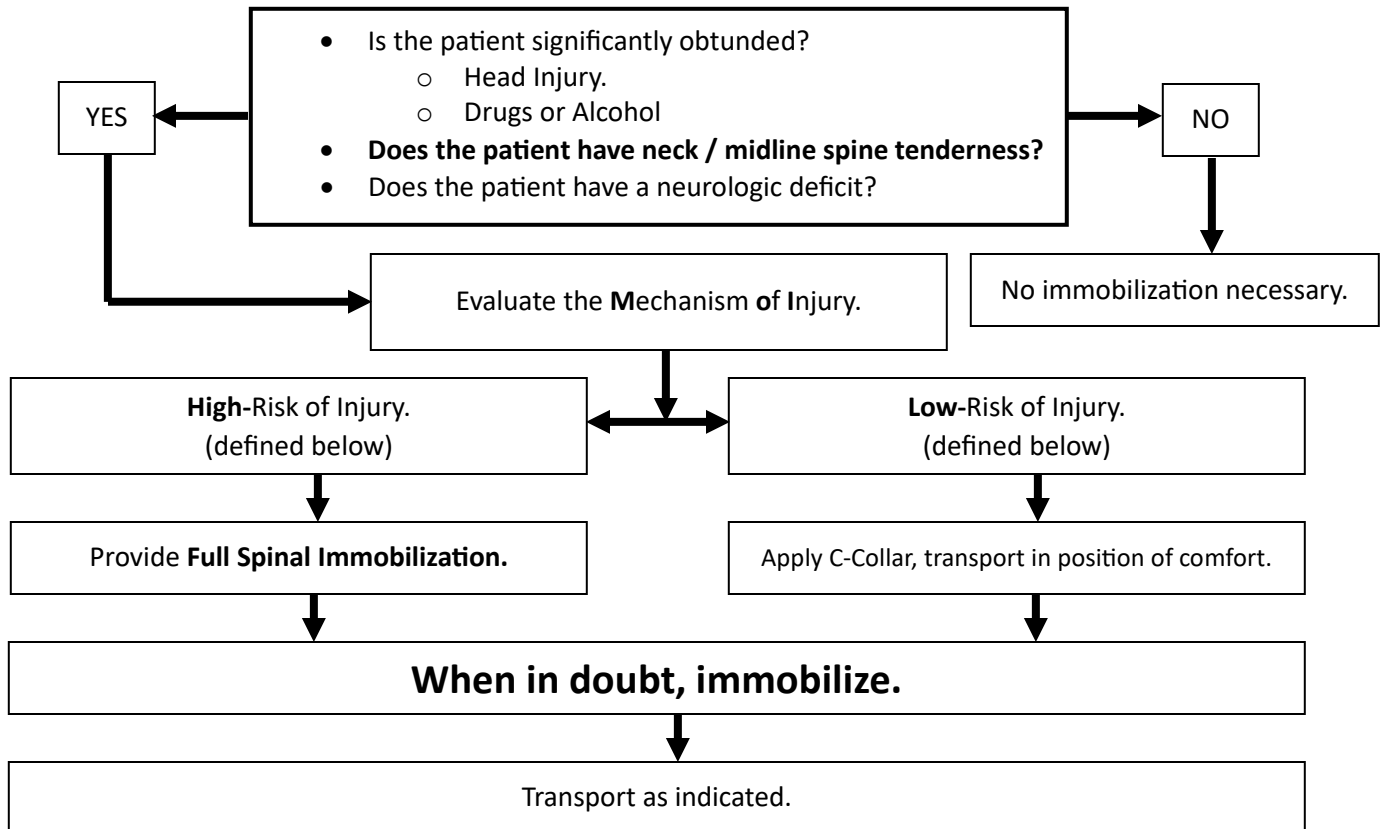


Transport without delay.

SELECTIVE SPINAL IMMOBILIZATION

Spinal immobilization should be performed on the basis of mechanism of injury and the patient's symptoms.

- If the patient has major mechanism of injury (MOI) or MOI cannot be ruled out, then always immobilize.

**Examples of High Risk Mechanism of Injury:**

- High-speed MVC.
- Rollover MVC.
- Intrusion into patient compartment.
- Death in the vehicle.
- Fall 2x patient height.
- Fall onto head.
- Auto vs. Pedestrian.
- Diving Injury.
- Penetrating trauma with motor/sensation deficits.

Examples of Low-Risk Mechanism of Injury:

- Low speed MVC.
- Fall from a standing position.
- Penetrating trauma without motor/sensation deficits.
- Ambulatory at the scene.

Pediatric Patient's Low-Risk Injury:**No board or collar if:**

- Fall from standing.
- Fall from a bed.
- MVA in a car seat.
- Neurologically normal for age.
- No apparent serious injury.

SPECIAL CONSIDERATION IN PREGNANT PATIENTS:

If spinal immobilization is required, attempt to tilt the patient 15 – 30 degrees to her left side in order to keep the baby in vitro from compressing the patient's inferior vena cava, causing hypotension.

If the patient refuses immobilization, all risks are to be explained to the patient and documented in the narrative along with a witness' signatures.

Any deviation from this protocol requires contact with Online-Line Medical Control.

SPINAL INJURY

- Assess ABC's, stabilize as needed, using C-Spine precautions as necessary.
- Pulse oximetry.

Oxygen as indicated.

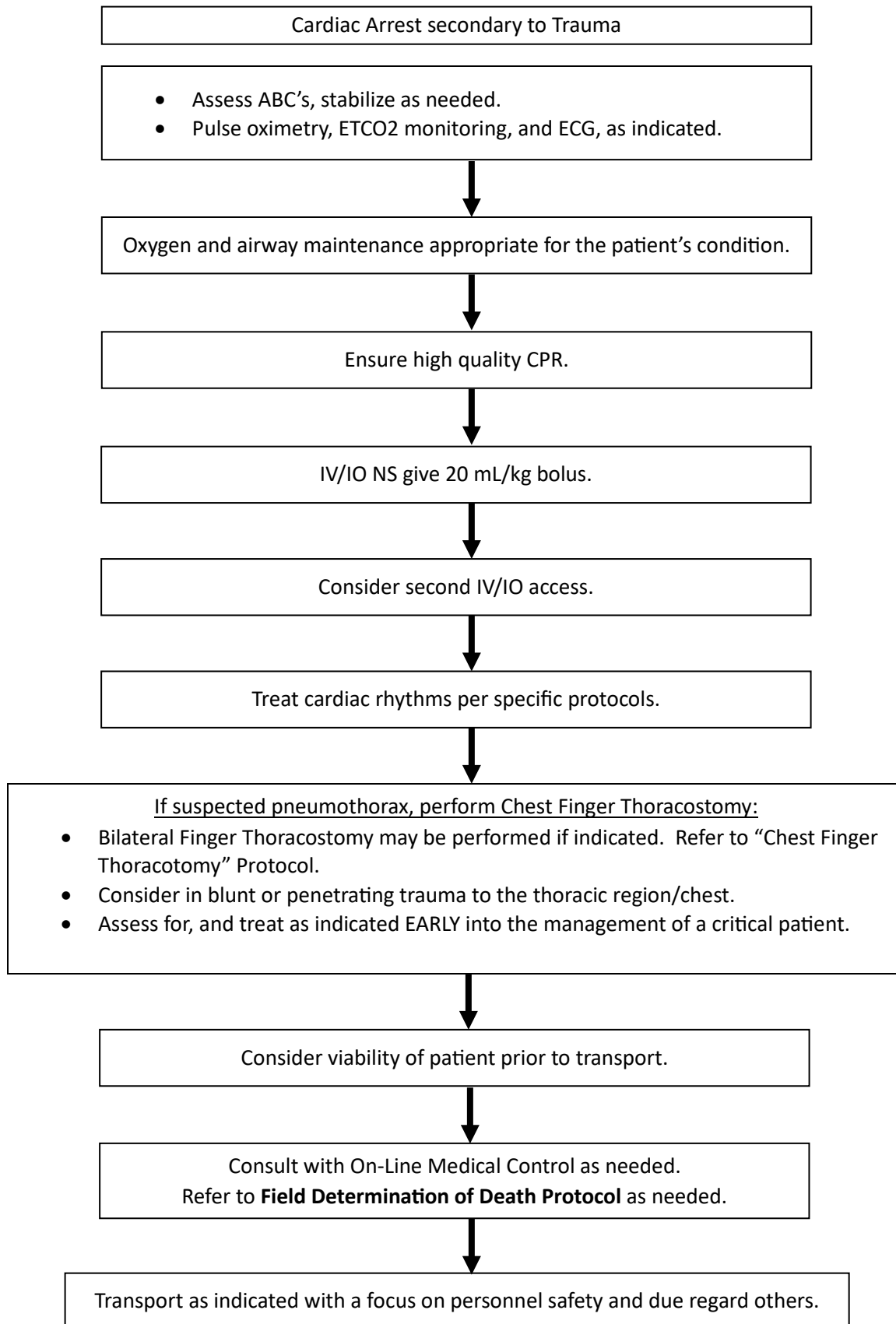
Cardiac Monitor.

Obtain IV access.

Treat for shock, if present – see Shock Protocol.

Reassure the patient.

Transport as indicated.

TRAUMATIC CARDIAC ARREST

UNCONTROLLABLE EXTREMITY BLEED / EXSANGUINATING HEMORRHAGE

BSI, SAFETY!!!

GLOVES AND GOGGLES TO PROTECT YOU FROM POTENTIAL EXPOSURE.Exsanguinating Hemorrhage?

- Bright red, heavy bleeding from arterial exposure.
- Amputation, de-gloving injuries when acute blood vessel exposure is noted.
- Penetrating Trauma with suspected internal hemorrhaging evidenced by acute swelling.

YES

Immediately **apply the arterial tourniquet** (TQ) – CAT, SOF-T, SWAT, ETC...

For upper extremities you will use 1 TQ.

For lower extremities, you may need 2 TQ's if so, apply them side by side using the same pressure.

CAT and SOF-T Tourniquets:

Apply tourniquets as proximal on the extremity as you can get them; make them tight as possible before turning the rods (windlass), turn approximately 3 times.

Then lock the rod in place, note the time applied, and recognize the need for pain management/sedation ASAP once hemorrhage is controlled. When effectively applied, the tourniquet is painful and the patient may attempt to remove it.

NO

Heavy Dark Red Oozing Bleeding from Suspected Venous Exposure?

1. Direct Pressure.
2. Elevate the Extremity.
3. Consider Tourniquet application as primary means of hemorrhage control if hemorrhage is significant or refractory to above treatment.

Consider use of **Hemostatic (Quick Clot) dressing/gauze as needed**. If needed, "Pack" the wound with the Hemostatic Gauze then bandage as needed.

- May be useful for junctional wounds.
- *These items are located in the RTF kit.*

Reference the **Pain Management** Protocol as needed... Fentanyl may be used in conjunction with Ketamine for severe pain.

TRANSPORT WITHOUT DELAY TO THE CLOSEST APPROPRIATE FACILITY.

Note that amputations or partial amputations should be transported to a Level 1 Trauma Center when possible.

TRANEXAMIC ACID (TXA) ADMINISTRATION – UNSTABLE TRAUMA PATIENTS

INDICATIONS:

- **Hypotension**, and other signs of shock, associated with known or suspected blood loss.
- **Less than 3 hours** since time of injury.

CONTRAINDICATIONS:

- More critical interventions need to be done for your patient (Do not delay to give TXA).
- Isolated head injury.
- Longer than 3 hours since time of injury.

Adult Dose (16 years and older):

Add 2 gram to 100 mL NS (or D5W) and infuse IV over 10 minutes.

ALL PATIENTS RECEIVING TXA SHOULD BE TRANSPORTED TO A FACILITY THAT IS CAPABLE OF MAINTAINING THE TXA INFUSION.

Both LifeFlight and AirEvac now have TXA that can be given in flight if needed, TXA can also be given in conjunction with blood products.

TRANEXAMIC ACID (TXA) ADMINISTRATION – STABLE COMPLICATED BLEEDING

INDICATIONS:

- Controlling Non-Arterial bleeding that is difficult to control with direct pressure, but not needing a tourniquet. (Example: Dialysis Fistula, etc.)
- Uncontrolled Epistaxis (Nose Bleed).

Application to a Wound

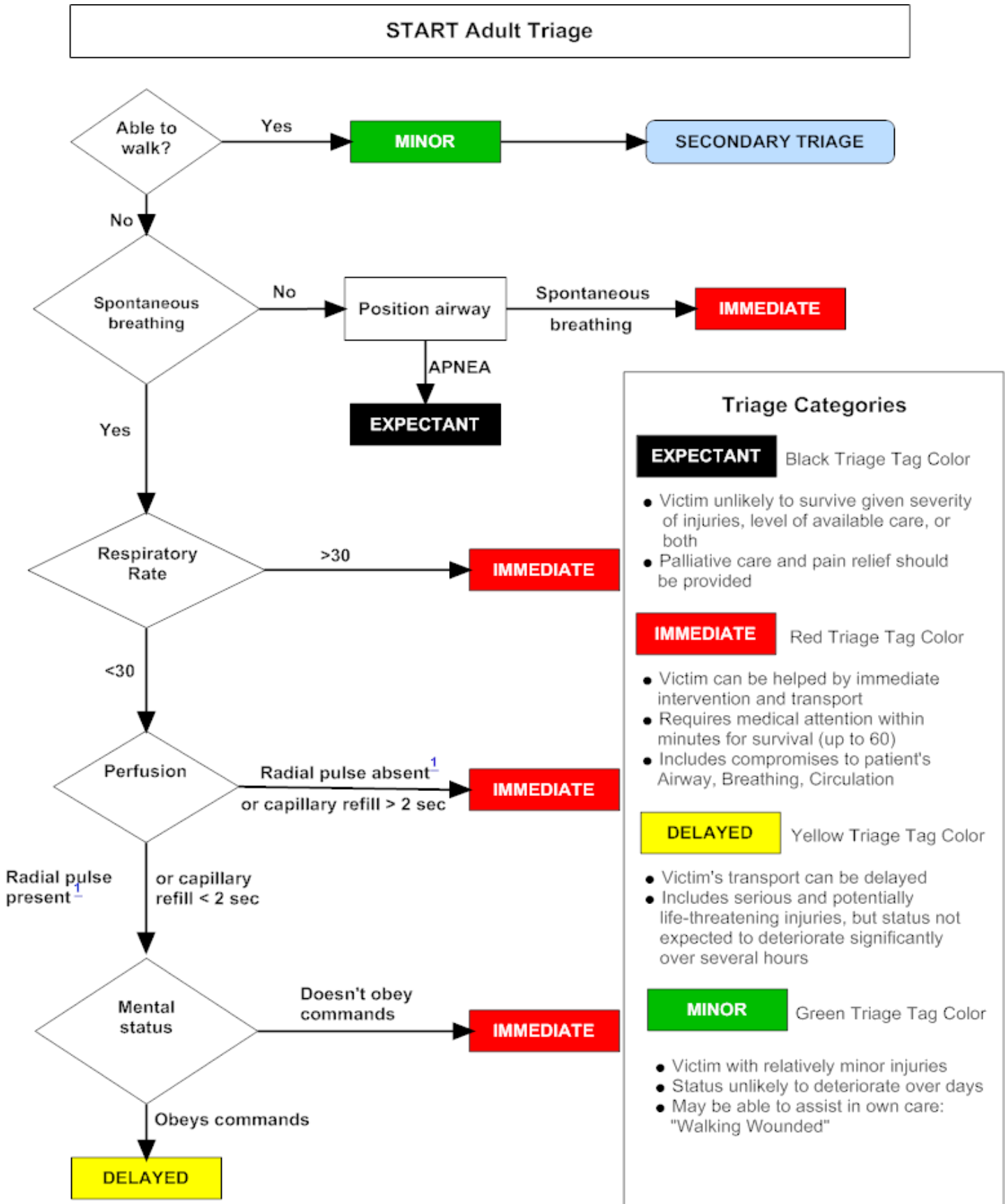
- Draw TXA from the vial, and saturate a sterile gauze pad with it.
- Apply directly over the wound with direct pressure.

Application for Epistaxis

- Administer 0.5-1 mL into the nostril by atomizer.
- Continue pinching the nose together on the soft tissue just below the bridge of the nose.

Application to Post-Dental Procedure

- Draw TXA from the vial, and saturate a sterile gauze pad with it.
- Apply directly over the wound.
- Be sure to instruct the patient to NOT swallow the gauze pad.

START ADULT TRIAGE

OB / GYN COMPLAINTS (NON – DELIVERY OR GYNECOLOGICAL ONLY)

- Patient Para (number of live births) and Gravida (number of pregnancies)?
- Term of pregnancy in weeks, EDC, Multiple births expected or history?
- Vaginal bleeding (how long and approximate amount)?
- Possible miscarriage/products of conception?
- Pre-natal medications, problems, and care?
- Last menstrual cycle?
- Any trauma prior to onset?
- Lower extremity edema?

1. Patient positioning appropriate for condition.
2. Oxygen and airway maintenance appropriate to the patient's condition.
3. Control hemorrhage as appropriate.
4. **IV Lactated Ringers (LR) TKO unless signs of shock, then 20 mL/kg fluid bolus of LR, consider Glucose check.**

Abruptio Placenta:

- Multiparity
- Maternal hypertension
- Trauma
- Drug use
- Increased maternal age
- History
- Vaginal bleeding with no increase in pain
- No bleeding with low abdominal pain

1. Position Patient in the left lateral recumbent position.
2. **Pregnant patients in 2nd and 3rd trimesters with blunt trauma (MVA-seatbelt), should never refuse transport, as they are at risk of abruption placenta.**

Placenta Previa:

- Painless bleeding which may occur as spotting or recurrent hemorrhage
- Bright red vaginal bleeding usually after 7th month
- History
- Multiparity
- Increased maternal age
- Recent sexual intercourse or vaginal exam
- Patient para (number of live births) and gravida (number of pregnancies)
- Term of pregnancy in weeks
- Pre-natal medications, problems, and care
- History of bed rest
- Placenta protruding through the vagina

1. Oxygen and airway maintenance appropriate to the patient's condition.
2. Position of comfort.

NOTE:

Any painless bleeding in the last trimester should be considered Placenta Previa until proven otherwise. If there are signs of eminent delivery membrane rupture is indicated followed by delivery of the baby. The diagnosis of eminent delivery depends on the visual presence of the baby's body part through the membrane.

OBSTETRIC EMERGENCIES – ACTIVE / IMMINENT DELIVERY

- Assess ABC's, stabilize as needed.
- Pulse oximetry.
- Cardiac monitor.

Apply oxygen 100% via NRB.

Notify the receiving facility ASAP
Ask pertinent history, as defined below.

Begin transport without delay, using caution to not risk the safety of EMS personnel and patients while in transport.

Obtain IV access – at least a 20 ga.
Between wrist and AC is preferred, LR bolus as needed to maintain normal BP.

Perineal examination (DO NOT perform an internal vaginal exam):

- If in active labor with no bleeding or crowning, transport as indicated.
- If vaginal bleeding and/or signs of shock, transport emergency.

Use gentle pressure to control delivery.

- Prevent an explosive delivery.

When head delivers, suction airway:

- Using bulb syringe, suction mouth, then nose!
- Suction any meconium from the airway ASAP!

Check for nuchal cord (umbilical cord around the neck)

- Carefully remove cord from neck if needed.

Check APGAR at 1 and 5 minutes post-delivery:

- REFERENCE on next page.

Allow placenta to deliver:

(May take up to 20 minutes normally)

- **Massage uterine fundus** (lower abdomen)
- Observe and treat signs of shock with increased deliver of oxygen and IV fluids.

- Reassess for postpartum hemorrhage.
- Reassess cord for bleeding, if bleeding add additional clamp and continue to monitor.
- Check glucose in baby's heel, **not finger.**
 - Normal neonate glucose level is >50.
- **Refer to the Neonatal Resuscitation Protocol if needed.**

Transport as indicated.

If delivery is imminent:

(Contractions q 3-5 minutes, lasting 30-60 sec)

- Prepare area for delivery (OB kit)
- Keep ambulance as warm as possible
- Prepare mother for delivery, preserve dignity.

*****Keep the baby level with the mother*****

Following delivery of the baby, ensure to:

- **Dry vigorously to stimulate breaths.**
- Maintain airway.
- Protect baby from fall risk.
- Protect from risk of hypothermia.
- Wrap baby in blanket to keep warm.
- Apply a head cover to baby to preserve body heat.
- Waiting 1-2 minutes before clamping the umbilical cord.
- **Clamp the umbilical cord at 8 and at 10 inches from the baby and cut between clamps once.**
- Consider allowing baby to nurse if mother is willing, and if there is no history of drug use.
- **Consult on-line medical control as needed.**

Pertinent history – gather SAMPLE, and:

- History of problems with pregnancy.
- Last menstrual period and due date.
- Number of pregnancies and deliveries.

OBSTETRIC EMERGENCIES – ACTIVE / IMMINENT DELIVERY (continued)Considerations:

1. The greatest risks to the newborn infant are airway obstruction and hypothermia. Keep the infant warm (silver swaddler or space blanket), dry, covered, and the infant's airway maintained with bulb syringe. Always remember to squeeze the bulb prior to insertion into the infant's mouth or nose.
2. The greatest risk to the mother is post-partum hemorrhage. Watch closely for signs of hypovolemic shock and excessive vaginal bleeding.
3. Spontaneous or induced abortions may result in copious vaginal bleeding. Reassure the mother, elevate legs, treat for shock, and transport.
4. Record a blood pressure and the presence or absence of edema in every pregnant woman you examine, regardless of chief complaint.

*****Complete patient care reports on BOTH mother and child.*****

OB EMERGENCIES – ABNORMAL / COMPLICATED DELIVERY

Please give report to the receiving facility as soon as possible.

Consult with On-line Medical Control as needed – High Risk OB patients, or those with complications may need to be transport to facilities that can provide specialty care. (*Vanderbilt, Centennial Women's, St. Thomas Mid-Town*)

Amniotic Sac Presentation:

1. Place patient in a position of comfort.
2. Amniotic sac –
 - if no fetus or head of fetus is visible, cover presenting sac with moist, sterile dressing.
3. Contact on-line medical control ASAP.

Nuchal Cord (umbilical cord around the neck):

1. Carefully remove cord from around baby's neck by slipping cord over the baby's head if possible.
2. Attempt to prevent creating a knot in the cord.
3. Prevent cord from strangulating the neonate during delivery.

Breech Presentation (Buttocks or body, not head coming out):

1. Place patient in best possible position and **transport EMERGENCY**.
2. Allow the delivery to progress spontaneously – DO NOT PULL!
3. Support the infant's body as it delivers.
4. If the head delivers spontaneously, deliver the infant as noted in 'Normal Delivery'.
5. If the head does not deliver within 3 minutes, insert a gloved hand into the vagina an airway for the infant.
6. DO NOT remove your hand until relieved by a Higher Medical Authority.
7. Contact on-line medical control ASAP.

Limb Presentation:

1. Position the mother in a supine position with the head lowered and pelvis elevated, **transport EMERGENCY**.

Meconium Staining:

1. Do not stimulate respiratory effort before suctioning the oropharynx.
2. Suction the **mouth then the nose (using a meconium aspirator)** while simultaneously providing Oxygen 100% by blow by method and while maintaining the airway appropriate to the patient's condition.
3. Obtain and APGAR score after airway treatment priorities. Score one minute after delivery and at five minutes after delivery. (Time permitting)

Prolapsed Cord:

1. Position the mother with hips elevated.
 - Knees to chest.
 - Transport in a supine position with her hips elevated as much as possible on pillows.
2. Instruct mother to pant with each contraction, which will prevent her from bearing down.
3. Check for a pulse in the cord.
 - **If no pulse** – insert a gloved hand into the vagina and gently push the infant's head off the cord. While pressure is maintained on the head cover the exposed cord with a sterile dressing moistened in saline. Transport immediately and DO NOT remove your hand until relieved by hospital staff.
 - **If pulse present** – cover exposed cord with moist dressing.
4. Contact on-line medical control ASAP if time and patient condition allows.

APGAR Score Reference Chart

APGAR Test Scoring			
	Score 0	Score 1	Score 2
A ppearance			
	Blue all over	Blue only at extremities	No blue coloration
P ulse	No pulse	<100 beats/min.	>100 beats/min.
G rimace			
	No response to stimulation	Grimace or feeble cry when stimulated	Sneezing, coughing, or pulling away when stimulated
A ctivity			
	No movement	Some movement	Active movement
R espiration	No breathing	Weak, slow, or irregular breathing	Strong cry

Score of 0-3 = Severely depressed, RESUSCITATE!!!

Score of 4-7 = Moderately depressed, STABILIZE

ASSESS A.P.G.A.R. SCORES AT 1 MINUTE AND 5 MINUTES POST PARTUM!

NEONATAL (NEWBORN) RESUSCITATION

This applies to term and pre-term newborn patients who fail to respond to initial stimulation and are in need of stabilization or resuscitation efforts. The standing order here applies to neonatal patients in general, which shall include patients who are less than 1 month of age.

Within the first 30 seconds:

- As soon as the baby is born: vigorously dry the infant and provide warmth (heat on high, cover with blanket).
- Position the infant to open the airway (sniffing position, careful not to hyperextend the neck).
- Clamp and cut the cord per OB/delivery protocol.
- If excessive secretions AND signs of compromise are present, clear with airway with a bulb syringe.
 - Routine suctioning of the oropharynx and nasal pharynx is no longer recommended.
 - If meconium staining is present AND the newborn is not vigorous (weak or absent respiratory efforts, weak or absent muscle tone, heart rate less than 100/min), tracheal suctioning may be considered.
- Stimulate breathing (flicking the soles of the baby's feet or rubbing the baby's back).



Assess respirations:

- If inadequate or gasping respirations are present, assist ventilations at a rate of 40 to 60 breaths per minute using an appropriately sized BVM attached to high flow O₂. (Careful to not over-inflate).
- If neonate is premature, titrate O₂ flow, initiate on room air, do not exceed 65% FiO₂ (6 lpm)
- If the respirations are shallow and slow, attempt a 1-minute period of stimulation while administering oxygen via blow-by method.
 - If respirations do not increase, initiate positive pressure ventilations as directed above.



Assess heart rate:

- If heart rate is less than 60 beats per minute, begin chest compressions, ventilate with 100% FiO₂.
 - Compression-to-ventilation ratio of 3:1 in neonatal resuscitation, compress at 120/min.
 - Compressions should be discontinued when heart rate is higher than 60 beats/min (with pulse).



Advanced Resuscitation:

- ❖ Consider advanced airway (one attempt only) for:
 - ❖ Persistent apnea
 - ❖ Central cyanosis
 - ❖ Bradycardia (HR < 100)
- ❖ If HR persistently < 60:
 - ❖ Continue CPR.
 - ❖ Initiate IV/IO normal saline.
 - ❖ Administer 1:10,000 Epinephrine 0.02 mg/kg (0.2 mL/kg) IV/IO every 3-5 minutes as needed.
- ❖ Obtain blood glucose level (perform heel stick, not finger), if < 50, administer Dextrose 10% 2-4 mL/kg IV/IO.
- ❖ Consider IV isotonic crystalloid fluid (NS, LR, D5W, etc.) administration at 10 mL/kg as needed.

PRE-ECLAMPSIA / ECLAMPSIA

- Assess ABC's, stabilize as needed.
- Pulse Oximetry.

Signs and Symptoms:

- Headache.
- Double-vision or seeing spots.
- Blood Pressure greater than 140/90.
- Generalized swelling of the face, arms, and legs.

Administer oxygen and begin external warming?

Monitor cardiac rhythm:

- Treat life-threatening arrhythmias.

Obtain IV access, consider 2 lines as patient's condition presents.

Try to have the patient relax as much as possible.

YES

Is systolic BP > 140 and/or diastolic BP > 90?

NO

Is the patient > 20 weeks pregnant OR < 2 weeks post-delivery without a history of seizures?

YES

NO

Mix 4 grams of Magnesium Sulfate in a 100 mL bag of NS and infuse IV/IO over 20 minutes.
This can be given in conjunction with Versed and Labetalol, if the patient becomes eclamptic and starts having seizures.

Administer **Labetalol** 10 mg slow IV.

May repeat at 20 mg slow IV every 10 minutes as needed.

Transport as indicated.

MATERNAL ACLS / CARDIAC ARREST IN PREGNANCY

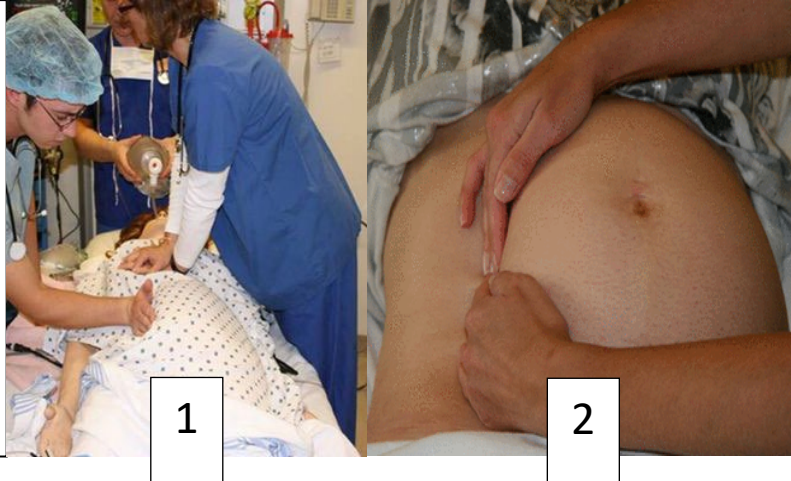
Management of the Unstable Pregnant Patient:

- The patient should be placed in a full left lateral decubitus position to relieve aortocaval compression.
- Intravenous access should be established above the diaphragm to ensure that the intravenously administered therapy is not obstructed by the gravid uterus.
- Precipitating factors should be investigated and treated.
- Pregnant patients are at an **increased risk of aspiration**, hypoxemia and pulmonary edema.
- Systemic hypotension can overwhelm compensatory mechanisms, which attempt to maintain uterine blood flow.

"Manual LUD"

Left Uterine Displacement

1. One-hand technique
2. Two-hand technique



Chest Compressions in Pregnancy:

- Guidelines for hand placement, rates, and ratios are the same as with non-pregnant adult patients.
- Keep supine, implement manual LUD during compressions, starting at ≈ 12 to 14 weeks of gestational age.
- There is no literature examining the use of mechanical CPR devices in pregnancy, this isn't advised at this time.

Transport / Operations:

- Transport to a facility that can perform a cesarean delivery may be required when indicated (For out-of-hospital cardiac arrest or cardiac arrest that occurs in a hospital not capable of cesarean delivery).
- If available, transport should be directed toward a facility that can perform PMCD, but transport should not be prolonged by >10 minutes to reach an ER with more capabilities. (PMCD = perimortem cesarean delivery).

Electrical Therapy (Defibrillation) Issues During Pregnancy:

- Current defibrillation protocols should be used in the pregnant patient as in the non-pregnant patient.
- The lateral defibrillator pad/paddle should be placed under the breast tissue when possible.

Breathing and Airway Management in Pregnancy:

- Oxygen reserves are lower and metabolic demands are higher; earlier ventilatory support may be necessary.
- Intubation starting with an ETT with a 6.0- to 7.0-mm inner diameter is recommended.
- Upper airway edema and friability (weakening of tissue) occur as a result of hormonal effects and may reduce visualization during laryngoscopy and increase the risk of bleeding.
- Renal blood flow increases, stressing the kidneys, increasing the tendency for pulmonary edema to develop.

Emergency (ACLS) Medications in Pregnancy:

- In the setting of cardiac arrest, no medication should be withheld because of concerns about fetal teratogenicity.
- Drug metabolism is altered in pregnancy. However, there are no current recommendations for altering dosages of routine ACLS drugs. **Use standard drugs and doses when indicated.**

FIELD DETERMINATION OF DEATH

Resuscitation is not to be attempted or continued, if a patient appears to be deceased, having the following signs:

- Rigor Mortis.
- Dependent Lividity.
- Decomposition of body tissue.
- Devastating, non-survivable injury(s) clearly incompatible with life:
 - Decapitation.
 - Incineration.
 - Brain matter visible.

Resuscitation may not be attempted or continued, if the following conditions are

A valid DNR order, advanced directive, P.O.S.T., P.O.L.S.T., P.O.A., or patient advocacy paperwork is present or produced and the patient is in full cardiac arrest.

- If the family states that the patient has a DNR but they cannot produce the paperwork, then On-Line Medical Control needs to be contacted to get an order to stop CPR.

An on-scene physician with appropriate identification or Medical Control issues an order to stop CPR.

Blunt traumatic cardiac arrest and the patient is in asystole.

Unwitnessed arrest with unknown amount of down time with asystole in 2 or more leads.

The paramedic may choose not to perform an EKG if obvious death is noted, because attempts need to be made to preserve any potential crime scene evidence.

If family member(s) / bystander(s) want resuscitative measures started, then begin and transport.

When in doubt, start CPR!

NOTE:

CPR prior to crew arrival can be stopped by the crew if the death is obvious. However, if the crew begins CPR, then it can only be stopped with direction for Medical Control. You should feel free to start or continue CPR when in doubt or if family insists.

SUPRAGLOTTIC AIRWAY (iGel) PROTOCOL**ADULT AND PEDIATRIC**

EMR	EMT	AEMT	PARAMEDIC
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Any licensed EMR, EMT-B, AEMT, Paramedic, or CCEMTP that operates under Sumner County EMS protocols and standing orders as an employee of EMS or a responder with an affiliated first responder agency may perform this skill if they have been properly trained, credentialed and documented.

Device Models:

Current inventory contains the iGel. A soft-tip suction catheter or OG tube can be inserted here to provide gastric suctioning / decompression.

Latex-Free Device:

The iGel is 100% latex free and should be considered safe to use on patients with latex sensitivity.

INDICATIONS:

- Adult or pediatric patients in need of prolonged ventilatory support that are not improving with 100% O₂ via BVM.
- Absence of a gag reflex.
- To be used as an alternate airway device in the absence of a paramedic or the inability to perform intubation.
- Can be used in medical or trauma patients.
- The iGel airway should be used as a first-line airway for adults with suspected COVID.

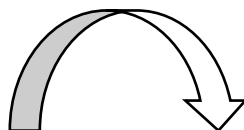
CONTRAINDICATIONS:

- Do not use the iGel airway to initially secure the airway for purposes of RSI.
- Do not use in responsive patients with an in-tact gag reflex.
- Do not use in patients who have ingested caustic substances.
 - Acids, Alkalis, Petroleum products.
- Do not use if the patient has a history of liver cirrhosis or esophageal varices.
- Do not use in patients with trismus, limited mouth opening, airway mass or abscess.

PRECAUTIONS:

- The iGel airway is not proven to protect the airway from the effects of regurgitation and aspiration. Orotracheal intubation by a paramedic remains the standard of care for all patients, adult or pediatric.
- Be sure to select the most appropriately sized iGel airway for the specific patient being treated. Reference the chart on this protocol to help choose the best size for each patient.

CONTINUED ON NEXT PAGE...



SUPRAGLOTTIC AIRWAY (iGel) PROTOCOL (continued)

ADULT AND PEDIATRIC

Directions for Insertion of the iGel Airway:

1. Stabilize ABC's as needed, Hyperoxygenate the patient with a BVM attached to high flow O₂, insert oral airway if appropriate, and suction as needed.
2. Prepare Equipment: Select the Appropriately Sized iGel Airway.

Size	Patient Criteria (Weight / Height)
1	Newborn (2-5 kg)
1.5	5-12 kg
2	10-25 kg
2.5	25-35 kg
3	30-60 kg
4	50-90 kg
5	90 + kg

Pediatric Color-Coded / Length Based Measurement Tape Identification for iGel Airways

Gray Size 1	Pink Size 1.5	Red Size 1.5	Purple Size 2	Yellow Size 2	White Size 2	Blue Size 2	Orange Size 2 or 2.5	Green Size 2.5 or 3
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3. Apply a water-based lubricant (KY jelly, etc.) to the beveled distal tip and posterior aspect of the tube. Be careful to not introduce lubricant in or near the ventilatory openings.
4. Open airway with appropriate maneuver, place patient's head in a sniffing position.
 - a. Use caution not to hyperextend the neck in patients with suspected C-spine injuries.
 - b. Pediatric patients with a proportionally larger head may need padding under the shoulder blades to help maintain positioning.
5. Insert the airway downward toward the feet. Without exerting excessive force, advance the airway until resistance is met and the integral bite block aligns with teeth or gums.
6. Confirm placement of the device using the following methods:
 - a. Colorimetric device (should turn yellow if getting exhaled CO₂ = good placement).
 - b. Bilateral chest rise and fall (condensation may be seen in the airway tube).
 - c. No epigastric sounds or gastric distention noted upon giving breaths.
 - d. Positive bilateral breath sounds heard upon giving breaths.
 - e. **Waveform capnography (AEMTs and Paramedics can perform) should be used as soon as possible to verify and monitor placement. AEMTs can only monitor the value of capnometry not the capnography waveform.**
7. Secure the device (ETT holder works well) and continue care / transport without delay.

Suction Catheter Sizes per Device

iGel Size	Catheter(s) Size
1.5	10 Fr
2	10-12 Fr
2.5	10-12 Fr
3	10-12 Fr
4	10-12 Fr
5	14 Fr

INTERFACILITY TRANSPORTATION OF BLOOD PRODUCTS

PURPOSE:

Patients may require transportation to another, Medical/Trauma Center with blood or blood products infusing. A paramedic or registered nurse must accompany these patients in the patient compartment, or physician trained in these procedures. The paramedic shall be knowledgeable in the administration of blood, blood products, adverse reactions, and all necessary equipment used in administering and regulating the blood products. Emergency medical technicians who have IV certification are not authorized to transport patients with admixtures, blood, or blood products. Prior to initiating transportation, the physician will provide the paramedic with written medical orders for the treatment of any adverse reaction(s) the patient might have. When transporting the patient, at least epinephrine, Benadryl, in the patient compartment. If the transporting paramedic has not received specific training, the paramedic may refuse to transport the patient with blood or blood products infusing. The ambulance service must maintain a record of all personnel completing this specialized training.

PROCEDURE:

- Blood should be infused within a 4-hour period, otherwise there is risk of clotting in the bag.
- Carefully check blood type for compatibility with the patient BEFORE beginning the transfusion.
- Check vital signs prior to the transfusion.
- The blood should be run through at least a 20-gauge IV catheter or larger with the blood hung three to four feet above the patient.
- The IV line should be flushed with Normal Saline prior to beginning the transfusion. **Blood should be administered only with Normal Saline IV fluid.** Dextrose causes red cells to clump, swell, and hemolyze; calcium (Lactated Ringer) may cause blood to clot.
- The transfusion should be initiated at a rate of 50 mL/hr, for the first ten minutes then as ordered by the referring physician.
- Patient condition and vital signs should be monitored closely during the transfusion.
- The blood should be mixed during the transfusion by inverting the bag occasionally.

After, the transfusion is completed, flush the IV tubing until clear with Normal Saline and maintain the IV as ordered by the referring physician. If a reaction occurs during the transfusion, terminate the transfusion immediately. Initiate the treatment ordered by the transferring physician and establish medical control as soon as possible. Save the donor blood for testing at the receiving facility.

ADVERSE REACTIONS:

- Circulatory Overloading: Dyspnea, increase in blood pressure, and jugular vein distention.
- Febrile Reaction: Chilling, fever, headache, flushing, tachycardia and anxiety.
- Septic Reaction: Chilling, fever, headache, tachycardia, and hypotension.

IMMUNOLOGIC REACTION:

- Flushing, itching, rash, urticaria, and asthmatic wheezing.
- Acute Hemolytic Reaction:
 - Severe reaction which may cause back pain.
 - Dyspnea, Hypotension.
 - Diaphoresis.
 - Cold skin.
 - Jugular Vein Distention.
 - Disseminated Intravascular Coagulation.
 - Death.

IF PROBLEMS OCCUR:

- Discontinue blood administration and flush tubing with normal saline.

CALL MEDICAL CONTROL TO CONSIDER:

- Benadryl 25 mg IV, if allergic signs and symptoms are present.
- Fluids if hypotensive.

CENTRAL VASCULAR ACCESS / INDWELLING CATHETER ACCESS

- *PICC lines or Midline caths may be accessed in unstable patients that IV/IO access cannot be rapidly achieved.*
- *Indwelling caths may be accessed in stable patients with poor IV access options or patient preference.*
- *If this access site has not been used (accessed, flushed) in the past 5 days, it should not be accessed at the time for EMS needs. The risk for clot development is too acute.*

PROCEDURE / PROTOCOL

Always access lines utilizing needless connectors or injection caps.

- Direct access breaches the system and increases chance of infection.



Always cleanse the injection cap with alcohol prior to access.

- Wipe vigorously for 30 seconds and allow drying.



How to assess for patency of catheter:

1. Unclamp the line, attempt to withdraw at least 10 ml of blood from the line, DISCARD that blood, do not re-infuse...
2. Attempt to flush the line with NS, using nothing smaller than a 10 ml syringe...
3. Re-clamp the line as you are completing the flush.
4. All lines should be clamped when not in use.



Treat PICC line / Mid-line Cath access as you would treat saline locks.



If you feel that you have breached the system for any reason:

- Inform the patient's primary in-hospital care giver.
- Document all actions taken.

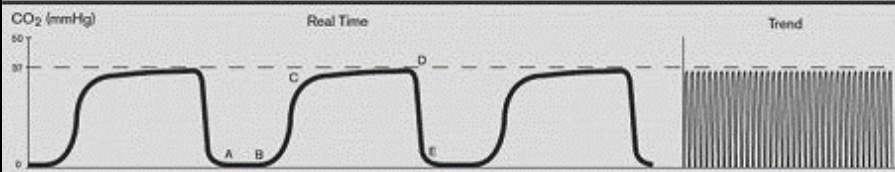


Document reason for PICC line / Mid-line access.

There are NO standing orders for accessing portacaths.

There are also NO standing orders for accessing Vas-Cath devices commonly seen in dialysis patients.

- **Do not attempt accessing these devices in the pre-hospital setting.**

CAPNOGRAPHY QUICK REFERENCE**Normal Capnogram****Normal EtCO₂: 35 – 45 mmHg**

The "normal" capnogram is a waveform which represents the varying CO₂ level throughout the breath cycle.

Waveform Characteristics:

A-B	Baseline	D	End-Tidal Concentration
B-C	Expiratory Upstroke	D-E	Inspiration
C-D	Expiratory Plateau		

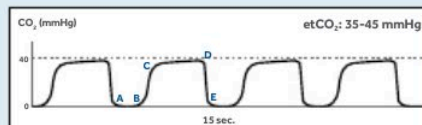
CAUSES OF ELEVATED EtCO ₂	CAUSES OF DECREASED EtCO ₂
METABOLISM Pain Hyperthermia Shivering	METABOLISM Hypothermia Metabolic acidosis
RESPIRATORY SYSTEM Respiratory insufficiency Respiratory depression COPD Analgesia/sedation	RESPIRATORY SYSTEM Alveolar hyperventilation Bronchospasm Mucus plugging
CIRCULATORY SYSTEM Increased cardiac output	CIRCULATORY SYSTEM Hypotension Sudden hypovolemia Cardiac arrest Pulmonary emboli
MEDICATIONS Bicarbonate administration	

NORMAL AND ABNORMAL etCO₂/CAPNOGRAPH WAVEFORMS**Normal Capnogram**

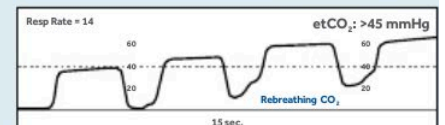
The normal capnogram is a waveform which represents the varying CO₂ level throughout the breath cycle.

Waveform Characteristics:

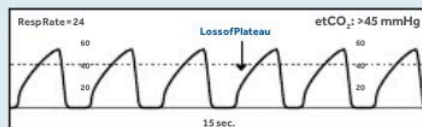
A-B: Baseline	D: End-Tidal Concentration
B-C: Expiratory Upstroke	D-E: Inspiration
C-D: Expiratory Plateau	

**Rebreathing CO₂****Other Possible Causes:**

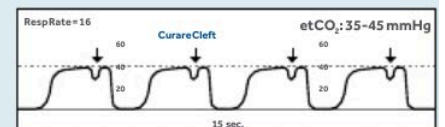
- Faulty expiratory valve
- Inadequate inspiratory flow
- Partial rebreathing
- Insufficient expiratory time

**Bronchospasm/Asthma****Other Possible Causes:**

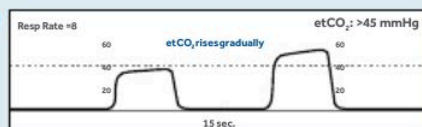
- Bronchospasm/COPD
- Obstruction in the expiratory limb of the breathing circuit
- Presence of a foreign body in the upper airway
- Partially kinked or occluded artificial airway

**Curare Cleft****Other Possible Causes:**

- Patient is mechanically ventilated
- Depth of cleft is proportional to degree of muscle relaxants

***Increasing etCO₂ (Hypoventilation)****Other Possible Causes:**

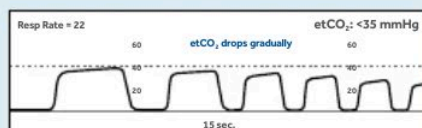
- Decrease in respiratory rate
- Decrease in tidal volume
- Increase in metabolic rate
- Rapid rise in body temperature (malignant hyperthermia)

**Cardiac Arrest****Other Possible Causes:**

- Decreased or absent cardiac output
- Decreased or absent pulmonary blood flow
- Sudden decrease in CO₂ values

***Decreasing etCO₂ (Hyperventilation)****Other Possible Causes:**

- Increase in respiratory rate
- Increase in tidal volume
- Metabolic acidosis
- Fall in body temperature

**Return of Spontaneous Circulation****Other Possible Causes:**

- Increase in cardiac output
- Increase in pulmonary blood flow
- Gradual increase in CO₂ production



*Assumes adequate circulation and alveolar gas exchange

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SPECIAL EVENT / FIRST – AID PROTOCOL (USE OF SPECIALTY ITEMS)**Overview:**

This protocol provides medical direction / standing orders for the usage of specialty items in the application of general patient care and first-aid. This protocol applies to patients being transported by Sumner County EMS and to any person receiving care for minor injuries during public events, stand-bys, and special operations.

Protocol for the following specialty items:

Waterjel brand burn gel - This item contains active ingredient Lidocaine HCL 2%, which is an external analgesic. This product may be used to provide temporary pain relief for superficial burns affecting less than 10% body surface area only (minor burns). If the patient was to develop rash or other suspected skin irritation, immediately wipe away with a dry cloth then flush with water.

Sting-Swabs - This item contains active ingredients Benzocaine 20% and Menthol 1%; both are topical analgesics. This product may be used to provide pain relief from insect bites and stings. Carefully monitor the patient for any indications of anaphylaxis. If the patient does not want EMS transport, complete a patient care report to include refusal of transport if the patient has any signs/symptoms indicating allergic reaction. Do not use in the eyes.

Bacitracin Ointment - This item contains active ingredient Bacitracin zinc 500 units, used as a first-aid antibiotic. This product may be used to prevent minor skin infections caused by small cuts, scrapes, or burns. Bacitracin works by stopping the growth of certain bacteria. It belongs to a class of drugs known as antibiotics. This product is intended for temporary / first-aid use only, and some wounds may require further medical treatment by a physician. The patient should be educated on the possible need for an updated tetanus shot and/or sutures for more serious soft tissue injuries. Minor wounds from cuts and scrapes should be cleansed appropriately prior to application of this product then covered with a sterile dressing/bandage.

Petroleum Gauze - This item contains a petroleum gauze strip, and may be used to dress minor wounds where clots or other fluids may adhere to regular gauze pads and dressings. This is to prevent the tissue/wound from sticking to the dressing and promote removal of the first-aid dressing when further medical attention is provided. This may also work well for other exudate (draining) wounds that require dressing/bandaging.

Hemostatic Gauze/Dressing – “Quick Clot” Hemostatic Gauze may be used to "pack" or cover heavily bleeding wounds. This product is for temporary external use to control traumatic bleeding. These specialty dressings should be avoided in minor head lacerations that seem to be bleeding heavily, yet are easily controlled with appropriate direct pressure. This is due to likely coagulation into the hair that will possibly complicate further medical treatment at the hospital.

SOG: Standard Medical Abbreviations

Effective Date: October 1, 2016, reviewed August 14, 2023, approved for continued use.

Approved: Chief of EMS

Rationale:

To assist in maintaining accurate and consistent documentation by utilizing abbreviations, the following policy shall be followed.

Standard Operating Guideline:

When using medical abbreviations in reports, personnel may only utilize abbreviations from Sumner County Emergency Medical Service's standard medical abbreviation list. Personnel shall refrain from using abbreviations not on this list. The list may be updated from time to time, so please look at the version number and date to ensure you have the most current list.

<u>ABBREVIATION</u>	<u>DEFINITION</u>
&	AND
@	AT
X	TIMES
—	NEGATIVE
+	PRESENT, YES, OR POSITIVE
#	NUMBER
=	EQUALS
>	GREATER THAN
<	LESS THAN
%	PERCENT

Abbreviation	Meaning
<u>A's</u>	
AAO	Awake, Alert, and Oriented
ABC	Airway, Breathing, and Circulation
ABD	Abdominal
AC	Antecubital
ACLS	Advanced Cardiac Life Support
AE	Air Evac (Aeromedical)
AED	Automated External Defibrillator
AEMT	Advanced Emergency Medical Technician
AFIB	Atrial Fibrillation
AHA	American Heart Association
AICD	Automatic Internal Cardiac Defibrillator
A.I.D.S	Acquired Immune Deficiency Syndrome
AKA	Above the Knee Amputation
AMA	Against Medical Advice
AMI	Acute Myocardial Infarction
Amt	Amount
AOS	Arrival On Scene
ARDS	Adult Respiratory Distress Syndrome
ASA	Aspirin
ASAP	As soon as possible

AV	Atrioventricular
AVPU	Alert, Verbal, Painful, Unresponsive
<u>B's</u>	
BBB	Bundle Branch Block
bid	Twice a day
BKA	Below the Knee Amputation
BLS	Basic Life Support
BM	Bowel Movement
BP	Blood Pressure
BPM	Beats per minute OR Breaths per minute
BRADY	Bradycardia
BS	Blood Sugar
BVM	Bag Valve Mask
<u>C's</u>	
C1 – C7	Cervical Vertebrae 1 - 7
CA	Cancer
CABG	Coronary Artery Bypass Grafting
CAD	Coronary Artery Disease
CCEMT-P	Critical Care Paramedic
C-COLLAR	Cervical Collar
C/C	Chief Complaint
CCU	Critical Care Unit
CEBBS	Clear Equal Bilateral Breath Sounds
C.H.A.R.T	Complaint, History, Assessment, Rx, Tx
CHF	Congestive Heart Disease
CID	Cervical Immobilization Device
cm	Centimeter
CMC	Centennial Medical Center
CNS	Central Nervous System
C/O	Complaining of
COPD	Chronic Obstructive Pulmonary Disease
C.P.	Cerebral Palsy
CP	Chest Pain
CPR	Cardiopulmonary Resuscitation
CSF	Cerebrospinal Fluid
CT	Computer Aided Tomography
CTFD	Cottontown Fire Department (squad 7)
CV	Cardiovascular
CVA	Cerebral Vascular Accident
D/C	Discontinue
<u>D's</u>	
D25	Dextrose 25%
D50	Dextrose 50%
DM	Diabetes Mellitus
DNR	Do Not Resuscitate
DOA	Dead On Arrival
DT	Delirium Tremens
<u>E's</u>	
ECG	Electrocardiogram
ED	Emergency Department
EEG	Electroencephalogram
EJ	External Jugular

EKG	Electrocardiogram
EMA	Emergency Management Agency
EMS	Emergency Medical Services
EMT	Emergency Medical Technician
EMT-IV	Emergency Medical Technician – Intravenous Therapy
EMT-P	Emergency Medical Technician – Paramedic
EOR	End of Report
ER	Emergency Room
ETA	Estimated time of arrival
etc	And so forth
ETOH	Ethly Alcohol
ETT	Endotracheal Tube
<u>F's</u>	
F	Fahrenheit
Fe	Iron
fl. oz.	Fluid Ounce
FNP	Family Nurse Practitioner
Fx	Fracture
<u>G's</u>	
GCFD	Gallatin Community Fire Department (squad 6)
GFD	Gallatin Fire Department
GI	Gastrointestinal
gm	Gram
GOA	Gone on arrival
GPD	Gallatin Police Department
Gravida	Number of Pregnancies
GSW	Gunshot Wound
GTT	Drop
GVFD	Goodlettsville Fire Department
GVPD	Goodlettsville Police Department
Gyn	Gynecology
<u>H's</u>	
H	Hour
HA	Headache
HBV	Hepatitis B
HEENT	Head, Eyes, Ears, Nose, and Throat.
HFD	Hendersonville Fire Department
HIFD	Highland Fire Department (squad 1)
HIV	Human Immunodeficiency Virus
HMC	Hendersonville Medical Center
HPD	Hendersonville Police Department
HR	Heart Rate
HTN	Hypertension
HX	History
<u>I's</u>	
IC	Incident Command
ICP	Intracranial Pressure
ICS	Intercostal Space
ICU	Intensive Care Unit
ID	Intradermal
IDDM	Insulin Dependent Diabetes Mellitus
IM	Intramuscular

IO	Intraosseous
IV	Intravenous
IVP	Intravenous Push
<u>J's</u>	
JVD	Jugular Vein Distention
K.E.D.	Kendrick Extrication Device
<u>K's</u>	
K+	Potassium
KCL	Potassium Chloride
Kg	Kilogram
KVO	Keep Vein Open
<u>L's</u>	
L1 – L5	Lumbar Vertebrae 1 - 5
L	Left
L&D	Labor and Delivery
Lab	Laboratory
LAD	Left Anterior Descending Artery
LBBB	Left Bundle Branch Block
Lbs.	Pounds
LF	Life Flight (Aeromedical)
Lg	Large
LLE	Left Lower Extremity
LLL	Left Lower Lobe
LLQ	Left Lower Quadrant
LMP	Last Menstrual Period
LOC	Level of Consciousness OR Loss of Consciousness
lpm	Liters Per Minute
LPN	Licensed Practical Nurse
LR	Lactated Ringer
LRS	Lactated Ringer Solution
LSB	Long Spine Board
LUE	Left Upper Extremity
LUL	Left Upper Lobe
LUQ	Left Upper Quadrant
<u>M's</u>	
m	Meter
MAEW	Moves All Extremities Well
mcg	Microgram
MCL	Mid-clavicular Line
M.D.	Medical Director
MD1	Medical Director (Ray Pinkston, MD)
MD2	Medical Director (Duane Harrison MD)
ME	Medical Examiner
MEDS	Medication
mEq	Milliequivalent
MFD	Millersville Fire Department
mg	Milligram
MGSO4	Magnesium Sulfate
MI	Myocardial Infarction
mL	Milliliter
mm	Millimeter
MM	Mile Marker

MPD	Millersville Police Department
MR	Mental Retardation
MRI	Magnetic Resonance Imaging
MS	Multiple Sclerosis
MVA	Motor Vehicle Accident
MVC	Motor Vehicle Collision
<u>N's</u>	
N/A	Not Applicable
N/V	Nausea / Vomiting
N/V/D	Nausea / Vomiting / Diarrhea
NAD	No Apparent Distress
NC	Nasal Cannula
NEURO	Neurological
NFD	Nashville Fire Department
NGT	Nasogastric Tube
NIDDM	Non-insulin Dependent Diabetes Mellitus
NKDA	No Known Drug Allergies
NOFD	Number One Fire Department (squad 12)
NRB	Non-Rebreather
NS	Normal Saline
NSR	Normal Sinus Rhythm
NTG	Nitroglycerin
<u>O's</u>	
O2	Oxygen
OB	Obstetric
OGFD	Oak Grove Fire Department (squad 8)
OTC	Over The Counter
Oz	Ounce
<u>P's</u>	
PA	Physician Assistant
PAC	Premature Atrial Contraction
PARA	Number of Live Births
PAT	Pediatric Assessment Triangle
PCI	Percutaneous Coronary Intervention – Cardiac Cath
PCO2	Partial Pressure of Carbon Dioxide
PE	Pulmonary Embolism
PED	Pediatric
PERRL	Pupils, Equal, Round, and Reactive to Light
PFD	Portland Fire Department
PID	Pelvic Inflammatory Disease
PJC	Premature Junctional Contraction
PMC-ER	Portland Medical Center – Emergency Room
P.M.	Afternoon and Evening
PMHX	Past Medical History
PMS	Pulse, Movement, and Sensation
p.o.	By mouth
pO2 or P02	Partial Pressure of Oxygen
POC	Position of Comfort
POV	Personal Owned Vehicle
PPD	Portland Police Department
PRN	As Needed
PSVT	Paroxysmal Supraventricular Tachycardia

PT	Patient
PTA	Prior to Arrival
PTD	Prior to Departure
PUTS	Patient Unable to Sign
PVC	Premature Ventricular Contraction
PVD	Peripheral Vascular Disease
P/W/D	Pink, Warm, and Dry
<u>Q's</u>	
q	Every
qt	Quart
<u>R's</u>	
R	Right
RBBB	Right Bundle Branch Block
RLE	Right Lower Extremity
RLL	Right Lower Lobe
RLQ	Right Lower Quadrant
RML	Right Middle Lobe
RN	Registered Nurse
R/O	Rule Out
ROM	Range of Motion
RT	Respiratory Therapy
RUE	Right Upper Extremity
RUL	Right Upper Lobe
RUQ	Right Upper Quadrant
RX	Treatment, Prescribed for, Therapy
<u>S's</u>	
S1 – S5	Sacrum Vertebrae 1 - 5
SAMPLE	S/S, Allergies, Medications, Past Medical Hx, Last Oral Intake, Events Leading Up To
SaO2	Oxygen Saturation by blood analysis (blood gas)
SCSO	Sumner County Sheriff Office
SEFD	South East Fire Department (squad 3)
SIFD	Shackle Island Fire Department (squad 11)
SL	Sublingual
SLMC	Skyline Medical Center
SMC	Summit Medical Center
SOB	Short of Breath
SQ	Subcutaneous
SRMC	Sumner Regional Medical Center
SRO	School Resource Officer
S/S	Signs and Symptoms
ST	Sinus Tachycardia
STEMI	ST Elevation Myocardial Infarction
STMT	St. Thomas Mid-town Hospital (formerly Baptist)
STW	St. Thomas West Hospital
SVT	Supraventricular Tachycardia
<u>T's</u>	
T1 - T12	Thoracic Vertebrae 1 - 12
TBI	Traumatic Brain Injury
THP	Tennessee Highway Patrol
TIA	Transient Ischemic Attack
t.i.d.	Three Times A Day
TKO	To Keep Open

tPA	Tissue Plasminogen Activator
TSI	Total Spinal Immobilization
TVI	Total Volume Infused
Tx	Treatment
<u>U's</u>	
U	Unit
UMC	University Medical Center
UOA	Upon Our Arrival
URI	Upper Respiratory Infection
UTI	Urinary Tract Infection
<u>V's</u>	
VA	Veteran Affairs
VF	Ventricular Fibrillation
VT	Ventricular Tachycardia
VUMC	Vanderbilt University Medical Center
<u>W's</u>	
WCFD	White House Community Fire Department (51)
WFD	Westmoreland Fire Department
WHFD	White House Fire Department (City)
WHPD	White House Police Department
WNL	Within Normal Limits
WPD	Westmoreland Police Department
Wt	Weight
<u>X,Y,Z's</u>	N/A

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- Edited and updated by Sumner County EMS Protocol Review Committee with approval from Medical Director Ray Pinkston, MD and Co-Medical Director Duane E. Harrison, MD